MAVYN
PEPTIDES

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Peptide Training Packet

Comprehensive Rep Training — Clinical Research & Sales Guide

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Mavyn Peptide Training Library

Version 1.0 | Prepared by Fernanda | March 2026

Complete rep training sheets for all 18 Mavyn peptide products. Each sheet contains: mechanism of action, clinical research with citations, dosing protocols, side effects, contraindications, stacking notes, and a full FAQ section.

■ Training Sheet Index

File	Peptide	SKU(s)	One-Line Summary
[selank-training.md](./selank-training.md)	**Selank**	SK10	Synthetic anxiolytic peptide that inhibits enkephalinase, boosts BDNF, and calms anxiety without sedation or addiction — ideal for anxiety, focus, and immune support
[semax-training.md](./semax-training.md)	**Semax**	XA5	ACTH-derived nootropic with the strongest BDNF upregulation in its class; approved in Russia for stroke recovery; the "clean stimulant" for focus, drive, and neuroprotection
[tb500-training.md](./tb500-training.md)	**TB-500**	TB5, TB10	Synthetic Thymosin Beta-4 fragment that drives systemic angiogenesis, satellite cell recruitment, and anti-fibrotic healing — the broadest-spectrum tissue repair peptide available
[vip-training.md](./vip-training.md)	**VIP**	VIP10	Vasoactive Intestinal Peptide — a master anti-inflammatory neuropeptide targeting MCAS, autoimmune conditions, PAH, IBD, and circadian rhythm via Treg induction and NF-κB suppression
[retatrutide-training.md](./retatrutide-training.md)	**Retatrutide**	RT5, RT10, RT15, RT20	Eli Lilly's GLP-1 + GIP + Glucagon triple agonist; produced 24%+ body weight loss in Phase 2 trials and 81% reduction in liver fat — the most powerful metabolic peptide available
[nad-training.md](./nad-training.md)	**NAD+**	NJ500, NJ1000	Fundamental cellular energy coenzyme that activates sirtuins, fuels PARP DNA repair, and drives mitochondrial function; injectable delivery provides 10–25x bioavailability vs. oral NMN

[melanotan1-training.md](./melanotan1-training.md)	**Melanotan I**	MT1	FDA/EMA-approved (Scenesse) MC1R-selective peptide that produces safe sunless tanning via eumelanin induction with no sexual side effects — also treats EPP, vitiligo, and photoprotection
[melanotan2-training.md](./melanotan2-training.md)	**Melanotan II**	ML10	Non-selective melanocortin agonist producing potent tanning AND significant libido/erection effects via MC4R; pharmaceutical derivative PT-141 (Vyleesi) is FDA-approved for female HSDD
[motsc-training.md](./motsc-training.md)	**MOTS-c**	MS10, MS15	Mitochondria-encoded peptide that activates AMPK (the "longevity switch") — circulating levels correlate with human longevity; reverses age-related exercise decline in animal models
[ipamorelin-training.md](./ipamorelin-training.md)	**Ipamorelin**	IP5, IP10	The cleanest GH secretagogue available — GHS-R1a agonist producing selective, pulsatile GH release without cortisol, prolactin, or ACTH elevation; ideal for anti-aging, body composition, and sleep
[bpc157-training.md](./bpc157-training.md)	**BPC-157**	(Stack component)	15-amino acid gastric peptide with 150+ published studies; heals tendons, muscles, nerves, and gut via FAK-paxillin/NO/VEGF pathway; uniquely stable for oral delivery; core component of KLOW and GLOW
[ghkcu-training.md](./ghkcu-training.md)	**GHK-Cu**	CU50, CU100	Naturally declining copper tripeptide that modulates 4,000+ aging genes, rebuilds collagen/elastin, reduces fibrosis, promotes hair growth, and delivers copper to SOD-1/cytochrome C — 40+ years of research
[epithalon-training.md](./epithalon-training.md)	**Epithalon**	ET10, ET50	Pineal-derived tetrapeptide with the only published evidence of telomere elongation in human cells; 15–25% lifespan extension in animals; restores melatonin and sleep; 35 years of institutional research
[fst344-training.md](./fst344-training.md)	**FST-344**	FST344	Follistatin 344 — endogenous myostatin inhibitor; releases the genetic ceiling on muscle growth; Phase 1/2 clinical data in muscular dystrophy shows real-world muscle hypertrophy; also anti-fibrotic

[igf1lr3-training.md](./igf1lr3-training.md)	**IGF-1 LR3**	IGF	Long-acting analogue of Insulin-like Growth Factor-1 with 20–30 hour half-life; activates mTOR/satellite cells for muscle growth; FDA-approved (Mecasermin) in pharmaceutical form for GH insensitivity
[hgh-training.md](./hgh-training.md)	**HGH**	H10	FDA-approved recombinant human growth hormone (somatropin); drives body composition, bone density, cognition, cardiovascular health, and anti-aging via GH receptor signaling and IGF-1 production
[klow-stack-training.md](./klow-stack-training.md)	**KLOW Stack**	KLOW (80mg)	BPC-157 + GHK-Cu + TB-500 + KPV — premium 4-peptide recovery stack; KPV (α-MSH C-terminal fragment) adds MC1R/MC3R → NF-κB suppression and gut barrier protection; ideal for IBD/MCAS + injury
[glow-stack-training.md](./glow-stack-training.md)	**GLOW Stack**	GLOW (70mg)	BPC-157 + GHK-Cu + TB-500 — foundational 3-peptide recovery and anti-aging stack; covers all three phases of tissue repair: healing signal (BPC), vascular delivery (TB-500), collagen quality (GHK-Cu)

■ Quick Reference: Client Profile → Peptide

Client Need	Top Recommendation	Supporting Add-On
Weight loss (significant)	Retatrutide	Ipamorelin (muscle preservation), BPC-157 (GI)
Athletic injury recovery	GLOW or KLOW	Ipamorelin, IGF-1 LR3
Anti-aging (comprehensive)	Epithalon + GHK-Cu	NAD+, MOTS-c
Anxiety / brain fog	Selank + Semax	NAD+, BPC-157
Cognitive performance	Semax	Selank, NAD+
Gut issues (IBD, leaky gut)	KLOW	VIP, BPC-157 oral
Muscle building	FST-344 + IGF-1 LR3	Ipamorelin, HGH
Tanning (no sexual effects)	Melanotan I	GHK-Cu (skin quality)
Tanning + libido	Melanotan II	—
MCAS / mast cell	VIP	KLOW (KPV component)
Longevity / healthspan	Epithalon + MOTS-c + NAD+	GHK-Cu, Ipamorelin
Body recomposition	Retatrutide + HGH + FST-344	Ipamorelin, MOTS-c
Sleep issues	Epithalon	Selank, Ipamorelin
Bone density	Ipamorelin + HGH	FST-344, MOTS-c

Post-surgical recovery	KLOW	HGH, NAD+
Autoimmune conditions	VIP	KLOW (KPV), Selank

■ Reconstitution Quick Reference

SKU	Vial Size	Recommended Bac Water	Concentration
SK10	10mg	2 mL	5 mg/mL
XA5	5mg	2 mL	2.5 mg/mL
TB5	5mg	1 mL	5 mg/mL
TB10	10mg	2 mL	5 mg/mL
VIP10	10mg	1 mL	10 mg/mL
RT5–RT20	5–20mg	2 mL	2.5–10 mg/mL
NJ500	500mg	5 mL	100 mg/mL
NJ1000	1000mg	10 mL	100 mg/mL
MT1	10mg	2 mL	5 mg/mL
ML10	10mg	2 mL	5 mg/mL
MS10	10mg	1 mL	10 mg/mL
MS15	15mg	1.5 mL	10 mg/mL
IP5	5mg	2 mL	2.5 mg/mL
IP10	10mg	2 mL	5 mg/mL
CU50	50mg	5 mL	10 mg/mL
CU100	100mg	10 mL	10 mg/mL
ET10	10mg	1 mL	10 mg/mL
ET50	50mg	5 mL	10 mg/mL
FST344	1mg	1 mL	1 mg/mL (1000 mcg/mL)
IGF	1mg	1 mL	1 mg/mL (1000 mcg/mL)
H10	10 IU	1 mL	10 IU/mL
KLOW	80mg	3 mL	~27 mg/mL
GLOW	70mg	3 mL	~23 mg/mL
WA10	Bacteriostatic Water 10mL	—	Diluent

■■ Universal Contraindications (All Peptides)

1. **Active malignancy** — angiogenic peptides (BPC-157, TB-500, GHK-Cu, FST-344, IGF-1 LR3, HGH) are contraindicated; others require clinical judgment
2. **Pregnancy** — insufficient safety data for all peptides in this portfolio
3. **Nursing** — insufficient safety data
4. **History of medullary thyroid carcinoma or MEN2** — GLP-1 agonists (Retatrutide) are contraindicated

■ Key Differentiators: High-Margin Highlight

Per internal analysis, the following peptides represent high-margin products (~81.8% margin) and should be primary focus for reps:

- **Selank (SK10)**
- **Semax (XA5)**
- **Epithalon (ET10)**

These three are also highly unique, research-validated, and difficult for clients to source independently — ideal for client retention and recurring orders.

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Prepared by Fernanda | Mavyn Business Operations | March 2026

BPC-157 (Body Protection Compound 157) — Mavyn Rep Training Sheet

Category: Tissue Repair / GI Healing / Neuroprotective Peptide | (Component of KLOW & GLOW Stacks)

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1. What Is BPC-157?

BPC-157 (Body Protection Compound-157) is a synthetic pentadecapeptide (15 amino acids: GEPPPGKPADDAGLV) derived from a protective protein found in human gastric juice. It was discovered and extensively studied by Dr. Predrag Sikiric and his team at the University of Zagreb, Croatia, who have published over 150 peer-reviewed papers on the compound.

BPC-157 is stable in gastric acid (unlike most peptides) because of its origin in the gastric environment — it was literally discovered in the gastric protective protein fraction. This stability makes both oral and injectable routes viable.

Mechanism of Action (Molecular/Cellular Level)

Nitric Oxide (NO) System Modulation:

BPC-157 is one of the most potent modulators of the nitric oxide system discovered. It activates eNOS (endothelial nitric oxide synthase) via the FAK-paxillin pathway, producing highly localized NO at injury sites. This:

- Promotes angiogenesis (new blood vessel formation at wound sites)
- Vasodilates local blood supply to injured tissues
- Drives fibroblast migration and proliferation
- Activates local stem cell recruitment

FAK-paxillin Pathway Activation:

BPC-157 directly phosphorylates Focal Adhesion Kinase (FAK) and paxillin — cytoskeletal signaling proteins that regulate cell migration, proliferation, and extracellular matrix attachment. This is the primary mechanism by which BPC-157 accelerates tissue healing at virtually every site it reaches.

Tendon Healing — Tendon-to-Bone Integration:

BPC-157 upregulates both VEGFR2 (vascular endothelial growth factor receptor 2) and FGFR2 (fibroblast growth factor receptor 2) in tenocytes and tendon fibroblasts — the two most critical growth factor signaling pathways for tendon repair. Studies specifically show enhanced collagen fiber organization, improved tendon-to-bone insertion strength, and faster functional recovery.

Gut Healing (Cytoprotective):

In the GI tract, BPC-157:

- Stimulates growth of intestinal villus epithelium
- Increases expression of tight junction proteins (claudin, occludin, ZO-1) — repairing "leaky gut"
- Suppresses gastric acid hypersecretion
- Promotes ulcer healing by stimulating mucosal blood flow
- Activates enteric nervous system (ENS) repair

Nerve Regeneration:

BPC-157 promotes peripheral nerve regeneration by stimulating Schwann cell activity, axonal sprouting, and remyelination. In crush injury models, functional nerve recovery is dramatically accelerated. Also promotes neurogenesis in the CNS and has

antidepressant-like effects via dopamine system modulation.

Dopamine & Serotonin System Repair:

Uniquely, BPC-157 modulates both dopaminergic and serotonergic transmission — it can reverse or mitigate drug-induced neurotoxicity (amphetamine, cocaine damage), and has demonstrated antidepressant and anxiolytic properties in animal models. This is unrelated to its tissue repair properties and is an independent mechanism.

Anti-Inflammatory: Potently suppresses COX-2 and NF-κB, reducing PGE2 and inflammatory cytokines. However — like TB-500 — it is "pro-healing" rather than purely anti-inflammatory; it directs inflammation toward productive healing rather than simply suppressing it.

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2. Clinical Research & Trials

Study 1: Ulcerative Colitis (Animal / Human Pilot)

- **Extensive animal data:** BPC-157 consistently heals colitis, Crohn's-like intestinal lesions, and gastric ulcers in rat and mouse models — across dozens of published studies from Dr. Sikiric's lab at Zagreb
- **Mechanism:** Multiple: tight junction restoration, mucosal angiogenesis, ENS repair, anti-inflammatory
- **Human pilot data:** Small case series in IBD patients administered BPC-157 orally showed significant clinical improvement in Crohn's disease scores; formal RCT pending
- **Citation:** Sikiric P et al., *Journal of Physiology (Paris)*, 2000–2018 (extensive series)

Study 2: Tendon Healing — Medial Collateral Ligament (MCL) Transection

- **Model:** Rat MCL complete transection
- **Treatment:** BPC-157 10 mcg/kg/day SubQ for 2 weeks
- **Results:** Histological healing score significantly superior to controls at 2 and 4 weeks. Collagen fiber alignment normalized. Return to functional load-bearing significantly faster. Comparable results in Achilles tendon model.
- **Citation:** Sikiric P et al., *Journal of Orthopaedic Research*, 2003.

Study 3: Peripheral Nerve Crush Injury

- **Model:** Rat sciatic nerve crush
- **Results:** BPC-157 treated animals showed:
- Motor function recovery 40% faster
- Sensory function recovery 35% faster
- Myelination quality significantly improved at 28 days
- Schwann cell density at injury site doubled
- **Citation:** Gjurasin M et al., *Regulatory Peptides*, 2010.

Study 4: NSAID and Corticosteroid Toxicity Protection

- **Model:** Rat models of NSAID (indomethacin), aspirin, and corticosteroid-induced GI damage
- **Results:** BPC-157 completely prevented or reversed:
- NSAID-induced gastric ulcers
- Corticosteroid-induced muscle and tendon damage (a critical clinical application — steroid-treated athletes/patients)
- Alcohol-induced GI injury
- **Clinical implication:** Clients on long-term NSAIDs or corticosteroids can benefit from GI and tissue protection

Study 5: Muscle Healing — Crush and Incision Models

- **Results:** BPC-157 accelerated complete skeletal muscle healing — both crush and incision injuries — with measurably better force generation and reduced scar formation at 4 weeks vs. controls.
- **Citation:** Novinscak T et al., *Annals of Surgery*, 2008.

Study 6: Dopamine System — Antidepressant Effects

- **Model:** Haloperidol-induced catalepsy and depressive behavior in rats
- **Results:** BPC-157 reversed haloperidol-induced symptoms, suggesting dopamine D2 receptor modulation. Also showed antidepressant effects in forced swim test and tail suspension test comparable to fluoxetine.
- **Citation:** Sikiric P et al., *Progress in Neuro-Psychopharmacology*, 2011.

Study 7: Bone Healing — Femur Fracture

- **Results:** BPC-157 accelerated callus formation, improved bone mineral density at fracture site, and reduced time to complete cortical bridging by approximately 30%.

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3. Primary Uses & Benefits

Goal	Mechanism
Gut healing (IBD, ulcers, leaky gut)	Tight junction repair, mucosal angiogenesis, ENS support
Tendon/ligament repair	VEGFR2/FGFR2 activation, collagen remodeling
Muscle healing	Satellite cell activation, FAK pathway
Nerve regeneration	Schwann cell activation, remyelination
Joint healing	Angiogenesis, anti-inflammatory, collagen
Bone repair	Callus formation, osteoblast activity
Neuroprotection	Dopamine/serotonin modulation
GI protection (NSAID/steroid users)	Mucosal cytoprotection

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4. Symptoms & Conditions It Targets

- Inflammatory bowel disease (Crohn's, UC)
- Gastric/duodenal ulcers
- Leaky gut / intestinal permeability
- IBS with inflammatory component
- GERD / esophagitis
- Tendinopathy / tendon tears
- Ligament injuries (ACL, MCL, etc.)
- Muscle strains and tears
- Neuropathy / nerve injuries
- Joint inflammation / synovitis
- Bone fractures
- NSAID/steroid-related GI damage

- Post-surgical healing support

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5. Key Selling Points (Rep Talking Points)

- ■ **The most research-backed healing peptide** — 150+ published studies by one of the most prolific peptide research groups globally; no other healing peptide has this evidence depth
- ■ **Oral AND injectable** — uniquely stable to gastric acid; oral capsules/tablets work for gut-specific issues; injections for systemic/musculoskeletal
- ■ **Gut + body simultaneously** — the only healing peptide with equally strong evidence for both GI repair and musculoskeletal repair; practitioners can address both systems at once
- ■ **Protects against NSAID/steroid damage** — patients on long-term anti-inflammatories who are simultaneously trying to heal; BPC-157 counteracts the GI toxicity of the very drugs they're taking
- ■ **Dopamine/mood benefits** — unexpected but significant; clients with injury-related depression or mood issues from chronic pain may notice improved mood alongside physical healing

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6. Common Protocols

Oral Protocol (Gut-Specific)

- **Dose:** 250–500 mcg BPC-157 orally, 2x/day (AM fasting, PM fasting)
- **Cycle:** 4–8 weeks for acute GI conditions; longer for IBD maintenance
- **Form:** Capsules (stable peptide survives gastric transit due to its acid-stable design)

SubQ Injection Protocol (Systemic / Musculoskeletal)

- **Dose:** 250–500 mcg SubQ (near injury site or abdomen), 1–2x/day
- **Loading:** 500 mcg/day for 2–4 weeks
- **Maintenance:** 250 mcg/day or EOD
- **Cycle:** 4–8 weeks acute; 8–12 weeks for chronic tendinopathy

Local Injection (Injury-Specific)

- **Dose:** 250–500 mcg injected near (not into) damaged tendon or joint
- **Frequency:** 1–2x/day
- **Targeted areas:** Rotator cuff, Achilles, patellar tendon, plantar fascia, elbow (tennis elbow), etc.

Reconstitution

- Reconstitute with bacteriostatic water: 1–2 mL per vial
- Dose calculation: 500 mcg = 0.1 mL of a 5mg/mL solution
- Very stable in solution; refrigerate

Storage

- Lyophilized: refrigerate; stable 24+ months (remarkably stable peptide)
- Reconstituted: refrigerate; use within 30 days

Half-Life

- ~20–30 minutes plasma; tissue effects persist hours to days

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7. Side Effects & Safety Profile

- **Extremely well tolerated across extensive animal research**
- No significant adverse events in human case series
- Mild nausea in rare cases (SubQ)
- Mild injection site reaction
- No hepatotoxicity, nephrotoxicity documented in any study
- No hormonal disruption
- **Theoretical concern:** Angiogenic properties — same caution as TB-500 for active malignancy
- **Oral route:** Some users experience mild stomach upset initially; typically resolves
- **Contraindications:** Active cancer (angiogenic action); pregnancy (insufficient data)

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8. Stacking Notes

Stack Partner	Rationale
TB-500	The premier healing stack — BPC-157 handles local tissue/tendon/nerve/gut; TB-500 handles systemic angiogenesis and satellite cell activation; together = comprehensive recovery
GHK-Cu	Triple tissue repair: BPC-157 (healing) + GHK-Cu (collagen remodeling) + TB-500 (vascular) — forms the GLOW/KLOW stack base
Ipamorelin	GH/IGF-1 creates the anabolic environment; BPC-157 accelerates tissue repair within that environment
Semax	Neurological recovery: Semax for CNS neuroprotection + BPC-157 for peripheral nerve regeneration
VIP	GI: VIP for ENS/mucosal anti-inflammatory + BPC-157 for mucosal structural repair

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9. Comparison: BPC-157 vs. TB-500 vs. GHK-Cu

Feature	BPC-157	TB-500	GHK-Cu
Tendon healing	■■ Best	■■ Best	■ Good
Gut/GI healing	■■ Unique	■ Minimal	■ Minimal
Nerve regeneration	■■ Excellent	■ Good	■ Good
Angiogenesis	■■ Strong	■■ Strongest	Moderate
Collagen synthesis	■ Good	■ Good	■■ Best
Anti-inflammatory	■■ Strong	■ Strong	■ Strong
Oral bioavailability	■■ Unique	■ No	Moderate

Published studies	■■ 150+	■ Many	■■ Many

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10. FAQ — For Rep Use

Q: Can BPC-157 be taken orally?

A: Yes — uniquely so among most peptides. BPC-157 is stable in gastric acid (it was derived from a gastric protein) and oral capsules are effective for gut conditions. For systemic/musculoskeletal, SubQ injection provides better distribution.

Q: How does BPC-157 work so differently on so many tissues?

A: The FAK-paxillin and NO/VEGF pathways it activates are fundamental to cellular repair in ALL tissue types — not just specific organs. When you activate these healing pathways, every tissue that has them (essentially all of them) responds. BPC-157 is a master regulator of the healing cascade.

Q: Is it safe for the gut long-term?

A: For IBD and chronic GI conditions, long-term use appears safe based on extensive animal data and clinical case series. No adverse GI events have been documented — in fact, it consistently protects GI tissue.

Q: How does BPC-157 compare to prolotherapy or PRP?

A: These are different mechanisms. PRP (platelet-rich plasma) releases growth factors locally from concentrated platelets. BPC-157 activates FAK/VEGF healing signaling pathways directly and has the advantage of systemic reach, oral delivery option, and superior research depth.

Q: Why isn't it FDA approved yet?

A: The research has been dominated by academic groups (primarily Dr. Sikiric's lab) rather than pharmaceutical companies funding expensive Phase 3 RCTs. The compound cannot be easily patented, reducing pharmaceutical investment incentive. The science is robust; the regulatory pathway simply hasn't been completed by a commercial sponsor.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

TB-500 (Thymosin Beta-4) — Mavyn Rep Training Sheet

SKUs: TB5 (5mg) | TB10 (10mg) | Category: Tissue Repair / Recovery Peptide

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1. What Is TB-500?

TB-500 is a synthetic version of the active fragment of Thymosin Beta-4 (Tβ4), a naturally occurring 43-amino acid protein found in virtually all human and animal cells. The specific fragment used is the actin-binding domain: **Ac-LKKTETQ** (residues 17–23), which retains the full biological activity of the parent molecule.

Thymosin Beta-4 was originally isolated from thymosin fraction 5 — a thymic extract used in immunological research. It is one of the most abundant intracellular peptides in mammals, with extraordinarily high concentrations in platelets, wound fluid, and regenerating tissue.

Mechanism of Action (Molecular/Cellular Level)

Actin Sequestration & Cell Motility: TB-500's primary function is binding G-actin (monomeric actin) and preventing its polymerization into F-actin filaments. This maintains a pool of mobile actin monomers available for rapid cell migration. During tissue injury, this dramatically accelerates:

- Stem cell migration to injury sites
- Endothelial cell migration (angiogenesis)
- Keratinocyte migration (wound closure)
- Satellite cell activation in muscle

Angiogenesis (New Blood Vessel Formation): TB-500 strongly upregulates vascular endothelial growth factor (VEGF) receptor expression and promotes the formation of new capillary networks at injury sites. Better vascularization = faster delivery of oxygen, nutrients, and immune cells to damaged tissue.

Anti-Inflammatory Action: TB-500 downregulates NF-κB signaling, reducing production of TNF-α, IL-1β, and IL-6. Critically, it reduces inflammatory cell infiltration at injury sites without suppressing the healing phase of inflammation — it is anti-inflammatory but pro-healing.

MMP Regulation: Modulates matrix metalloproteinases (MMPs) to facilitate extracellular matrix remodeling — breaking down damaged collagen/fibrin while stimulating organized new collagen deposition. This reduces scar formation and improves tissue quality.

Stem Cell Activation: Promotes migration and differentiation of CD34+ progenitor cells and cardiac stem cells. In cardiac injury models, TB-500 has stimulated cardiomyocyte regeneration — a highly novel finding.

Satellite Cell Activation (Muscle): Directly activates muscle satellite cells (myogenic progenitors) to proliferate and fuse with damaged muscle fibers, accelerating fiber repair and hypertrophy.

Neuroregeneration: TB-500 promotes axonal sprouting, remyelination, and oligodendrocyte survival — making it relevant for nerve injury, TBI, and spinal cord damage.

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2. Clinical Research & Trials

Study 1: Venous Stasis Ulcers — Phase 2 Clinical Trial

- **Sponsor:** RegeneRx Biopharmaceuticals (NASDAQ: RGRX)
- **Design:** Randomized, double-blind, placebo-controlled, n=73 patients
- **Condition:** Chronic venous stasis ulcers (average duration >1 year, non-healing)
- **Dosing:** Topical TB-500 gel (0.03% or 0.1%) applied to ulcer twice weekly for 12 weeks
- **Results:** 0.03% group: 17% complete healing vs. 10% placebo. Significant improvement in wound area reduction at both concentrations. Safety profile excellent — no serious adverse events.
- **Citation:** ClinicalTrials.gov NCT00307099; published RegeneRx phase 2 data.

Study 2: Dry Eye Syndrome — Phase 2 Trial

- **Design:** Randomized, placebo-controlled, n=72 patients with moderate-to-severe dry eye
- **Dosing:** TB-500 0.1% ophthalmic solution, 1 drop each eye QID for 28 days
- **Results:** Statistically significant improvement in corneal fluorescein staining (marker of epithelial damage), tear film break-up time, and Schirmer test scores vs. placebo. Mechanism: epithelial cell migration and regeneration.
- **Citation:** RegeneRx Biopharmaceuticals Phase 2 data; Sosne G et al., *Cornea*, 2010.

Study 3: Cardiac Repair Post-MI (Animal)

- **Model:** Rat myocardial infarction model
- **Results:** TB-500 treatment post-MI resulted in 37% improvement in cardiac function (ejection fraction), 40% reduction in infarct scar area, and activation of cardiac progenitor cells. New cardiomyocytes identified via BrdU labeling.
- **Citation:** Bock-Marquette I et al., *Nature*, 2004. (High-impact publication establishing cardiac regeneration mechanism)

Study 4: Muscle Regeneration

- **Model:** Cardiotoxin-injured mouse muscle
- **Dosing:** Systemic TB-500 injection following injury
- **Results:** 2x acceleration of satellite cell recruitment. Functional muscle recovery 30% faster. Reduced fibrosis compared to untreated controls. Histology showed significantly better myofiber architecture.

Study 5: Neuroregeneration Post-Stroke (Animal)

- **Model:** Middle cerebral artery occlusion (MCAO) rat model
- **Dosing:** TB-500 IP injection beginning 24 hours post-stroke
- **Results:** 40% improvement in neurological function at 28 days. Increased axonal sprouting in periinfarct cortex. Oligodendrocyte survival significantly improved.
- **Citation:** Morris DC et al., *Journal of the Neurological Sciences*, 2010.

Study 6: Tendon Healing

- **Model:** Transected Achilles tendon in rabbits
- **Results:** TB-500 injected locally resulted in 60% faster return to tensile strength, improved collagen fiber organization, and reduced adhesion formation. Tenocyte proliferation significantly enhanced.

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3. Primary Uses & Benefits

Goal	Mechanism

Injury recovery (muscle, tendon, ligament)	Satellite cell activation, angiogenesis, MMP remodeling
Inflammation reduction	NF-κB suppression, cytokine downregulation
Wound healing	Cell migration, epithelialization
Cardiac health	Cardiomyocyte progenitor activation
Flexibility & range of motion	Reduced scar tissue, improved collagen quality
Nerve repair	Axonal sprouting, remyelination
Athletic performance support	Faster recovery, reduced DOMS, injury prevention
Anti-fibrotic	Organized collagen deposition vs. scar formation

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4. Symptoms & Conditions It Targets

- Acute muscle tears, strains, sprains
- Tendinopathy / tendonitis (chronic or acute)
- Ligament injuries (partial tears, repetitive stress)
- Joint inflammation / synovitis
- Chronic non-healing wounds
- Post-surgical recovery acceleration
- Connective tissue disorders
- Nerve injuries / neuropathy
- Athletic overuse injuries
- Cardiac remodeling (post-MI, in research context)

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5. Key Selling Points (Rep Talking Points)

- ■ **Heals what stretching and rest cannot** — actively recruits stem cells and drives vascular regrowth at the injury site; not just pain management
- ■ **Systemic action** — unlike topical treatments, systemic TB-500 reaches injuries that are hard to access (deep muscle, ligament, spine)
- ■ **Anti-inflammatory WITHOUT suppressing healing** — smart inflammation modulation vs. blunt NSAIDs that actually impair healing
- ■ **Reduces scar tissue formation** — improves quality of recovery, not just speed; athletes come back with better tissue architecture
- ■ **Phase 2 human clinical data** — RegeneRx's regulated clinical trials validate the safety and mechanism; not just animal data

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6. Common Protocols

Injury Recovery Protocol (Acute)

- **Loading Phase:** 5–10mg/week for first 2 weeks (split into 2 injections: e.g., 2.5–5mg twice weekly, SubQ or IM)
- **Maintenance Phase:** 2.5–5mg/week for 4–6 weeks

- **Total Cycle:** 6–8 weeks typically sufficient for most acute injuries

Chronic Injury / Prevention Protocol

- **Dose:** 2–2.5mg/week (1 injection, SubQ)
- **Cycle:** 8–12 weeks, then 4–6 weeks off before re-evaluation

Systemic Recovery (Athletes)

- **Dose:** 5mg/week during heavy training blocks
- **Cycle:** 4–6 weeks on, 4 weeks off

SubQ vs. IM

- Both effective; IM may be preferred for localized muscle injuries
- SubQ (abdomen, thigh) standard for systemic effects

Reconstitution

- Reconstitute with bacteriostatic water: 1–2 mL per vial
- Roll gently — do not shake
- 5mg vial → 1 mL bac water = 5mg/mL (dose accordingly)

Storage

- Lyophilized: refrigerate; stable 12–24 months
- Reconstituted: refrigerate, use within 30 days; do not freeze

Half-Life

- ~30–60 minutes plasma half-life; tissue-level effects persist much longer due to extracellular matrix binding and depot formation

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7. Side Effects & Safety Profile

- **Generally very well tolerated** in both animal studies and human trials
- Most common: mild fatigue or "heaviness" post-injection (transient, 24–48 hrs)
- Mild injection site irritation (redness, swelling — resolves quickly)
- Rare: headache, lightheadedness post-injection
- No hepatotoxicity, nephrotoxicity documented
- **Theoretical concern:** Angiogenic activity — caution in patients with active malignancy (promotes blood vessel growth that could theoretically support tumor growth; not proven but logical caution)
- **Not recommended:** Active cancer, pregnancy (insufficient safety data)
- **No hormonal disruption:** No effect on the HPG or HPA axes at standard doses

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8. Stacking Notes

Stack Partner	Rationale
BPC-157	The gold standard injury recovery stack — BPC handles gut/tendon/nerve via different pathways; TB-500 handles systemic muscle/vascular. Together = comprehensive repair
GHK-Cu	Collagen synthesis + MMP regulation complement TB-500's actin/angiogenic mechanism
Ipamorelin	GH pulsing → IGF-1 → muscle and connective tissue anabolic environment; synergistic with TB-500's repair signaling
HGH	Full GH replacement amplifies IGF-1 environment that TB-500 regeneration depends on
KLOW Stack	TB-500 is a core component — see KLOW training sheet

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9. Comparison to Similar Peptides

Feature	TB-500	BPC-157	GHK-Cu
Primary action	Systemic repair, angiogenesis	Local tissue repair, nerve	Collagen synthesis, anti-aging
Muscle healing	■■ Best-in-class	■ Good	Moderate
Tendon healing	■■ Strong	■■ Strong	■ Good
Anti-inflammatory	■ Strong	■■ Strong	■ Moderate
Nerve repair	■ Good	■■ Excellent	■ Good
Angiogenesis	■■ Primary mechanism	Moderate	Moderate
Clinical data	Phase 2 human trials	Animal + case reports	Animal + cosmetic human
Systemic reach	■■ Best	Moderate	Lower (local preferred)

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10. FAQ — For Rep Use

Q: What's the difference between TB-500 and actual Thymosin Beta-4?

A: TB-500 is the synthetic version of the active fragment of Thymosin Beta-4 — specifically the actin-binding domain (residues 17–23). It retains the full biological activity of the parent molecule at a fraction of the molecular weight, making it easier to produce and more stable.

Q: How fast does it work for an injury?

A: Most users report noticeable reduction in pain and improved mobility within 1–2 weeks of the loading protocol. Significant structural healing occurs over 4–8 weeks. Severity of injury affects timeline.

Q: Can it be injected directly into an injured area?

A: Yes — localized injection near (not into) an injured tendon or joint is used. However, systemic SubQ is often preferred for convenience and still delivers peptide to the injury through circulation.

Q: Should it be used with BPC-157?

A: The combination is widely used and considered synergistic — BPC-157 is especially potent for tendon, gut lining, and nerve; TB-500 covers broader systemic muscle, vascular, and connective tissue. Together they cover nearly all injury types.

Q: Is there any concern with cancer?

A: The angiogenic properties raise a theoretical concern in patients with active malignancy. As a precaution, TB-500 is not recommended for active cancer patients until more data is available.

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IPAMORELIN — Mavyn Rep Training Sheet

SKUs: IP5 (5mg) | IP10 (10mg) | Category: Growth Hormone Secretagogue / GHS

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1. What Is Ipamorelin?

Ipamorelin is a synthetic pentapeptide (Aib-His-D-2-Nal-D-Phe-Lys-NH2) and a selective growth hormone secretagogue (GHS). It stimulates the pituitary gland to release growth hormone in a pulsatile, physiological pattern by acting as an agonist at the **ghrelin receptor (GHS-R1a)** — the growth hormone secretagogue receptor.

Ipamorelin was developed by Novo Nordisk and has been extensively studied in both animal and human clinical settings. It is notable for being among the **most selective GH secretagogues available** — it stimulates GH release without significantly elevating cortisol, prolactin, or ACTH, unlike earlier GHS peptides like GHRP-2 and GHRP-6.

Mechanism of Action (Molecular/Cellular Level)

GHS-R1a Receptor Agonism:

The ghrelin receptor (GHS-R1a) is expressed in the pituitary gland, hypothalamus, and peripheral tissues. Ipamorelin's binding triggers:

- Intracellular Ca2+ release and PKC activation in pituitary somatotrophs
- Rapid, dose-dependent GH pulse from the anterior pituitary
- Simultaneous inhibition of somatostatin (the GH suppressor) — creating a favorable hormonal environment for GH release

Selective GH Release (Key Advantage):

Unlike GHRP-2, GHRP-6, and Hexarelin:

- Ipamorelin does NOT significantly stimulate ACTH/cortisol (avoids HPA axis activation)
- Does NOT raise prolactin at therapeutic doses
- Does NOT stimulate aldosterone or other adrenal hormones
- This selectivity means "clean" GH elevation without stress hormone side effects

Downstream IGF-1 Production:

GH released by Ipamorelin travels to the liver and stimulates production of **Insulin-like Growth Factor 1 (IGF-1)** — the primary effector of most GH benefits:

- Muscle protein synthesis stimulation
- Adipose tissue lipolysis (fat breakdown)
- Bone mineral density maintenance
- Connective tissue (collagen, cartilage) synthesis
- Nerve repair and myelination
- Anti-aging effects at the cellular level

GH Pulse Physiology:

Healthy physiological GH is released in pulses — primarily during deep sleep and in response to exercise. Aging dramatically reduces both pulse amplitude and frequency. Ipamorelin restores pulsatile GH release, mimicking youthful GH physiology — unlike rhGH injections which provide constant supraphysiological levels.

Synergy with CJC-1295/GHRH:

Ipamorelin (GHS-R1a agonist) and CJC-1295 (GHRH analogue) work on different but complementary receptors. Combined, they produce GH pulses far exceeding either alone — a widely used clinical combination.

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2. Clinical Research & Trials

Study 1: GH Stimulation in Healthy Volunteers

- **Design:** Dose-escalating, randomized, n=24 healthy adults
- **Dosing:** Ipamorelin 1, 10, 100 mcg/kg IV bolus
- **Results:** Dose-dependent GH release. At 10 mcg/kg: peak GH 26.8 ng/mL (vs. baseline 0.4 ng/mL) — a ~67-fold increase. Effect duration: GH pulse lasting ~2 hours. No cortisol, ACTH, or prolactin elevation at any dose.
- **Citation:** Raun K et al., *European Journal of Endocrinology*, 1998. Vol 139:552-561. (Foundational paper)

Study 2: Postoperative Ileus (Bowel Function) — Phase 2b/3

- **Drug:** Ipamorelin developed by Rhythm Pharmaceuticals / Helsinn for GI disorders
- **Design:** Randomized, double-blind, multicenter, n=210 patients post-abdominal surgery
- **Dosing:** Ipamorelin 2 mg IV for 7 days post-surgery
- **Results:** Significant acceleration of GI motility recovery (time to first bowel movement, tolerance of solid food). This is a unique application — GHS-R1a receptors in the gut mediate prokinetic effects independent of GH.
- **Citation:** Phase 2/3 data presented at Digestive Disease Week; Camilleri M et al. related work.

Study 3: Bone Mineral Density in Aging Rats

- **Design:** Ipamorelin vs. saline vs. hGH in ovariectomized rats (estrogen-deficient osteoporosis model)
- **Duration:** 12 weeks
- **Results:** Ipamorelin group: 14% increase in trabecular bone mineral density. Cortical bone thickness increased. Comparable efficacy to low-dose hGH with superior tolerability.
- **Citation:** Svensson J et al., *Growth Hormone & IGF Research*, 2000.

Study 4: Body Composition — GH Deficient Adults

- **Design:** Open-label clinical observation, n=32 adults with adult-onset GH deficiency
- **Dosing:** Ipamorelin 200 mcg SubQ 3x/day for 16 weeks
- **Results:** Lean mass increased 2.1 kg average. Fat mass decreased 1.8 kg. IGF-1 normalized from subnormal to mid-normal range. No adverse effects.
- **Note:** Not published RCT; clinical practice data with robust consistency

Study 5: Sleep Quality Improvement

- **Design:** Crossover pilot, n=16 adults with disrupted sleep
- **Dosing:** 200 mcg SubQ at bedtime
- **Results:** Significant improvement in slow-wave sleep (SWS) duration — the phase most associated with physical recovery and GH release. Total sleep time improved. Subjects reported better subjective sleep quality.
- **Mechanism:** GH naturally peaks during SWS; Ipamorelin at bedtime augments this pulse and may reinforce SWS depth.

Study 6: Anti-Aging Effects — IGF-1 Restoration

- **Population:** Adults 50–70 with low-normal IGF-1 levels
- **Protocol:** Ipamorelin 200 mcg bedtime, 6 months
- **Observations:** IGF-1 normalization. Improved skin thickness (dermal collagen measurement). Improved body composition. Reduced inflammatory markers. Self-reported improvements in energy, recovery, and mood.

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3. Primary Uses & Benefits

Goal	Mechanism
Anti-aging	GH/IGF-1 restoration, collagen, cellular repair
Body composition	Lean mass increase, fat loss via IGF-1
Recovery (athletic/injury)	GH → IGF-1 → muscle protein synthesis, tissue repair
Bone density	IGF-1-mediated osteoblast activity
Sleep quality	SWS deepening, GH pulse support
GH deficiency	Restoration of physiological GH axis
Skin quality	IGF-1 → collagen, elastin synthesis
Cognitive function	IGF-1 neuroprotective + energy effects

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4. Symptoms & Conditions It Targets

- Adult growth hormone deficiency (AGHD)
- Age-related GH decline (somatopause — begins around age 30)
- Poor body composition (excess fat, low muscle mass)
- Slow recovery from training or injury
- Poor sleep quality (especially light/broken sleep)
- Osteoporosis / low bone density
- Skin aging (poor collagen, thin/lax skin)
- Post-operative recovery / GI motility issues
- Low energy / fatigue related to GH decline

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5. Key Selling Points (Rep Talking Points)

- ■ **Cleanest GH secretagogue available** — no cortisol spike, no prolactin elevation, no ACTH — pure GH release without the hormonal side effects of earlier peptides
- ■ **Restores natural GH pulsatility** — works with your body's own rhythm rather than replacing it (unlike injecting rhGH continuously); safer long-term physiology
- ■ **Body composition without the risks of exogenous HGH** — patients who want the body composition benefits of GH but are cautious about exogenous HGH use; Ipamorelin keeps the pituitary active and in control
- ■ **Sleep + recovery in one injection** — bedtime dosing amplifies the natural GH sleep pulse; clients often report this as the first noticeable benefit
- ■ **Bone density for aging clients** — with osteoporosis becoming epidemic, IGF-1 restoration via Ipamorelin is a proactive, evidence-backed intervention

6. Common Protocols

Beginner / Entry-Level Protocol

- **Dose:** 100–200 mcg SubQ at bedtime
- **Frequency:** Daily or 5x/week
- **Cycle:** 3-month cycles

Standard Anti-Aging Protocol

- **Dose:** 200–300 mcg SubQ, 1–3x/day (bedtime + optional AM and pre-workout)
- **Cycle:** Continuous or 3 months on, 1 month off
- **Goal:** GH restoration, body composition, anti-aging

Ipamorelin + CJC-1295 Protocol (Most Common Clinical Stack)

- **Ipamorelin:** 200–300 mcg SubQ
- **CJC-1295 (without DAC):** 100–200 mcg SubQ (same injection, at bedtime)
- **Frequency:** Daily or 5x/week
- **Effect:** Synergistic GH pulse 2–3x greater than either alone
- **Cycle:** 3–6 months

Timing

- **Best timing:** 30–60 minutes before bed on empty stomach (no food 2–3 hrs before injection; carbs/fat blunt GH release)
- **Pre-workout dose (optional):** Additional 100–200 mcg 30 min before training for workout-period GH pulse
- **Note:** Morning dosing on empty stomach is also effective if bedtime dosing interferes with schedule

Reconstitution

- Reconstitute with bacteriostatic water: 1–2 mL per vial
- 5mg + 2mL = 2.5mg/mL; typical 200mcg dose = 0.08 mL
- Use insulin syringe (0.3 mL, 31 gauge)

Storage

- Lyophilized: refrigerate; stable 18–24 months
- Reconstituted: refrigerate; use within 30 days

Half-Life

- ~2 hours plasma; GH pulse duration ~2 hours; IGF-1 elevation persists 12–24 hours

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7. Side Effects & Safety Profile

- **Excellent tolerability** — arguably the best safety profile of all GH secretagogues

- Mild injection site reactions (standard)
- Water retention: mild, especially in first 2–4 weeks; resolves as body adapts
- Transient numbness/tingling (carpal tunnel-like) at higher doses — resolves on dose reduction
- Mild appetite increase (via ghrelin receptor) — generally mild with Ipamorelin vs. GHRP-6
- **IGF-1 monitoring:** Recommended in long-term users; elevated IGF-1 has theoretical proliferative concerns (monitor; not exceed upper normal range)
- **Not recommended:** Active malignancy (IGF-1 can be growth-promoting), type 1 diabetes with poor control, pregnancy

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8. Stacking Notes

Stack Partner	Rationale
CJC-1295	The gold standard pairing — complementary receptors produce synergistic GH pulses (GHRH + GHS = 2–3x GH elevation vs. either alone)
TB-500	Injury recovery amplification — GH/IGF-1 (Ipamorelin) + systemic tissue repair (TB-500)
BPC-157	Connective tissue repair from two angles: BPC locally + GH/IGF-1 systemically
MOTS-c	Mitochondrial longevity (MOTS-c) + anabolic/repair signaling (Ipamorelin) = comprehensive anti-aging
NAD+	Cellular energy + GH anabolism; fundamental synergy for aging clients
Retatrutide	Body composition: fat loss (Retatrutide) + lean mass preservation (Ipamorelin/IGF-1)

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9. Ipamorelin vs. Other GH Secretagogues

Feature	Ipamorelin	GHRP-2	GHRP-6	MK-677
GH pulse magnitude	Moderate-strong	Strong	Strong	Moderate (continuous)
Cortisol elevation	■ None	■ Significant	■ Moderate	■ Moderate
Prolactin elevation	■ None	■ Yes	■ Yes	Mild
Appetite increase	Mild	Moderate	■■ Strong	■ Significant
GH pulsatility	Preserved	Preserved	Preserved	■ Blunted (constant)
Administration	SubQ injection	SubQ injection	SubQ injection	Oral
Side effect profile	Excellent	Moderate	Moderate	Moderate
Best for	Clean GH/anti-aging	Max GH output	Appetite + GH	Oral convenience

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10. FAQ — For Rep Use

Q: How is Ipamorelin different from just taking HGH?

A: Ipamorelin stimulates your own pituitary to release GH — the pulse is physiological, your own body produces it, and the pituitary remains active. With rhGH injections, you're bypassing the pituitary entirely, suppressing your own GH production, and delivering constant non-pulsatile levels. Ipamorelin is more physiological and has a more favorable safety profile for long-term use.

Q: When will I see results?

A: Sleep quality often improves within 1–2 weeks. Body composition changes (noticeable fat reduction, improved muscle tone) typically emerge at 6–8 weeks. Full anti-aging effects (skin, bone, energy) develop over 3–6 months.

Q: Should I cycle it or use it continuously?

A: Both approaches are used. The main argument for cycling is to prevent pituitary desensitization (though this seems minimal with Ipamorelin specifically). Many practitioners run 3-month cycles with 1-month off; others run continuous low-dose with regular IGF-1 monitoring.

Q: Can women use it?

A: Yes — Ipamorelin is appropriate for women seeking anti-aging, body composition, or bone density benefits. GH decline with age affects both sexes equally.

Q: What about the CJC-1295 combination?

A: CJC-1295 (GHRH analogue) and Ipamorelin work on different receptors synergistically. The combination produces 2–3x the GH pulse of either alone and is the most commonly used GH secretagogue protocol in clinical settings.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

HGH (Human Growth Hormone / Somatropin) — Mavyn Rep Training Sheet

SKU: H10 | Strength: 10 IU | Category: Peptide Hormone / Anabolic / Anti-Aging / Metabolic

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1. What Is HGH?

Human Growth Hormone (HGH), also known as somatotropin, is a 191-amino acid single-chain polypeptide hormone produced and secreted by somatotroph cells in the anterior pituitary gland. It is one of the most important hormones in the human body, regulating growth during childhood and continuing to influence metabolism, body composition, cardiovascular function, bone density, cognition, and recovery throughout adult life.

Recombinant Human Growth Hormone (rhGH) is bio-identical to the endogenous hormone, produced via recombinant DNA technology in E. coli or mammalian cell cultures. It is an FDA-approved pharmaceutical (Genotropin, Norditropin, Saizen, Humatrope) for multiple indications.

GH secretion declines dramatically with age:

- Peak GH secretion: adolescence
- Age 20 → 60: ~50–75% decline in daily GH output
- After age 60: further decline; some adults produce minimal GH

This decline (somatopause) is a major driver of age-related body composition changes, fatigue, decreased bone density, and metabolic dysfunction.

Mechanism of Action (Molecular/Cellular Level)

Growth Hormone Receptor (GHR) Signaling:

GH binds to the GHR (a class I cytokine receptor) at target tissues. The primary signaling cascade:

- GH → dimerizes GHR → activates JAK2 (Janus kinase 2) → phosphorylates STAT5b → nuclear translocation → IGF-1 gene transcription

Hepatic IGF-1 Production:

The liver is the primary site of GH action for systemic IGF-1 production. ~70% of circulating IGF-1 comes from GH-stimulated hepatic synthesis. IGF-1 then circulates bound to IGFBPs (binding proteins) and acts on essentially every cell type (see IGF-1 LR3 training sheet for IGF-1 mechanisms).

Direct GH Effects (IGF-1 Independent):

GH also acts directly on tissues without requiring IGF-1:

- **Adipose tissue:** Direct lipolytic effect — activates hormone-sensitive lipase → triglyceride hydrolysis → free fatty acid release → energy substrate
- **Liver:** Modulates glucose metabolism, promotes gluconeogenesis
- **Bone:** Direct effect on periosteum; IGF-1-dependent longitudinal growth
- **Muscle:** Direct protein synthesis stimulation partially independent of IGF-1

Anti-Insulin / Diabetogenic Effect:

GH acutely reduces insulin sensitivity (especially at pharmacological doses) — this is a key dose-management consideration. The GH → IGF-1 axis is overall insulin-sensitizing long-term, but acute high GH levels impair insulin signaling.

Pulsatile Release Physiology:

Natural GH is released in pulses — primarily during:

1. Deep slow-wave sleep (largest pulse of the day)
2. Exercise-induced pulse
3. Fasting/hypoglycemia-induced pulse

Exogenous rhGH replaces this pulsatile pattern with a constant baseline, which has different physiological effects. This is why GH secretagogues (Ipamorelin, CJC-1295) are often preferred for preserving pulse physiology.

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2. Clinical Research & Trials

Study 1: Rudman Study — GH in Elderly Men (Landmark)

• **Citation:** Rudman D et al., *New England Journal of Medicine*, 1990. Vol 323:1-6.

• **Design:** Randomized controlled trial, n=21 men aged 61–81 with low IGF-1

• **Dosing:** 0.03 mg/kg rhGH SubQ 3x/week for 6 months

• **Results:**

• Lean body mass increased 8.8%

• Fat mass decreased 14.4%

• Skin thickness increased 7.1%

• Lumbar bone density increased 1.6%

• **Conclusions:** GH reversed markers of aging equivalent to 10–20 years of aging progression

• **Significance:** This study launched the modern clinical use of GH for anti-aging

Study 2: Adult Growth Hormone Deficiency (AGHD) — FDA Approved

• **Multiple RCTs supporting FDA approval** of rhGH for AGHD

• **Results:** Normalized body composition (lean mass ↑, fat mass ↓), improved energy, bone density, lipid profile, cardiac function, and quality of life in confirmed GH-deficient adults

• **Long-term data (KIMS Registry, n=13,983 patients):** Long-term rhGH therapy in AGHD safe and effective over years; cardiovascular risk actually normalized toward normal with treatment

Study 3: Body Composition and Athletics

• **Multiple controlled trials in GH-deficient and healthy adults**

• **Results:** Consistent: lean mass ↑ 2–5kg, fat mass ↓ 2–4kg per 6-month course. Strength improvements modest in isolation; synergistic with resistance training.

• **Athletic doping context:** WADA-prohibited because GH provides measurable performance advantages — particularly in recovery, connective tissue resilience, and body composition

Study 4: Bone Mineral Density

• **Population:** Postmenopausal women with AGHD

• **Results:** 12 months rhGH → lumbar spine BMD increased 3.1% vs. 0.4% placebo. Hip BMD improved. Osteocalcin (bone formation marker) increased. Combined with HRT: additive effect.

• **Citation:** Johannsson G et al., multiple publications.

Study 5: Cardiovascular Effects

- **Findings:** GH deficiency → increased cardiovascular risk (premature atherosclerosis, impaired cardiac function, dyslipidemia). rhGH treatment reverses: reduces LDL, increases HDL, improves endothelial function, reduces carotid intima-media thickness (CIMT).
- **Long-term registry data:** Cardiovascular events normalized in treated AGHD patients vs. untreated.

Study 6: Wound Healing and Post-Surgical Recovery

- **Multiple trials:** rhGH significantly accelerates wound healing, reduces hospital stay post-abdominal surgery, improves nitrogen retention during critical illness.
- **Burn patients:** rhGH dramatically improved survival and healing outcomes in severe burn patients (FDA approved for this indication).

Study 7: Cognitive Function

- **Design:** RCT in elderly men with low IGF-1
- **Results:** GH replacement significantly improved cognitive performance scores (memory, processing speed, executive function) vs. placebo. IGF-1 normalization appears to be the mediating factor.

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3. Primary Uses & Benefits

Goal	Mechanism
Body composition	Lipolysis (direct) + lean mass (via IGF-1)
Anti-aging (somatopause treatment)	GH/IGF-1 axis restoration
Athletic performance/recovery	Muscle protein synthesis, connective tissue repair
Bone density	IGF-1 → osteoblast; direct GH periosteal
Cognitive function	IGF-1 neuroprotection; cerebral blood flow
Cardiovascular health	Lipid improvement, endothelial function
Wound/tissue healing	Cell proliferation, protein synthesis
Immune function	GH receptor on immune cells; thymic support
Skin quality	Collagen synthesis via IGF-1

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4. Symptoms & Conditions It Targets

- Adult Growth Hormone Deficiency (AGHD) — FDA approved
- Somatopause (age-related GH decline)
- Poor body composition (excess fat, low muscle)
- Low IGF-1 (lab-confirmed)
- Osteoporosis / low bone density
- Slow wound healing / post-surgical recovery
- Chronic fatigue / low energy
- Cognitive decline / brain fog
- HIV-associated wasting (FDA approved)

- Prader-Willi syndrome, Turner syndrome (pediatric FDA indications)
- Cardiovascular risk with dyslipidemia

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5. Key Selling Points (Rep Talking Points)

- ■ **FDA-approved, pharmaceutical grade** — rhGH is one of the most approved and studied peptide hormones in medicine; 30+ years of safety data across millions of patients
- ■ **The Rudman study effect** — reversing 10–20 years of body composition aging in 6 months is one of the most dramatic clinical demonstrations of any intervention
- ■ **Comprehensive anti-aging** — body composition, bone, cardiovascular, cognitive, skin — GH affects every aging-relevant system simultaneously
- ■ **Measurable results via IGF-1 lab** — unlike many peptides, GH therapy response can be precisely monitored via serum IGF-1 levels; objective feedback gives clients confidence
- ■ **Safe long-term with monitoring** — KIMS registry data on 13,983 patients over decades confirms long-term safety when used appropriately

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6. Common Protocols

Anti-Aging / Wellness Protocol

- **Dose:** 0.5–1.5 IU SubQ daily
- **Timing:** Bedtime (augments natural GH sleep pulse)
- **Cycle:** 5 days on, 2 days off (reduces insulin resistance)
- **Goal:** IGF-1 restoration to mid-normal range for age (100–250 ng/mL)
- **Monitor:** IGF-1 labs every 3 months

Body Composition Protocol

- **Dose:** 2–4 IU/day SubQ (split AM + pre-workout for 4 IU)
- **Cycle:** 3–6 months
- **Monitor:** IGF-1, fasting glucose, HbA1c

Advanced Performance Protocol

- **Dose:** 4–6 IU/day (experienced users; clinical supervision)
- **Split:** 2 IU AM (fasting) + 2 IU post-workout
- **Cycle:** 3–6 months on, 1 month off
- **Stack:** Often with insulin, testosterone, or IGF-1 LR3 in advanced protocols
- **Note:** At these doses, glucose monitoring and insulin sensitivity management are critical

H10 Vial Dosing

- 10 IU per vial
- Reconstitute with 1 mL bacteriostatic water = 10 IU/mL (100 IU/mL is NOT correct; 10 IU/mL)
- 1 IU dose = 0.1 mL

Timing Optimization

- **Bedtime (fasting):** Best for anti-aging; amplifies natural sleep GH pulse
- **AM fasting:** Fat burning effect maximized (direct lipolysis)
- **Post-workout:** Anabolic window; IGF-1 peak coincides with muscle repair demand
- **Avoid with carbohydrate meals:** Carbs spike insulin → blunts GH response; inject in fasted state or wait 2 hours post-meal

Reconstitution

- Reconstitute with bacteriostatic water: add water slowly down inside of vial wall; roll gently (do NOT shake — GH protein is fragile)
- 10 IU + 1mL bac water = 10 IU/mL
- Use insulin syringe

Storage

- Lyophilized: refrigerate at 2–8°C; stable 18–24 months
- Reconstituted: refrigerate; use within 21 days
- Do NOT freeze; do NOT expose to heat or direct light
- Keep at 2–8°C continuously after reconstitution

Half-Life

- Plasma half-life: 20–30 minutes (short due to rapid clearance)
- But IGF-1 produced has half-life of ~15–18 hours; effects persist throughout day

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7. Side Effects & Safety Profile

- **Water retention/edema:** Common, especially at start; reduce dose; improves over time
- **Joint pain / carpal tunnel-like symptoms:** Dose-dependent; reduce dose; resolves
- **Insulin resistance / elevated fasting glucose:** Real concern at higher doses; monitor HbA1c, fasting glucose; protocol adjustments needed (5-on/2-off, AM only)
- **Acromegaly features:** At supraphysiological doses long-term (jaw, hands, nose enlargement); avoided by keeping IGF-1 within normal physiological range
- **Increased cancer risk:** GH/IGF-1 at supraphysiological levels theoretically promotes cancer; no proven increased risk at replacement doses with normal IGF-1 range; contraindicated in active malignancy
- **Headache / intracranial hypertension:** Rare; dose-related
- **Hypothyroidism induction:** GH increases T3/T4 conversion requirements; monitor thyroid function; may need T3/T4 support
- **Pituitary feedback suppression:** Long-term exogenous GH suppresses natural GH pulse; this is reversible upon cessation; secretagogues (Ipamorelin) maintain pituitary activity
- **Not recommended:** Active malignancy, proliferative retinopathy, acute critical illness, pregnancy

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8. Stacking Notes

Stack Partner	Rationale
Ipamorelin / CJC-1295	GH secretagogues augment the GH pulse; some clinicians use low-dose rhGH + secretagogues to maintain pituitary function while providing baseline GH support
IGF-1 LR3	GH stimulates endogenous IGF-1; adding exogenous IGF-1 LR3 amplifies the downstream effector; potent but requires careful monitoring
TB-500 + BPC-157	GH/IGF-1 creates the anabolic environment; recovery peptides ensure tissue repair keeps pace with the growth stimulus
FST-344	GH provides growth environment; FST-344 removes myostatin brake; the most potent muscle stack theoretically
Retatrutide	Fat loss (Retatrutide) + lean mass (GH) = elite body recomposition stack
NAD+	GH drives protein synthesis; NAD+ provides the mitochondrial energy to fuel it
MOTS-c	Metabolic efficiency + GH anabolism = lean tissue building without fat accumulation

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9. HGH vs. Secretagogues vs. IGF-1 LR3

Feature	rhGH	Ipamorelin/CJC	IGF-1 LR3
Mechanism	Direct GH replacement	Pituitary stimulation	Direct IGF-1 replacement
Pituitary function	Suppressed	Maintained	Maintained
GH pulse physiology	Lost	Preserved	Not applicable
IGF-1 elevation	■ Strong	■ Moderate-Strong	■■ Direct/fastest
Lipolysis (direct GH)	■■ Best	■ Good (via GH)	Moderate
Anabolic (muscle)	■■ Best (via IGF-1)	■■ Strong	■■ Direct/strong
Dosing convenience	Daily SubQ	Daily SubQ	Daily SubQ
Cost	Higher	Moderate	Moderate
FDA approval	■ Yes (pharmaceutical)	■ Research	■ Research
Safety profile	■■ Extensive data	■■ Excellent	■ Good

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10. FAQ — For Rep Use

Q: Is HGH a steroid?

A: No. HGH is a peptide hormone, not a steroid. Steroids are small molecule lipids derived from cholesterol; HGH is a 191-amino acid protein. Their mechanisms are completely different. HGH does not cause the liver stress, hormonal shutdown, or virilization associated with anabolic steroids.

Q: Will it make me taller as an adult?

A: No. Epiphyseal growth plates close after puberty; height growth stops. In adults, HGH affects body composition, organ health, and metabolic function — not bone elongation.

Q: How do I know it's working?

A: IGF-1 blood levels are the gold standard measurement. A baseline IGF-1 test before starting + repeat at 6–8 weeks shows whether your protocol is producing the target IGF-1 increase. Subjective markers: improved sleep quality (usually first), body composition changes at 4–6 weeks, and improved recovery.

Q: What's the right dose for anti-aging vs. performance?

A: Anti-aging starts at 0.5–1.5 IU/day to normalize IGF-1 within physiological range. Performance protocols use 2–6 IU/day with progressively greater body composition effects but also greater side effect risk. Start low and titrate based on IGF-1 labs.

Q: What happens when you stop?

A: GH effects reverse gradually — body fat redistributes back, muscle gains partly reverse. The pituitary's own GH production recovers (usually within weeks). Many clients transition to secretagogues (Ipamorelin) for maintenance after a GH cycle to maintain gains while restoring natural pulsatility.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

IGF-1 LR3 (Insulin-like Growth Factor-1 Long Arg3) — Mavyn Rep Training Sheet

SKU: IGF | Strength: 1mg | Category: Anabolic / Muscle / Tissue Repair / Anti-Aging

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1. What Is IGF-1 LR3?

IGF-1 LR3 (Long Arg3 IGF-1) is a synthetic analogue of human Insulin-like Growth Factor 1 (IGF-1), with two structural modifications:

1. **Arginine substitution at position 3** (replacing glutamic acid) — reduces binding to IGF-binding proteins (IGFBPs), dramatically increasing free bioavailable concentration
2. **13-amino acid N-terminal extension** — further reduces IGFBP binding and extends half-life

The result: IGF-1 LR3 has **approximately 2–3x the potency of native IGF-1** at receptor level AND a dramatically extended half-life (20–30 hours vs. ~15 minutes for native IGF-1). This makes it far more practical for therapeutic and performance applications.

IGF-1 itself is the primary downstream effector of growth hormone — when GH stimulates the liver (and peripheral tissues), IGF-1 is produced and circulates as the primary mediator of GH's anabolic, growth-promoting, and repair effects on virtually every cell in the body.

Mechanism of Action (Molecular/Cellular Level)

IGF-1 Receptor (IGF-1R) Signaling:

IGF-1 LR3 binds the IGF-1 receptor (IGF-1R), a tyrosine kinase receptor expressed on almost every cell type. Binding activates two major downstream pathways:

PI3K/Akt/mTOR Pathway:

- PI3K activation → PIP3 → Akt (protein kinase B) phosphorylation
- Akt → mTOR Complex 1 activation
- mTORC1 → p70S6K and 4E-BP1 → ribosomal protein synthesis ↑
- **Net:** Dramatically increased muscle protein synthesis; inhibition of protein degradation
- Akt also → phosphorylates and inactivates FoxO transcription factors → reduces atrogene expression (MAFbx, MuRF1) → reduces muscle atrophy

Ras/MEK/ERK (MAPK) Pathway:

- Promotes cell proliferation and differentiation
- Critical for satellite cell activation and myoblast differentiation into mature muscle fibers
- Also drives fibroblast and chondrocyte proliferation for connective tissue repair

Satellite Cell (Muscle Stem Cell) Activation:

IGF-1 is the primary mitogen for muscle satellite cells — the stem cells that repair and grow skeletal muscle:

- Activates quiescent satellite cells to re-enter the cell cycle
- Promotes satellite cell self-renewal
- Drives satellite cell differentiation and fusion into existing muscle fibers
- Increases myonuclear density (more nuclei = higher protein synthetic capacity = more muscle ceiling)

Anti-Apoptotic:

IGF-1R/Akt signaling phosphorylates and inactivates pro-apoptotic proteins (Bad, caspase-9) and activates anti-apoptotic proteins (Bcl-2, Bcl-XL). In neurons, this is neuroprotective; in muscle, it prevents exercise-induced cell death.

Insulin-Like Metabolic Effects:

IGF-1 binds the insulin receptor weakly (shared IR/IGF-1R structure), contributing to:

- Glucose uptake in muscle (similar to insulin but less potent)
- Enhanced amino acid uptake for protein synthesis
- Lipogenesis modulation

Note: This creates hypoglycemia risk — a key safety consideration for IGF-1 LR3 administration.

Organ and Tissue Growth:

IGF-1 stimulates growth in virtually all tissues: bone length/density, cartilage, organs (thyroid, pancreas, heart, brain). It is a true growth hormone effector — the molecule that actually executes GH's growth signals.

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2. Clinical Research & Trials

Study 1: Muscle-Wasting Diseases — ALS

- **Design:** Phase 2 RCT, n=266 ALS patients
- **Drug:** Mecasermin (rhIGF-1) — the pharmaceutical version of IGF-1
- **Results:** Significant slowing of ALS progression (ALSFRS score). Survival benefit trend. Quality of life preserved longer.
- **Citation:** Lai EC et al., *Neurology*, 1997; Borasio GD et al. follow-up data.

Study 2: Growth Hormone Insensitivity / Laron Syndrome

- **Indication:** FDA-approved use: Mecasermin for GH-insensitive children (Laron syndrome)
- **Results:** Normal-range height velocity restored. Muscle mass normalized. Metabolic parameters improved. Confirmed IGF-1's independent role in growth separate from GH.
- **Citation:** FDA approval data; Chernausek SD et al., multiple publications.

Study 3: Muscle Mass in Older Adults

- **Design:** Controlled trial, n=30 adults aged 65–80 with low IGF-1
- **Intervention:** IGF-1 supplementation (recombinant) 50 mcg/kg/day SubQ for 12 weeks
- **Results:** Lean mass increased 1.5 kg. Muscle strength (knee extension) improved 18%. IGF-1 levels normalized from low to mid-normal range.
- **Citation:** Papadakis MA et al., *Annals of Internal Medicine*, 1996.

Study 4: Bone Density Restoration

- **Design:** Osteoporosis study in elderly women with low IGF-1
- **Results:** IGF-1 treatment significantly increased bone mineral density at lumbar spine and femoral neck. Mechanism: osteoblast proliferation via IGF-1R; inhibition of osteoclast activity via Akt/FoxO.
- **Citation:** Ghiron LJ et al., *Journal of Bone and Mineral Research*, 1995.

Study 5: Traumatic Brain Injury Recovery

- **Design:** Animal TBI models + Phase 2 human pilot
- **Results:** IGF-1 treatment post-TBI reduced neuronal death, improved cognitive outcomes, and promoted neural regeneration. IGF-1R activation in neurons is directly neuroprotective and regenerative.
- **Citation:** Hatton J et al., *Journal of Neurosurgery*, 1997.

Study 6: Wound Healing

- **Results:** IGF-1 significantly accelerates wound healing — proliferative phase (cell division), remodeling phase (ECM). Fibroblast and keratinocyte proliferation both enhanced. Wound closure time reduced 30–40% in controlled studies.

IGF-1 LR3 Specific Research

- Most direct IGF-1 LR3 studies are in animal models and in vitro; its half-life advantage (20–30 hrs vs. 15 min for native IGF-1) and reduced IGFBP binding are pharmacologically confirmed
- Sport performance community has extensive real-world data on body composition effects; clinical trials specifically for IGF-1 LR3 are pending
- The clinical data on native IGF-1/Mecasermin is directly applicable to LR3 given identical receptor mechanism; LR3 is simply more potent and longer-acting

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3. Primary Uses & Benefits

Goal	Mechanism
Muscle hypertrophy	Satellite cell activation, mTOR, protein synthesis
Anti-catabolic (muscle preservation)	Akt/FoxO → reduced atrogene expression
Fat loss	IGF-1 → lipolysis, metabolic shift
Bone density	Osteoblast activation via IGF-1R
Neuroprotection / TBI recovery	Anti-apoptotic, neural regeneration
Wound/tissue healing	Fibroblast/keratinocyte proliferation
Growth hormone deficiency effects	IGF-1 restoration
Anti-aging	Cellular maintenance, protein synthesis, tissue quality

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4. Symptoms & Conditions It Targets

- Low IGF-1 (lab-confirmed deficiency)
- Sarcopenia / muscle wasting
- ALS / neuromuscular disease (adjunct)
- Growth hormone insensitivity
- Osteoporosis / low bone density
- Athletic recovery and performance optimization
- Traumatic brain injury recovery
- Chronic wound healing
- Post-surgical tissue recovery
- General body composition optimization

5. Key Selling Points (Rep Talking Points)

- ■ **The downstream executor of GH** — IGF-1 is where GH effects are actually delivered; for clients whose GH → IGF-1 conversion is poor, direct IGF-1 LR3 bypasses the conversion issue entirely
- ■ **20–30 hour half-life vs. 15-minute native** — LR3 modification means one injection lasts all day; native IGF-1 would require continuous infusion for similar effect
- ■ **FDA approved in its pharmaceutical form (Mecasermin)** — the receptor mechanism and safety profile are fully validated by FDA drug approval; IGF-1 LR3 is the research version of an established therapeutic
- ■ **Muscle + bone + brain simultaneously** — IGF-1R is expressed on all three tissue types; systemic IGF-1 LR3 reaches all simultaneously
- ■ **Natural GH pathway** — not a hormone itself; stimulates the natural IGF-1 axis; pituitary remains active; reversible if discontinued

6. Common Protocols

Body Composition Protocol

- **Dose:** 20–50 mcg SubQ or IM, once daily (post-workout is common)
- **Cycle:** 4 weeks on, 4 weeks off minimum (receptor desensitization concern)
- **Note:** Longer cycles may reduce receptor sensitivity; rotating sites helps

Anti-Aging / GH Deficiency Protocol

- **Dose:** 10–20 mcg/day (lower, more sustained dose)
- **Timing:** AM or bedtime
- **Cycle:** 8 weeks on, 6 weeks off
- **Goal:** IGF-1 normalization, anti-aging, tissue quality

Advanced Performance Protocol

- **Dose:** 50–80 mcg/day (experienced users only)
- **Split:** May divide into AM + post-workout doses
- **Important:** Always have glucose/food available; hypoglycemia risk at higher doses
- **Cycle:** 4 weeks max at higher doses

Timing Guidance

- **Post-workout:** Maximizes satellite cell recruitment during anabolic window; most common timing
- **Pre-sleep:** Allows IGF-1 to act during the natural GH/growth repair window
- **Always with a meal nearby:** Critical to prevent hypoglycemia

Reconstitution

- 1mg + 1–2 mL bacteriostatic water = 0.5–1mg/mL
- 20 mcg dose = 0.02–0.04 mL from 1mg/mL solution
- Use insulin syringe (31 gauge, 0.3mL)

• Very careful dose measurement required at mcg doses

Storage

• Lyophilized: refrigerate; stable 12–18 months
• Reconstituted: refrigerate; use within 30 days; protect from heat/light
• More fragile than many peptides once reconstituted; handle carefully

Half-Life

• IGF-1 LR3: ~20–30 hours (vs. native IGF-1: 15–20 minutes)
• This is the defining pharmacokinetic advantage of the LR3 modification

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7. Side Effects & Safety Profile

• **Hypoglycemia:** Most important and immediate risk. IGF-1 has insulin-like activity. Administer with food or glucose available. Signs: sweating, shaking, confusion, lightheadedness. Especially at doses >40 mcg in sensitive individuals.
• **Water retention / edema:** Mild, especially in first 1–2 weeks; resolves
• **Joint pain:** At higher doses (carpal tunnel-like symptoms) — dose reduction resolves
• **Jaw/facial growth concern:** Long-term IGF-1 elevation can cause acromegaly features; avoid sustained supraphysiological levels (4-week cycles mitigate this)
• **Organ growth:** IGF-1 drives all tissue growth including potentially organs; long-term excessive use has theoretical organ hypertrophy concern
• **Cancer:** IGF-1 is a growth factor; higher IGF-1 levels associated with some cancers (prostate, colon, breast) in epidemiological data. Active malignancy is an absolute contraindication.
• **Not recommended:** Active cancer, pregnancy, type 1 diabetes (may cause unstable glucose), diabetic retinopathy

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8. Stacking Notes

Stack Partner	Rationale
FST-344	The elite anabolic stack: FST-344 removes myostatin brake + IGF-1 LR3 drives satellite cell proliferation = synergistic muscle growth from two independent mechanisms
Ipamorelin / HGH	Stimulate GH → more endogenous IGF-1; together with exogenous IGF-1 LR3 = comprehensive GH axis optimization
BPC-157 / TB-500	Recovery peptides prevent injury during intensive training driven by IGF-1 anabolic environment
MOTS-c	AMPK metabolic efficiency + IGF-1 anabolic signaling = optimized nutrient partitioning
NAD+	Cellular energy ensures cells can actually execute the protein synthesis IGF-1 is demanding

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9. IGF-1 LR3 vs. Native IGF-1 vs. Des(1-3)IGF-1

Feature	Native IGF-1	IGF-1 LR3	Des(1-3)IGF-1
Half-life	15–20 min	**20–30 hours**	~20–30 min
IGFBP binding	High	Reduced (~100x)	Very low (local)
Receptor potency	1x	~2–3x (free form)	10x (very potent)
Systemic distribution	Moderate	■■ Best	Limited (local)
Hypoglycemia risk	Moderate	Moderate	Higher (more potent)
Best for	Baseline reference	Systemic anabolic	Local injection

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10. FAQ — For Rep Use

Q: How is IGF-1 LR3 different from regular HGH?

A: HGH stimulates the liver to produce IGF-1. IGF-1 LR3 IS IGF-1 (the active effector molecule). For clients whose GH → IGF-1 conversion is inefficient (common with aging), IGF-1 LR3 delivers the effector directly. They're complementary: HGH (via Ipamorelin) + IGF-1 LR3 amplifies the entire axis.

Q: What does hypoglycemia feel like and what do I do?

A: Shakiness, sweating, racing heart, confusion. Have fast-acting glucose ready (glucose tablets, juice, gummy bears). Always inject after/with a meal, especially when starting. At 20–30 mcg doses in fed state, hypoglycemia is rare but possible.

Q: Can it be used for injury recovery?

A: Yes — IGF-1 is a key driver of satellite cell activation for muscle repair, fibroblast proliferation for connective tissue, and neural regeneration. It's excellent stacked with TB-500 and BPC-157 for comprehensive recovery.

Q: How long before seeing results?

A: Body composition changes typically emerge at 3–4 weeks. Most users report improved pumps and recovery within 1–2 weeks. Full muscle growth response: 4–8 weeks of consistent training + adequate protein.

Q: Is the cancer concern serious?

A: For healthy individuals at therapeutic doses and cycle protocols, the risk appears manageable. The correlation between high IGF-1 and cancer risk is epidemiological (observational) — not proven causal. The recommendation: avoid use if you have active cancer or a strong family history of IGF-1-sensitive cancers (prostate, breast, colon); get IGF-1 labs periodically to ensure levels remain within normal physiological range.

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FST-344 (Follistatin 344) — Mavyn Rep Training Sheet

SKU: FST344 | Strength: 1mg | Category: Myostatin Inhibitor / Muscle / Anti-Fibrotic

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1. What Is FST-344?

Follistatin-344 (FST-344) is one of three isoforms of follistatin — a naturally occurring glycoprotein that functions as a potent endogenous inhibitor of activin, myostatin, and other TGF-β superfamily members. FST-344 is the predominant form found in circulating plasma and is produced primarily in the liver, ovaries, and pituitary gland.

The number "344" refers to the amino acid length — 344 amino acids — which corresponds to the most biologically active circulating isoform. The other major isoforms are FST-288 (tissue-bound, gonadal dominant) and FST-315 (also circulating).

FST-344 became the subject of intense research and popular attention when studies in animals with myostatin inhibition (genetic knockouts or follistatin overexpression) showed extraordinary muscle hypertrophy — the "mighty mouse" and "double-muscled" cattle phenomena are related to myostatin deficiency.

Mechanism of Action (Molecular/Cellular Level)

Myostatin (GDF-8) Inhibition — Primary Mechanism:

Myostatin is the body's primary "muscle brake" — a TGF-β superfamily member that limits muscle growth by:

- Inhibiting satellite cell (myogenic progenitor) activation
- Suppressing myoblast proliferation and differentiation
- Reducing protein synthesis signaling
- Promoting muscle protein catabolism

FST-344 binds myostatin with extremely high affinity (Kd ~1 nM), forming a stable, inert complex that neutralizes myostatin activity. Without myostatin brake:

- Satellite cells are released from inhibition → proliferate → fuse with muscle fibers
- Protein synthesis signaling (mTOR, Akt) operates at higher capacity
- Muscle hypertrophy and hyperplasia can proceed
- Net: muscle grows beyond its "natural ceiling"

Activin A/B Inhibition:

FST-344 also potently inhibits activin A and B (other TGF-β members):

- In muscle: activin A promotes catabolism and inhibits growth (similar to myostatin); blocking it amplifies the anabolic effect
- In reproductive tissue: activin regulates FSH; FST inhibition has hormone-relevant effects
- In bone: activin inhibition promotes bone formation
- Anti-fibrotic: activin A is a key driver of fibrosis; FST-344's inhibition of activin A reduces fibrotic scar formation in muscle and other tissues

BMP (Bone Morphogenic Protein) Modulation:

FST-344 also binds select BMPs (GDF-11, BMP-2, BMP-4, BMP-7):

- Modulation of BMPs affects bone formation, cardiac muscle, and neural development

- GDF-11 binding is particularly relevant: GDF-11 is debated as an "aging factor" in circulation; FST-344's inhibition of GDF-11 may have anti-aging implications

Anti-Fibrotic Action:

Beyond muscle, FST-344's inhibition of TGF-β family members (particularly activin A and GDF-8) reduces pathological fibrosis in:

- Muscle (replacing fibrosis with functional tissue)
- Liver (anti-fibrotic in NASH models)
- Lungs (anti-fibrotic in pulmonary fibrosis models)
- Heart (anti-fibrotic post-MI)

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2. Clinical Research & Trials

Study 1: Gene Therapy with Follistatin — Muscular Dystrophy

- **Design:** Phase 1/2 clinical trial, n=6 patients with Becker Muscular Dystrophy (BMD)
- **Intervention:** AAV-mediated follistatin gene delivery to quadriceps
- **Results:** All 6 patients showed increased muscle fiber cross-sectional area. 5/6 showed functional improvement in 6-minute walk test. Strength improvements persisted for 2+ years follow-up. No adverse events.
- **Citation:** Mendell JR et al., *Molecular Therapy*, 2015. (Landmark human gene therapy trial)

Study 2: Muscular Dystrophy — Phase 2 Extension

- **Continued follow-up of Mendell trial cohort**
- **5-year data:** Sustained muscle size improvements maintained. Histological analysis showed increased muscle fiber diameter and reduced fibrosis. Satellite cell density improved.
- **Clinical significance:** Confirms long-term efficacy and safety of follistatin-mediated myostatin inhibition in humans.

Study 3: Post-Menopausal Bone/Muscle Loss

- **Design:** Animal model (ovariectomized mice = menopause model) + in vitro human cells
- **Results:** FST overexpression prevented ovariectomy-induced muscle atrophy and reduced bone loss. Mechanism: activin inhibition preserves both muscle and bone mass simultaneously.
- **Implication for clients:** Menopausal women losing muscle and bone simultaneously — a condition with no single approved drug solution — could benefit from FST-344's dual action.

Study 4: Cardiac Fibrosis Prevention

- **Model:** Mouse myocardial infarction
- **Results:** Follistatin significantly reduced cardiac fibrosis at 28 days, with better preserved cardiac function (higher ejection fraction) vs. controls. Anti-fibrotic mechanism via activin A inhibition in cardiac fibroblasts.

Study 5: Animal Muscle Hypertrophy — Foundational Research

- **Multiple models:** Mice overexpressing follistatin, including Sweeney lab (UPenn), Bhatt lab
- **Results:** Systemic follistatin overexpression produces 100–194% increases in skeletal muscle mass in mice — across all muscle groups, uniformly. These are the most dramatic muscle growth results ever achieved with any biological agent without exercise.
- **Citation:** Lee SJ et al., *PNAS*, 2004; multiple follow-up publications.

Study 6: Liver Fibrosis / NASH

- **Model:** Carbon tetrachloride (CCl_4) liver fibrosis model
- **Results:** FST treatment significantly reduced liver collagen deposition (fibrosis score) and normalized AST/ALT levels. Anti-fibrotic via activin A inhibition in hepatic stellate cells.

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3. Primary Uses & Benefits

Goal	Mechanism
Muscle mass increase	Myostatin inhibition → satellite cell release
Strength enhancement	Muscle hypertrophy + hyperplasia
Anti-catabolism	Myostatin/activin inhibition during caloric deficit
Bone density (adjunct)	Activin inhibition → osteoblast promotion
Anti-fibrotic (muscle, liver, heart)	Activin A inhibition
Muscular dystrophy support	Myopathy management (clinical data)
Body composition (lean mass preservation)	Myostatin inhibition during fat loss

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4. Symptoms & Conditions It Targets

- Sarcopenia (age-related muscle loss)
- Muscular dystrophy (Becker, Duchenne — clinical data)
- Muscle atrophy from immobilization / injury
- Low muscle mass despite training
- Genetic "hard gainer" plateau
- Menopausal muscle/bone loss
- Liver fibrosis / NASH
- Cardiac fibrosis post-MI
- Body composition optimization

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5. Key Selling Points (Rep Talking Points)

- ■ **Removes your genetic ceiling** — myostatin is literally the body's muscle growth limiter; FST-344 releases that brake; clients can grow muscle beyond their genetic potential
- ■ **Human clinical trial data in muscular dystrophy** — actual Phase 1/2 human data showing muscle growth and functional improvement; not just animal research
- ■ **The most potent anabolic peptide by mechanism** — no other non-hormonal peptide achieves 100–194% muscle mass increase in animal models; the mechanism is categorically different from any GH/IGF-1 approach
- ■ **Anti-fibrotic across multiple organs** — dual benefit: muscle gains + reduction of pathological scarring in muscle, liver, heart
- ■ **Bone AND muscle simultaneously** — activin inhibition prevents bone loss while building muscle; especially compelling for aging or menopausal clients

6. Common Protocols

Standard Body Composition Protocol

- **Dose:** 100 mcg SubQ, once daily for 10 days
- **Cycle:** 10 days on, 2–4 months off (important: long cycle breaks due to potent myostatin inhibition; continuous dosing may cause excessive tissue adaptation)
- **Note:** 1mg vial = 10 doses at 100 mcg/dose

Anti-Aging / Sarcopenia Protocol

- **Dose:** 50–100 mcg SubQ, once daily for 10 days
- **Frequency:** Every 3–4 months
- **Goal:** Maintained muscle mass during aging, anti-sarcopenia

High-Performance Protocol

- **Dose:** 200 mcg/day for 10 days
- **Cycle:** 10 days, 3–4 months off minimum
- **Note:** Higher doses have diminishing returns as receptor saturation approached

Reconstitution

- 1mg + 1mL bacteriostatic water = 1mg/mL (1000 mcg/mL)
- 100 mcg dose = 0.1 mL
- Use insulin syringe

Storage

- Lyophilized: refrigerate; stable 12–18 months
- Reconstituted: refrigerate; use within 30 days
- **Protein stability note:** FST-344 is a larger protein (344 aa); handle gently; do not vortex; roll to mix; avoid temperature extremes

Half-Life

- Plasma half-life: ~4–8 hours for free circulating form
- Effective inhibition of myostatin: 24–48 hours per dose due to high-affinity binding

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7. Side Effects & Safety Profile

- **Gene therapy studies show excellent safety profile** — the most direct human data indicates good tolerability
- Animal studies with chronic overexpression show:
- Some models: excessive muscle growth causing joint stress (at extreme doses / genetic overexpression — not therapeutic)

• Reproductive effects: FST inhibits activin → reduces FSH → may temporarily affect follicle development in women; monitor menstrual cycle

• **Testosterone**: Activin inhibition may modestly increase testosterone in men (activin normally reduces T production)

• No hepatotoxicity documented at therapeutic doses

• No nephrotoxicity

• **FSH suppression in women:** A real consideration — women planning pregnancy should avoid use; may temporarily suppress ovulation

• **Cardiac:** Theoretical benefit (anti-fibrotic) but monitor in patients with existing cardiomyopathy

• **Active cancer:** Anti-angiogenic and anti-fibrotic effects create complex theoretical picture — avoid until better understood in oncology context

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8. Stacking Notes

Stack Partner	Rationale
IGF-1 LR3	The premier anabolic stack — IGF-1 (satellite cell activation + protein synthesis) + FST-344 (myostatin inhibition) = anabolic signaling from two independent pathways
Ipamorelin / HGH	GH/IGF-1 environment supports the satellite cell proliferation FST-344 unleashes
BPC-157 / TB-500	Recovery support during intensive muscle building; preventing the injuries that hard training can cause
MOTS-c	Metabolic + anabolic: MOTS-c maintains mitochondrial efficiency while FST-344 drives muscle growth
Retatrutide	Body recomposition: fat loss (Retatrutide) + lean mass maximization (FST-344); potentially the most powerful recomp stack available

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9. FST-344 vs. Other Muscle/Anabolic Peptides

Feature	FST-344	IGF-1 LR3	HGH	TB-500
Myostatin inhibition	■■ Primary mechanism	■	■	■
Muscle hypertrophy	■■ Most potent	■■ Strong	■ Good	■ Good
Angiogenesis	Moderate	Moderate	Moderate	■■ Best
Fat loss	Indirect	■ Good	■■ Best	■
Anti-fibrotic	■■ Best	Moderate	Moderate	■ Good
Bone density	■ Good (activin)	■ Good	■ Good	Moderate
Human clinical data	■ Phase 1/2	Limited	■■ Extensive	Phase 2
Hormonal effects	Mild (FSH)	IGF-1 elevation	GH axis	None

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10. FAQ — For Rep Use

Q: Will FST-344 make me abnormally huge?

A: Not at therapeutic doses for short cycles. The extraordinary animal results (100–194% mass increase) came from genetic overexpression — not short peptide cycles. Therapeutic 10-day cycles produce meaningful but proportional muscle gains. The results depend heavily on training stimulus and protein intake.

Q: Why do cycles need to be so short (10 days)?

A: Because follistatin's inhibition of myostatin is so potent and comprehensive, extended dosing can cause adaptive effects. The 10-day cycle with 2–4 month breaks allows the body to adapt to the new muscle state before the next stimulus. It's also partly precautionary pending more human dosing data.

Q: How does it affect hormones in women?

A: Activin normally promotes FSH secretion. FST-344's inhibition of activin can transiently reduce FSH → potential menstrual irregularity or delayed ovulation. This is temporary and resolves after the cycle. Women trying to conceive should avoid use.

Q: Is it detectable in sports testing?

A: Potentially — WADA has methods for detecting follistatin isoform ratios. Athletes in tested sports should be aware of this. For non-tested fitness clients, not a concern.

Q: What's the best stack for body recomposition?

A: FST-344 + Retatrutide + Ipamorelin is the strongest theoretical recomp stack: Retatrutide drives fat loss and metabolic improvement, Ipamorelin provides GH/IGF-1 for lean mass support, and FST-344 removes the myostatin brake entirely. Extremely potent combination.

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RETATRUTIDE — Mavyn Rep Training Sheet

SKUs: RT5 (5mg) | RT10 (10mg) | RT15 (15mg) | RT20 (20mg) | Category: GLP-1/GIP/Glucagon Triple Agonist

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1. What Is Retatrutide?

Retatrutide (LY3437943) is a novel synthetic peptide developed by Eli Lilly as a **triple incretin receptor agonist** — the first compound to simultaneously activate all three key metabolic receptors:

- **GLP-1R** (Glucagon-Like Peptide-1 Receptor) — appetite suppression, insulin secretion, gastric emptying
- **GIPR** (Glucose-Dependent Insulinotropic Polypeptide Receptor) — synergistic insulin and glucagon modulation, fat metabolism
- **GCGR** (Glucagon Receptor) — energy expenditure, hepatic fat reduction, thermogenesis

This triple mechanism is what differentiates Retatrutide from:

- **Semaglutide** (Ozempic/Wegovy): GLP-1 only
- **Tirzepatide** (Mounjaro/Zepbound): GLP-1 + GIP dual agonist
- **Retatrutide**: GLP-1 + GIP + Glucagon — the most comprehensive metabolic action available

Mechanism of Action (Molecular/Cellular Level)

GLP-1 Receptor Agonism:

- Stimulates glucose-dependent insulin secretion from pancreatic beta cells
- Suppresses glucagon from alpha cells (reduces hepatic glucose output)
- Slows gastric emptying → prolonged satiety, reduced post-meal glucose spikes
- Acts on hypothalamic appetite centers (arcuate nucleus, nucleus tractus solitarius) to reduce hunger signals via pro-opiomelanocortin (POMC) pathway
- Reduces food reward/cravings via mesolimbic dopamine modulation
- Cardioprotective: reduces cardiovascular inflammation, improves endothelial function

GIP Receptor Agonism:

- Synergizes with GLP-1 for insulin secretion (additive effect at GIPR + GLP-1R co-stimulation)
- Promotes adipocyte lipid utilization and reduces adipogenesis
- Reduces nausea (counterintuitively; GIPR activation attenuates GLP-1-induced nausea)
- Bone protective effects (GIPR expressed on osteoblasts)
- Central appetite effects complementary to GLP-1

Glucagon Receptor Agonism (the key differentiator):

- Increases energy expenditure (thermogenesis) — glucagon receptor activation in brown adipose tissue and liver increases metabolic rate
- Drives hepatic fat oxidation — directly reduces liver fat (non-alcoholic fatty liver disease)
- Stimulates lipolysis in adipose tissue
- Increases basal metabolic rate — counteracts the metabolic adaptation that limits weight loss at plateau
- This explains why Retatrutide shows greater weight loss than dual agonists despite similar or lower GLP-1 component dosing

Long Half-Life Engineering: Retatrutide is engineered as a fatty-acid-conjugated peptide enabling once-weekly dosing (similar to semaglutide's approach but applied to triple agonism).

2. Clinical Research & Trials

Study 1: Phase 2 Dose-Ranging Trial — Obesity

• **Sponsor:** Eli Lilly

• **Design:** Randomized, double-blind, placebo-controlled, n=338 adults with BMI ≥27 with weight-related comorbidity or BMI ≥30

• **Duration:** 48 weeks

• **Dosing Arms:** 1mg, 4mg, 8mg, 12mg weekly SubQ injection

• **Results (at 48 weeks):**

• **1mg:** -7.9% body weight

• **4mg:** -17.3% body weight

• **8mg:** -22.8% body weight

• **12mg: -24.2% body weight** (unprecedented in any single agent at 48 weeks)

• Placebo: -2.1%

• **Comparator context:** Semaglutide 2.4mg (Wegovy) achieves ~15% at 68 weeks; tirzepatide achieves ~22.5% at 72 weeks

• **Additional findings:** 26% of 12mg group achieved ≥30% body weight reduction. Waist circumference reduced up to 18cm. HbA1c reduced significantly in diabetic participants.

• **Safety:** Adverse events primarily GI (nausea, diarrhea) — typical of GLP-1 class; no serious safety signals

• **Citation:** Jastreboff AM et al., *New England Journal of Medicine*, 2023, Vol 389:514-526. (Major publication)

Study 2: Non-Alcoholic Fatty Liver Disease (NAFLD/NASH) — Phase 2 Data

• **Design:** Subset analysis of Phase 2 trial + dedicated liver imaging cohort

• **Results:** MRI-PDFF (liver fat fraction) reduced by 81% in the 8mg group at 24 weeks — from mean 22% liver fat to 4.2%. Liver fibrosis markers (ALT, AST, FIB-4 index) normalized in majority. NASH resolution demonstrated.

• **Context:** No approved NASH drug achieved this liver fat reduction rate

• **Citation:** Lilly Phase 2 supplemental data; Jastreboff et al. NEJM 2023 supplementary.

Study 3: Type 2 Diabetes — Phase 2 (TRIUMPH-1)

• **Design:** n=281 T2DM patients on stable metformin

• **Duration:** 24 weeks

• **Results:** HbA1c reduced -2.02% (8mg arm) vs. -0.18% placebo. 87% of treated patients achieved HbA1c < 7%. Weight loss 16.9% at 24 weeks in 8mg arm (equivalent to many bariatric outcomes).

• **Citation:** Lilly TRIUMPH-1 trial data; presented at ADA 2023.

Study 4: Cardiovascular Risk Markers

• **Findings in Phase 2 data:** LDL-C reduced -18%; triglycerides reduced -42%; systolic BP reduced -5.6 mmHg. These improvements compound the direct cardiovascular benefit of GLP-1 receptor activation (anti-atherosclerotic, anti-inflammatory endothelial effects).

Phase 3 Status

• Retatrutide Phase 3 trials (TRIUMPH series) ongoing for obesity, T2DM, and NASH as of 2024–2025. FDA approval anticipated 2026–2027.

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3. Primary Uses & Benefits

Goal	Mechanism
Weight loss / obesity	Appetite suppression (GLP-1 + GIP), thermogenesis (GCGR)
Fat mass reduction	Lipolysis, adipogenesis inhibition, energy expenditure ↑
Blood sugar control	Insulin stimulation, glucagon suppression
Liver fat reduction	GCGR-driven hepatic fat oxidation
Cardiovascular protection	LDL/TG reduction, BP reduction, endothelial health
Metabolic syndrome reversal	All-in-one metabolic correction
Bodycomposition	Fat loss with potential lean mass preservation

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4. Symptoms & Conditions It Targets

• Obesity (BMI ≥27 with comorbidity or ≥30)
• Type 2 Diabetes Mellitus
• Prediabetes / insulin resistance
• Non-alcoholic fatty liver disease (NAFLD/NASH)
• Metabolic syndrome
• Hypertriglyceridemia
• Cardiovascular risk reduction
• Weight-related joint pain / sleep apnea

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5. Key Selling Points (Rep Talking Points)

• ■ **Most powerful weight loss peptide available** — 24% body weight loss at 48 weeks surpasses every approved medication; up to 30%+ reduction in significant responders
• ■ **Triple mechanism = results where others plateau** — the glucagon component adds thermogenesis that semaglutide and tirzepatide lack; clients who've "maxed out" on other GLP-1s often see further progress
• ■ **NASH/liver fat reversal** — 81% reduction in liver fat is extraordinary; for clients with fatty liver, this is potentially life-changing
• ■ **Comprehensive metabolic correction** — blood sugar, liver health, lipids, blood pressure, and weight all improve simultaneously from a single weekly injection
• ■ **Pipeline drug, cutting-edge access** — retatrutide is still in Phase 3 trials; clients accessing it now are ahead of the market; Lilly is projecting this to become the dominant metabolic drug

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6. Common Protocols

Beginner / Low-Dose Start (Recommended)

- **Week 1–4:** 2mg SubQ, once weekly (minimize GI side effects)
- **Week 5–8:** 4mg SubQ, once weekly
- **Assess tolerance before escalating**

Intermediate Protocol

- **Week 1–4:** 2mg → escalate to 4mg
- **Week 5–12:** 4mg → escalate to 8mg
- **Maintenance:** 8mg/week
- **Most clients hit significant results at 8mg** — mirrors the "sweet spot" in Phase 2 data

Advanced / Aggressive Protocol

- **Dose titration to 12mg/week** (per Phase 2 protocol)
- **Requires close monitoring:** GI tolerability, gallbladder health (long-term GLP-1 use → reduced gallbladder motility)
- Only for weight management under clinical supervision

Injection Protocol

- SubQ injection, abdomen, thigh, or upper arm
- Rotate sites weekly
- Once weekly, same day each week
- Use insulin-grade syringes (0.3–0.5 mL, 31 gauge)

Reconstitution

- Reconstitute with bacteriostatic water: 1–2 mL per vial
- 10mg vial + 2mL bac water = 5mg/mL solution
- Typical 8mg dose = 1.6 mL from 5mg/mL solution

Storage

- Lyophilized: refrigerate (2–8°C); stable 12–18 months
- Reconstituted: refrigerate; use within 28 days; do not freeze

Half-Life

- ~6–7 days (engineered for weekly dosing via fatty acid conjugation)

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7. Side Effects & Safety Profile

- **GI effects (most common):** Nausea (40–65% at higher doses), vomiting, diarrhea, constipation — typically front-loaded and improve after first 4–8 weeks; managed with slow titration
- **Reduced appetite:** Expected effect; some clients report excessive appetite suppression at higher doses
- **Injection site reactions:** Mild, transient
- **Gallbladder:** Long-term GLP-1 use associated with increased gallstone risk (reduced gallbladder motility); assess baseline gallbladder health in long-term users

- **Pancreatitis:** Very rare but class risk; contraindicate in patients with history of pancreatitis
- **Thyroid:** GLP-1 agonists carry a black box warning for medullary thyroid carcinoma risk in rodents; not confirmed in humans but contraindicated in MEN2 syndrome or personal/family history of MTC
- **Lean mass:** Some concern about muscle loss with rapid weight loss; pairing with resistance training and adequate protein critical
- **Hypoglycemia:** Low risk in non-diabetics; modest risk when combined with insulin or sulfonylureas

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8. Stacking Notes

Stack Partner	Rationale
Ipamorelin	GH pulsing supports lean mass preservation during caloric deficit; prevents muscle loss
BPC-157	Gut healing to manage GI side effects of GLP-1 class; reduces nausea/motility issues
MOTS-c	Mitochondrial metabolic activation synergistic with Retatrutide's insulin sensitization
NAD+	Supports mitochondrial energy production and metabolic efficiency during weight loss
GHK-Cu	Skin elasticity support as rapid fat loss can leave loose skin; GHK-Cu drives collagen remodeling
HGH	Lean mass preservation + fat loss amplification; GH + GLP-1 triple agonist is a high-performance body composition stack

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9. Comparison to GLP-1 Class

Metric	Semaglutide	Tirzepatide	Retatrutide
Receptors	GLP-1	GLP-1 + GIP	GLP-1 + GIP + GCGR
Weight loss (max)	~15%	~22.5%	**~24%+ (up to 30%)**
Thermogenesis	Minimal	Moderate	■ Strong (GCGR)
Liver fat	Moderate reduction	Good	**81% reduction**
Nausea	High	Moderate	Moderate (GIPR helps)
Dosing frequency	Weekly	Weekly	Weekly
Approval status	FDA approved	FDA approved	Phase 3 (pre-approval)
Best for	Weight + diabetes	Weight + diabetes	Maximum metabolic impact

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10. FAQ — For Rep Use

Q: How does this compare to Ozempic or Wegovy?

A: Retatrutide is a triple agonist vs. Ozempic's single GLP-1 mechanism. In head-to-head comparable data, Retatrutide produces nearly double the weight loss of semaglutide. It also adds a unique thermogenesis component that prevents

metabolic adaptation — a major limitation of GLP-1-only drugs.

Q: What about muscle loss?

A: Rapid weight loss with any GLP-1 class drug carries muscle loss risk. The mitigation strategy is: adequate protein intake (1.2–1.6g/kg lean mass), resistance training, and optionally stacking with Ipamorelin or low-dose HGH to preserve lean tissue.

Q: Is it safe long-term?

A: Phase 2 data shows a strong safety profile. The GLP-1 class concerns (pancreatitis, thyroid, gallbladder) apply, but incidence remains low. Phase 3 data will clarify long-term safety further.

Q: Can it be used for people who aren't obese?

A: Clinical interest is growing for metabolic optimization, NAFLD, and body composition goals. Clinically it's indicated for significant weight concerns, but wellness clients use it off-label for body composition.

Q: When will it be FDA approved?

A: Phase 3 trials are underway. Based on Phase 2 efficacy data, analyst projections suggest 2026–2027 approval if trials succeed. It is currently available as a research peptide.

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NAD+ (Nicotinamide Adenine Dinucleotide) — Mavyn Rep Training Sheet

SKUs: NJ500 (500mg) | NJ1000 (1000mg) | Category: Metabolic / Anti-Aging / Cellular Repair

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1. What Is NAD+?

NAD+ (Nicotinamide Adenine Dinucleotide) is a coenzyme found in every living cell — arguably the most fundamental molecule in cellular energy metabolism. It exists in two forms: NAD+ (oxidized) and NADH (reduced), cycling between them as it shuttles electrons in metabolic reactions.

NAD+ levels peak in youth and decline 40–60% between ages 20 and 60. This decline is increasingly recognized as a central driver of aging, metabolic dysfunction, and age-related disease. Restoring NAD+ levels is one of the most evidence-backed longevity interventions in modern geroscience.

Mechanism of Action (Molecular/Cellular Level)

Mitochondrial Energy Production (Primary):

NAD+ is the essential cofactor for the electron transport chain (ETC). In every cell, NAD+ accepts electrons from glycolysis and the TCA cycle to become NADH, which then donates electrons to Complex I of the mitochondrial ETC → generating ATP. Without NAD+, cells cannot produce energy efficiently — every organ system suffers.

Sirtuin Activation (Sirtuins 1–7):

Sirtuins are NAD+-dependent deacetylases — the "longevity enzymes." They require NAD+ as a substrate (not just a cofactor). Key functions:

- **SIRT1:** Gene expression regulation, autophagy induction, mitochondrial biogenesis (via PGC-1α)
- **SIRT3:** Mitochondrial protein deacetylation, ROS reduction, metabolic efficiency
- **SIRT6:** DNA repair, genomic stability, telomere maintenance
- **SIRT7:** Chromatin remodeling, stress response

As NAD+ declines with age, sirtuin activity declines — accelerating the hallmarks of aging: genomic instability, telomere attrition, mitochondrial dysfunction, cellular senescence.

PARP Activation (DNA Repair):

PARP-1 (Poly ADP-ribose polymerase) is the primary DNA repair enzyme in the nucleus. It consumes enormous amounts of NAD+ during DNA repair. Aging, UV radiation, inflammation, and oxidative stress cause persistent DNA damage, chronically depleting NAD+ through PARP hyperactivation — creating a vicious cycle of NAD+ depletion → impaired repair → more damage.

CD38 — NAD+ Consumer:

CD38 is an enzyme that degrades NAD+ to cyclic ADP-ribose (cADPR). CD38 expression increases dramatically with age and inflammation. It is responsible for a large portion of age-related NAD+ decline. This is why precursor supplementation must overcome the CD38 drain.

NMN/NR Pathway:

Intravenous/subcutaneous NAD+ bypasses the oral bioavailability issue — oral NAD+ is largely cleaved to precursors in the gut. Direct NAD+ injection provides immediate cellular replenishment without conversion steps, achieving 10–100x higher bioavailability than oral NMN or NR.

2. Clinical Research & Trials

Study 1: IV NAD+ — Addiction Recovery & Neurological Restoration

- **Setting:** Springfield Wellness Center (Louisiana) — the pioneering NAD+ IV clinic
- **Population:** Patients with opioid, alcohol, and stimulant dependence; also Parkinson's and chronic fatigue
- **Protocol:** 500–1500mg IV NAD+ over 4–8 hours/day for 10 days
- **Results:** Significant reduction in acute withdrawal symptoms. 70–80% report complete absence of cravings by day 7–10. Parkinson's patients reported improved motor function. Chronic fatigue patients showed energy restoration. No significant adverse effects at therapeutic doses.
- **Citation:** Braidy N, Grant R, Saied H — multiple publications; clinical case series data from Springfield Wellness Center.

Study 2: NAD+ and Muscle Function in Older Adults — RCT

- **Design:** Double-blind RCT, n=60 adults aged 60–80
- **Intervention:** NMN (oral NAD+ precursor) 250mg/day vs. placebo for 12 weeks [direct parallel to injectable NAD+]
- **Results:** Significant improvement in gait speed, grip strength, and chair stand test scores. NAD+ levels in PBMCs increased 38%. No adverse events.
- **Citation:** Yi L et al., *GeroScience*, 2022.

Study 3: NAD+ and Metabolic Syndrome

- **Design:** Crossover RCT, n=30 postmenopausal women with metabolic syndrome
- **Dosing:** NMN 250mg/day for 10 weeks
- **Results:** Skeletal muscle insulin sensitivity improved significantly (+25% vs. placebo by clamp studies). Muscle NAD+ levels confirmed elevated. Gene expression analysis: SIRT1, SIRT3, PGC-1α all upregulated.
- **Citation:** Yoshino M et al., *Science*, 2021. (High-impact study establishing clinical insulin sensitization)

Study 4: Cognitive Function & Neurodegeneration

- **Design:** Animal models (Alzheimer's, Parkinson's) + emerging human data
- **Key findings:** NAD+ precursor supplementation reduced amyloid-β accumulation, improved cognitive performance in transgenic AD mice, activated autophagy to clear protein aggregates. SIRT1 activation shown to reduce Tau phosphorylation.
- **Human pilot (n=22):** Mild cognitive impairment patients; NMN 600mg/day for 16 weeks showed trend improvements in MMSE and MOCA scores.

Study 5: DNA Repair & Anti-Cancer (Theoretical Benefit)

- **Findings:** NAD+ replenishment in aging cells restores PARP-1 activity $\rightarrow$ improved DNA double-strand break repair $\rightarrow$ reduced mutation accumulation. In animal models, this translated to reduced carcinogenesis rates and extended healthy lifespan.
- **Note:** This is a theoretical benefit with strong mechanistic support; no human RCT for cancer prevention specifically.

Study 6: IV NAD+ Bioavailability Study

- **Comparison:** IV NAD+ 500mg vs. oral NMN 500mg

• **Finding:** IV NAD+ produced plasma NAD+ levels 15–25x higher than oral NMN at matched doses. Tissue uptake (muscle, brain, liver) significantly superior via IV route.

Study 7: Longevity — Interventions Testing Program (ITP)

• **Model:** NMN supplementation in naturally-aging mice (C57BL/6)

• **Results:** 8% increase in median lifespan. Preserved vascular endothelial function, mitochondrial density in skeletal muscle, and fertility metrics in aging females.

• **Citation:** Mills KF et al., *Cell Metabolism*, 2016.

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3. Primary Uses & Benefits

Goal	Mechanism
Anti-aging / longevity	Sirtuin activation, telomere support, DNA repair
Energy & fatigue	Mitochondrial ATP production restoration
Brain health / cognition	Neuronal energy, SIRT1, neuroprotection
Metabolic health	Insulin sensitivity via SIRT1/3, PGC-1α
Athletic performance	Muscle mitochondrial density, recovery
Addiction recovery	Neurochemical restoration, neurotransmitter balance
DNA repair	PARP-1 reactivation
Cardiovascular health	Endothelial function, vascular NAD+ restoration

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4. Symptoms & Conditions It Targets

• Chronic fatigue / low energy / mitochondrial dysfunction
• Brain fog / cognitive decline
• Age-related metabolic decline
• Insulin resistance / prediabetes
• Poor post-exercise recovery
• Neurodegenerative disease (adjunct support)
• Addiction recovery (opioid, alcohol, stimulant)
• Inflammatory conditions (NAD+ modulates NF-κB)
• Depression / mood disorders (neurochemical energy basis)
• Skin aging (fibroblast SIRT1 activation)

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5. Key Selling Points (Rep Talking Points)

• ■ **The cellular energy molecule** — every cell in the body runs on NAD+; replenishing it is the most foundational intervention in the longevity toolkit

• ■ **Injectable = real replenishment** — oral NAD+ precursors (NMN, NR) have limited conversion; SubQ/IV NAD+ delivers directly into circulation at 10–25x higher bioavailability

- ■ **Proven in human clinical trials** — not just animal data; RCTs showing muscle function improvement, insulin sensitivity, and cognitive trends in older adults
- ■ **Sirtuin activation = the longevity pathway** — the same pathway activated by caloric restriction and exercise; NAD+ is the master key
- ■ **Addiction recovery evidence** — the Springfield protocol has helped hundreds of patients through withdrawal with NAD+ IV; injectable NAD+ is the closest practical equivalent

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6. Common Protocols

SubQ Injection Protocol (Standard for Mavyn Clients)

- **Dose:** 100–500mg per injection, SubQ
- **Frequency:** 3–5x/week (initial loading), then 2–3x/week maintenance
- **Loading Phase (2 weeks):** 250–500mg daily or EOD
- **Maintenance:** 500mg, 2–3x/week

High-Dose Anti-Aging Protocol

- **NJ1000:** 1000mg SubQ, 3x/week for 4 weeks (loading), then 500mg 2x/week
- **Best for:** Significant metabolic dysfunction, post-illness recovery, aggressive anti-aging protocols

Reconstitution

- Reconstitute with sterile water or bacteriostatic water
- 500mg vial + 5 mL bac water = 100mg/mL solution
- SubQ injection: abdomen or thigh

Storage

- **Lyophilized:** Refrigerate; stable 12–24 months
- **Reconstituted:** Refrigerate, use within 14 days — NAD+ is relatively unstable in solution; refrigerate and minimize light exposure
- Do NOT freeze reconstituted solution

Half-Life

- IV NAD+: peak plasma ~30 minutes; tissue incorporation (liver, muscle, brain) occurs rapidly
- Half-life of replenishment effect: cellular NAD+ levels remain elevated for 24–48 hours post-injection

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7. Side Effects & Safety Profile

- **SubQ:** Mild injection site discomfort; transient redness
- **IV (rapid infusion):** Nausea, tightness in chest, flushing — resolved by slower infusion rate. Must NOT be pushed as rapid IV bolus.
- **SubQ high doses:** Some users report warmth, flushing, or mild nausea transiently
- No hepatotoxicity documented at therapeutic doses
- No nephrotoxicity

• No hormonal disruption

• **Theoretical:** Very high-dose sustained supplementation may overclock PARP-1 in cancer cells — not proven at therapeutic doses but noted in literature

• **Pregnancy:** Insufficient data; avoid high-dose supplementation

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8. Stacking Notes

Stack Partner	Rationale
Retatrutide	NAD+ fuels the mitochondria that metabolic improvement depends on; fundamental synergy
Ipamorelin / HGH	GH stimulates mitochondrial biogenesis; NAD+ provides the fuel; deep anti-aging synergy
MOTS-c	MOTS-c activates AMPK; NAD+ is the downstream fuel; powerfully complementary metabolic stack
Epithalon	DNA repair (Epithalon) + DNA repair fuel (PARP-1 via NAD+) = comprehensive genomic longevity stack
Semax	Neuroprotection (Semax) + neuronal energy (NAD+) = brain optimization stack
GHK-Cu	Cellular regeneration from multiple angles: collagen/skin (GHK-Cu) + mitochondrial/genomic (NAD+)

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9. Comparison: NAD+ Delivery Methods

Method	Bioavailability	Onset	Convenience	Best For
Oral NMN/NR	Low–moderate	Slow	High	Daily maintenance
Sublingual NMN	Moderate	Moderate	Moderate	Improved oral
SubQ NAD+ injection	High	Fast	Moderate	Therapeutic protocols
IV NAD+	Highest	Very fast	Low (clinical)	Addiction, acute deficiency
Intranasal NAD+	Moderate	Fast	High	Brain-targeted delivery

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10. FAQ — For Rep Use

Q: Is injectable NAD+ better than NMN supplements?

A: Significantly. Oral NMN/NR must be converted through enzymatic pathways and absorb through the gut — with significant loss. SubQ NAD+ bypasses all of that and enters circulation directly, achieving blood levels 10–25x higher per equivalent dose.

Q: How fast will I feel a difference?

A: Most clients report increased energy, mental clarity, and improved sleep quality within 3–7 days of starting a loading protocol. More subtle metabolic improvements (insulin sensitivity, mitochondrial function) develop over 4–8 weeks.

Q: Is it safe to use long-term?

A: The human body produces NAD+ naturally and uses it continuously — restoring depleted levels is physiologically rational. Long-term use data from clinical programs (some patients on IV NAD+ for years) shows excellent safety. SubQ protocols are even better tolerated than IV.

Q: Who benefits most?

A: Anyone over 35 benefits from NAD+ restoration given universal age-related decline. Specifically: chronic fatigue, metabolic syndrome, brain fog, addiction recovery, and longevity-focused clients see the most dramatic results.

Q: Can it be combined with sirtuins-activating supplements like resveratrol?

A: Yes — resveratrol activates SIRT1 via allosteric modulation but requires NAD+ as a substrate. They're synergistic: resveratrol = the accelerator, NAD+ = the fuel.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

GHK-Cu (Copper Peptide / GHK-Copper) — Mavyn Rep Training Sheet

SKUs: CU50 (50mg) | CU100 (100mg) | Category: Tissue Remodeling / Anti-Aging / Regenerative

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1. What Is GHK-Cu?

GHK-Cu (Glycyl-L-Histidyl-L-Lysine–Copper) is a naturally occurring copper-binding tripeptide first isolated from human plasma in 1973 by biochemist Loren Pickart. It is found in plasma, saliva, and urine, and is released from tissue proteins during injury or remodeling as a wound healing signal.

GHK-Cu is one of the most thoroughly studied anti-aging peptides, with over four decades of research demonstrating remarkable effects on skin, hair, tissue repair, anti-inflammatory signaling, and gene expression modulation. It is distinct from other peptides in that it functions as a **copper chaperone** — delivering bioavailable Cu2+ ions to specific enzymes while simultaneously acting as a signaling peptide in its own right.

GHK-Cu levels decline significantly with age:

- Age 20: plasma ~200 ng/mL
- Age 60: plasma ~80 ng/mL
- Age 80: plasma ~20–30 ng/mL

This 60–80% decline correlates with reduced wound healing capacity, skin aging, increased inflammation, and cancer vulnerability in older adults.

Mechanism of Action (Molecular/Cellular Level)

Copper Delivery to Key Enzymes:

GHK-Cu delivers Cu2+ to copper-dependent enzymes:

- **Lysyl oxidase:** Cross-links collagen and elastin fibers → tensile strength of connective tissue
- **SOD-1 (Cu/Zn Superoxide Dismutase):** Primary cytoplasmic antioxidant enzyme
- **Cytochrome C oxidase (Complex IV):** Mitochondrial electron transport → ATP production
- **Ceruloplasmin:** Iron metabolism and antioxidant

Collagen & ECM Remodeling:

GHK-Cu is arguably the most potent stimulator of collagen synthesis in skin:

- Stimulates type I and type III collagen synthesis in fibroblasts
- Stimulates elastin production
- Stimulates glycosaminoglycan (hyaluronic acid, chondroitin sulfate) synthesis
- Simultaneously activates MMPs (collagenases) to remove damaged/cross-linked collagen AND their inhibitors (TIMPs) to modulate remodeling pace
- Net effect: old/damaged collagen degraded → replaced with new, organized collagen

Angiogenesis: Stimulates VEGF expression → new capillary formation at wound/remodeling sites → better oxygen and nutrient delivery

Anti-Inflammatory: Suppresses pro-inflammatory cytokines (IL-1β, TNF-α, TGF-β1). Particularly notable: reduces TGF-β1, which drives fibrosis — making GHK-Cu specifically anti-fibrotic.

Antioxidant Action:

- Delivers Cu to SOD-1 → dramatically increases oxidative stress defense
- Independently scavenges lipid peroxides
- Reduces oxidative damage markers (8-OHdG, protein carbonyls) in aging tissue

Gene Expression — "Systems Reset":

The most remarkable and recent discovery about GHK-Cu: large-scale gene expression analysis shows it modulates **4,000+ genes** — effectively reversing gene expression patterns associated with aging back toward younger profiles. Specifically:

- Upregulates 84% of genes that decrease with aging
- Downregulates 68% of genes that increase with aging
- Normalizes expression of genes related to: inflammation, oxidative stress, DNA repair, protein synthesis, and cancer suppression
- **Citation:** Pickart L, Vasquez-Soltero JM, Margolina A, *BioMed Research International*, 2015.

Stem Cell Activation:

GHK-Cu activates bone marrow-derived stem cell migration and differentiation — contributing to tissue repair from a regenerative standpoint.

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2. Clinical Research & Trials

Study 1: Wound Healing — Human Clinical Trials

- **Design:** Multiple randomized controlled trials by Dr. Loren Pickart and colleagues
- **Population:** Chronic wounds, post-surgical wounds, split-thickness skin grafts
- **Results:** GHK-Cu topically or via injection:
- Accelerated wound closure by 30–50% vs. controls
- Significantly improved collagen density and organization (histology)
- Reduced scar formation
- Improved skin graft take rates
- **Citation:** Pickart L, *Human Plasma Factor with Activity to Restore Replicative Viability of Aging Human Fibroblasts*, 1973 (original discovery); multiple follow-up wound healing RCTs.

Study 2: Skin Aging — Wrinkle Reduction

- **Design:** Double-blind RCT, n=67 women aged 50–65
- **Intervention:** GHK-Cu peptide cream vs. placebo for 12 weeks
- **Results:**
- Skin laxity improved (elasticity measurements)
- Wrinkle depth reduced 14% vs. placebo
- Skin density (ultrasound measurement) increased
- Dermal thickness increased by histological assessment
- Skin tightness (subjective + objective) improved significantly
- **Citation:** Multiple studies; Pickart L et al. skin aging series; Leyden JJ et al. cosmetic dermatology data.

Study 3: Hair Growth

- **Design:** Multiple studies — topical and systemic GHK-Cu

- **Results:** GHK-Cu significantly enlarged hair follicle size (follicle anchoring diameter) and stimulated hair shaft elongation. Proposed mechanism: stimulation of dermal papilla cells + anti-DHT inflammatory pathway modulation
- **Comparison to minoxidil:** GHK-Cu comparable or superior in follicle enlargement studies
- **Citation:** Pickart L, Margolina A, *Archives of Dermatological Research*, 2018.

Study 4: Anti-Cancer Gene Expression

- **Key finding:** GHK-Cu modulates expression of multiple tumor suppressor genes (PTEN, TP53 pathway genes) and oncogene suppression. In cancer cell lines, GHK-Cu treatment reduced proliferation and migration.
- **Population-level correlation:** GHK-Cu plasma levels inversely correlate with cancer risk in epidemiological data.
- **Citation:** Pickart L et al., gene expression analyses; Campbell JD et al., lung cancer gene expression database correlations.

Study 5: Lung Damage / COPD — Gene Expression

- **Notable finding:** GHK-Cu reverses gene expression patterns associated with COPD, emphysema, and metastatic lung cancer when applied to damaged lung tissue culture
- **Genes normalized:** Elastin synthesis, anti-inflammatory pathways, DNA repair pathways
- **Clinical trials ongoing** for inhaled GHK-Cu in COPD

Study 6: Collagen Synthesis Quantification

- **In vitro (human fibroblast cultures):**
- GHK-Cu at 1 nM: collagen synthesis increased 100–200%
- GHK-Cu at 10 nM: collagen synthesis increased 300–400%
- TGF-β1 induced collagen (scar-type, disorganized) simultaneously suppressed
- Net: more collagen, better quality, less scarring

Study 7: Neuroregeneration

- **Model:** GHK-Cu administration following nerve injury
- **Results:** Enhanced nerve regeneration, Schwann cell activity, and myelination comparable to BPC-157 in some models; GHK-Cu's SOD-1 enhancement protects neurons from oxidative damage

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3. Primary Uses & Benefits

Goal	Mechanism
Skin anti-aging (wrinkles, laxity)	Collagen/elastin synthesis, MMPs remodeling
Wound healing	Angiogenesis, fibroblast activation, ECM remodeling
Hair growth/thickness	Follicle enlargement, dermal papilla activation
Joint/connective tissue repair	Collagen + glycosaminoglycan synthesis
Anti-inflammatory	TGF-β1 suppression, cytokine modulation
Anti-cancer support	Tumor suppressor gene upregulation
Neuroprotection	SOD-1/antioxidant + nerve regeneration
Longevity/gene reset	4,000+ gene expression normalization

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4. Symptoms & Conditions It Targets

- Skin aging (wrinkles, sagging, poor elasticity)
- Hair thinning / androgenic alopecia
- Chronic wounds / poor wound healing
- Joint damage / arthritis / cartilage degeneration
- Connective tissue injuries
- Scarring / fibrosis
- COPD / pulmonary fibrosis (research stage)
- Post-surgical tissue optimization
- Inflammation-driven tissue damage
- Oxidative stress / mitochondrial aging

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5. Key Selling Points (Rep Talking Points)

- ■ **4,000 genes reset toward youth** — no other single molecule demonstrates this breadth of anti-aging gene expression modulation; GHK-Cu is the closest thing to a "youth restorer" at the molecular level
- ■ **Skin, hair, AND joints in one peptide** — unique tri-system value; clients dealing with multiple signs of aging get comprehensive benefit
- ■ **40+ years of research** — Dr. Pickart's work spans decades with consistent findings; robust evidence base for a peptide still considered cutting-edge
- ■ **Anti-fibrotic + pro-collagen simultaneously** — the sophisticated remodeling action (remove bad collagen, build good collagen) is superior to any topical treatment and unmatched by any single drug
- ■ **Cancer suppressor gene profile** — growing evidence of genuine cancer-protective properties via gene expression; a unique selling point for longevity clients focused on cancer prevention

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6. Common Protocols

SubQ Injection (Systemic Anti-Aging / Joint / Hair)

- **Dose:** 1–2mg SubQ, once daily
- **Cycle:** 8–12 weeks on, 4–6 weeks off
- **Best for:** Systemic anti-aging, joint repair, hair growth from inside

High-Dose Protocol (Significant tissue damage / intensive repair)

- **CU100:** 2–4mg/day SubQ for 4 weeks (loading), then 1–2mg/day maintenance
- **Use case:** Post-surgical, severe skin damage, significant joint degeneration

Topical Protocol (Skin / Hair)

- **Dose:** 0.1–1% solution applied to skin or scalp
- **Frequency:** Daily or twice daily
- **Reconstitute in:** Sterile water; mix with carrier serum/oil for topical application
- **Note:** Topical GHK-Cu is widely used in cosmetics (the active in many "copper peptide serums")

Combined Topical + Injectable

- Systemic SubQ for body/joint/hair + topical application to face/scalp
- Represents comprehensive protocol for appearance + longevity

Reconstitution

- Reconstitute with bacteriostatic water: 1–2 mL per vial
- CU50: 50mg + 5mL bac water = 10mg/mL
- Topical: dilute further in appropriate carrier

Storage

- Lyophilized: refrigerate; stable 24+ months (very stable)
- Reconstituted: refrigerate; use within 30 days; stable in solution with copper complex
- Protect from UV light

Half-Life

- GHK-Cu plasma half-life: not definitively established; tissue effects persist hours to days due to ECM binding

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7. Side Effects & Safety Profile

- **Outstanding safety record across 40+ years of study**
- Topical: occasional mild skin irritation at high concentrations; improves coloring (blue-green from copper complex in some preparations — harmless)
- SubQ: standard injection site reactions; transient mild redness
- No systemic toxicity documented at therapeutic doses
- Copper toxicity: not a concern at therapeutic GHK-Cu doses — the chelated form is tightly regulated; NOT the same as free copper supplementation
- No hormonal disruption
- No hepatotoxicity, nephrotoxicity
- **Pregnancy:** Insufficient data; avoid
- **Interaction:** High-dose zinc supplementation may compete with copper absorption (systemic zinc) — monitor in clients on high zinc

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8. Stacking Notes

Stack Partner	Rationale
BPC-157	The foundational tissue repair stack — BPC for healing/angiogenesis/nerve + GHK-Cu for collagen quality/ECM remodeling/anti-fibrotic
TB-500	Comprehensive repair: TB-500 (actin/vascular/satellite cells) + GHK-Cu (collagen/ECM remodeling) covers all phases of healing
Epithalon	Longevity synergy: Epithalon (telomere/DNA) + GHK-Cu (gene expression normalization/anti-aging) = deep anti-aging stack

Ipamorelin	GH/IGF-1 provides anabolic environment + GHK-Cu builds optimized collagen within that environment
Retatrutide	Fat loss clients: Retatrutide removes fat → GHK-Cu helps collagen remodel the skin and connective tissue after rapid fat loss
KLOW/GLOW	GHK-Cu is a core component of both stacks

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9. Comparison: GHK-Cu vs. Other Skin/Repair Peptides

Feature	GHK-Cu	BPC-157	TB-500	Epithalon
Collagen synthesis	■■ Best-in-class	■ Good	■ Good	Moderate
Anti-fibrotic	■■ Best (TGF-β1↓)	■ Good	Moderate	Moderate
Anti-aging (gene expression)	■■ Remarkable	Moderate	Moderate	■■ Strong
Hair growth	■■ Documented	■	■	■
Angiogenesis	■ Good	■■ Strong	■■ Strongest	Minimal
Nerve regeneration	■ Good	■■ Excellent	■ Good	Moderate
Topical route	■■ Works well	Limited	■	■
Research history	40+ years	25+ years	25+ years	25+ years

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10. FAQ — For Rep Use

Q: Is GHK-Cu safe if it uses copper?

A: Yes. The copper in GHK-Cu is tightly chelated — it's in a biologically controlled complex, not free ionic copper. The body uses this form to deliver Cu2+ to specific enzymes. There is no risk of copper toxicity at therapeutic doses; the amounts are trace quantities.

Q: How does injectable GHK-Cu compare to copper peptide skincare products?

A: Topical products are effective for skin surface but have limited penetration into deeper dermis or systemic distribution. SubQ injection delivers GHK-Cu systemically to joints, scalp, connective tissue, and internal organs — topical can't reach these sites.

Q: Is GHK-Cu only for skin/cosmetics?

A: Not at all — that's a common misconception because it's widely marketed in cosmetics. The injectable research shows profound effects on joints, lungs, neurological tissue, gene expression, and cancer suppressor pathways. It is one of the most multi-system peptides available.

Q: How long before seeing skin results?

A: Most clients notice improved skin texture and hydration within 4–6 weeks. Visible wrinkle reduction and elasticity improvements typically emerge at 8–12 weeks of consistent use.

Q: Can men use it too?

A: Absolutely. GHK-Cu is not sex-specific. Men particularly benefit from the joint/connective tissue, hair growth, and systemic anti-aging applications.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

EPITHALON (Epitalon) — Mavyn Rep Training Sheet

SKUs: ET10 (10mg) | ET50 (50mg) | Category: Anti-Aging / Telomere / Pineal / Longevity

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1. What Is Epithalon?

Epithalon (also spelled Epitalon) is a synthetic tetrapeptide (Ala-Glu-Asp-Gly) derived from Epithalamin — a polypeptide isolated from the bovine pineal gland. It was developed and extensively studied over 35 years by Professor Vladimir Khavinson and his team at the St. Petersburg Institute of Bioregulation and Gerontology, one of Russia's foremost longevity research institutions.

Epithalon is best known as the first compound proven to **elongate telomeres in human cells** — a finding published in peer-reviewed journals that catapulted it to the center of longevity research. It is also a pineal gland regulator, stimulating melatonin production and restoring circadian rhythm function that declines with aging.

Mechanism of Action (Molecular/Cellular Level)

Telomerase Activation — The Core Mechanism:

Telomeres are the protective caps at the ends of chromosomes. With each cell division, they shorten. When telomeres become critically short, cells enter replicative senescence or apoptosis. Telomere length is one of the most validated biomarkers of biological aging.

Epithalon activates **telomerase (hTERT — human Telomerase Reverse Transcriptase)** — the enzyme that rebuilds telomeres. This is:

- Normally active only in embryonic cells, stem cells, and cancer cells
- In somatic cells: essentially dormant
- Epithalon reactivates telomerase in somatic human cells, measurably extending telomere length

Published finding (Khavinson VKh et al., 2003): Epithalon increased telomere length in human fetal fibroblasts in culture. Cells that had undergone 80+ divisions (near Hayflick limit) showed telomere elongation and restored replicative capacity after Epithalon treatment.

Pineal Gland / Melatonin Stimulation:

The pineal gland produces melatonin — the master circadian hormone and a potent antioxidant. Pineal function declines dramatically with age (calcification, reduced melatonin output). Epithalon:

- Stimulates pinealocyte activity directly
- Increases melatonin synthesis and secretion
- Restores circadian melatonin rhythms toward youthful patterns
- Improves sleep architecture (especially deep/slow-wave sleep)
- Melatonin's antioxidant effects complement Epithalon's direct telomere protection

DNA Repair Enhancement:

Epithalon activates chromatin remodeling and DNA repair gene expression, particularly for double-strand break repair (DSBR) — a critical defense against mutagenesis. Aging cells show reduced DSBR capacity; Epithalon partially restores this.

Tumor Suppressor Activation:

Epithalon modulates expression of tumor suppressor genes (p53, p16, p21) — increasing their expression to appropriate baseline levels without causing excessive growth arrest. It also normalizes oncogene expression.

Anti-Oxidant Gene Expression:

Upregulates antioxidant defense genes: Nrf2 pathway targets, SOD, catalase, glutathione peroxidase. Reduces oxidative DNA damage markers (8-OH-dG) in aging animals.

Neuroendocrine Axis Restoration:

As a peptide derived from the pineal gland, Epithalon has broad regulatory effects on the hypothalamic-pituitary axis:

- Normalizes FSH, LH, cortisol, insulin, and thyroid hormone rhythms in aging animals
- Restores feedback sensitivity in the HPA and HPG axes

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2. Clinical Research & Trials

Study 1: Telomere Elongation — Key Publication

- **Citation:** Khavinson VKh, Bondarev IE, Butyugov AA, *Bulletin of Experimental Biology and Medicine*, 2003, Vol 135:590-592.
- **Design:** In vitro human fetal fibroblasts, standard aging model
- **Results:** Epithalon treatment produced statistically significant telomere elongation. Cells maintained proliferative capacity beyond expected Hayflick limit. Telomerase activity confirmed as mechanism.
- **Significance:** First published evidence of telomere elongation in human cells by any non-cancer agent.

Study 2: Lifespan Extension — Animal Studies

- **Model:** Multiple rat and mouse cohorts (Khavinson lab, 1990s–2010s)
- **Results:** Consistent 15–25% increase in maximum lifespan across multiple animal cohort studies. Significant reduction in spontaneous tumor development (carcinomas, sarcomas). Improved immune function in aged animals.
- **Citation:** Multiple publications by Khavinson VKh et al., *Gerontology* and *Biogerontology* journals.

Study 3: Cancer Incidence Reduction — Human Longitudinal Study

- **Design:** Long-term prospective observational study, n=266 patients in geriatric care
- **Intervention:** Annual courses of Epithalon for 3 years vs. untreated control group
- **Follow-up:** 12 years
- **Results:** Epithalon group showed significantly reduced all-cause mortality. Lung cancer incidence reduced. Colon polyp development reduced. Cardiovascular events reduced. Immune function better preserved.
- **Citation:** Khavinson VKh et al., *Biogerontology*, 2002–2010 series.

Study 4: Sleep & Melatonin Restoration

- **Design:** Clinical study, n=40 elderly patients (avg age 75) with sleep dysfunction
- **Dosing:** Epithalon 10mg/day IM for 10 days
- **Results:** Melatonin production increased to near-normal levels (confirmed by urinary 6-sulfatoxymelatonin). Sleep quality improved significantly (Pittsburgh Sleep Quality Index). Circadian rhythm phase restored.
- **Citation:** Khavinson VKh et al., *Advances in Gerontology* (Russian), 2004.

Study 5: Retinal Function Preservation

- **Condition:** Age-related retinal degeneration in elderly subjects (n=61)

- **Results:** Visual acuity and electroretinogram function improved or stabilized vs. control. Photoreceptor density preservation on histology in animal model. Mechanism: antioxidant protection + cellular regeneration in retina.
- **Citation:** Multiple Khavinson publications on retinal effects of Epithalon.

Study 6: Reproductive Function Restoration

- **Model:** Old female rats (estrous cycle cessation)
- **Results:** Epithalon restored estrous cycles in 27 of 33 aged rats. Ovarian follicle development resumed. LH/FSH ratio normalized. Proposed mechanism: hypothalamic-pituitary axis re-synchronization via melatonin/pineal restoration.

Study 7: COVID-19 Severity Reduction (Emerging Data)

- **Observation:** Cohort studies suggesting that patients with higher melatonin levels (older patients on Epithalon protocols) had better COVID outcomes — consistent with melatonin's known anti-inflammatory and ACE2 modulation properties.

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3. Primary Uses & Benefits

Goal	Mechanism
Longevity / anti-aging	Telomerase activation, telomere elongation
Sleep restoration	Pineal stimulation, melatonin normalization
Cancer prevention	Tumor suppressor upregulation, DNA repair
Circadian rhythm regulation	Melatonin restoration, SCN normalization
Hormonal balance	HPA/HPG axis re-synchronization
Eye health	Retinal cell protection
Immune enhancement	Thymic support, NK cell function
Neurological health	Pineal-brain axis, neuroprotection

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4. Symptoms & Conditions It Targets

- Age-related biological aging (primary indication)
- Insomnia / circadian disruption
- Age-related hormonal decline
- High cancer risk / cancer prevention
- Low melatonin / pineal dysfunction
- Retinal degeneration / visual decline
- Age-related immune dysfunction
- Post-menopausal HPA dysregulation
- Telomere shortening (measurable)
- General healthspan optimization

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5. Key Selling Points (Rep Talking Points)

- ■ **The only peptide proven to elongate telomeres in human cells** — this is the most coveted property in longevity science; Epithalon can make that claim with published peer-reviewed data
- ■ **35 years of research from a premier longevity institute** — Professor Khavinson's body of work is among the most extensive in geropeptide science; not fringe research
- ■ **15–25% lifespan extension in animals** — consistently reproduced across multiple cohorts; one of the strongest animal longevity effects of any compound
- ■ **Cancer and longevity simultaneously** — reduces cancer incidence in human long-term study data WHILE extending lifespan; extremely rare dual effect
- ■ **Sleep restoration is immediate** — clients notice sleep quality improvement within the first 10-day course; a fast "proof of concept" that builds trust for the long-term protocol

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6. Common Protocols

Standard Anti-Aging Course

- **Dose:** 5–10mg/day SubQ or IM
- **Course length:** 10–20 days
- **Frequency:** 2–3 courses per year (every 4–6 months)
- **Most common:** 10mg/day × 10 days = 100mg per course

Extended Longevity Protocol

- **ET50 vials:** 50mg total per course (5mg/day × 10 days, or 10mg/day × 5 days)
- **Frequency:** 3–4 courses/year for intensive anti-aging

Maintenance / Annual Protocol

- **Dose:** 10mg/day for 10 days, once annually
- **For:** Clients who want protective longevity support without intensive cycling

Reconstitution

- Reconstitute with bacteriostatic water: 1–2 mL per vial
- 10mg + 1mL = 10mg/mL; 50mg + 5mL = 10mg/mL
- Dose per injection: 0.5–1 mL

Storage

- Lyophilized: refrigerate; stable 18–24 months
- Reconstituted: refrigerate; use within 30 days; protect from light

Half-Life

- Plasma half-life not definitively established for tetrapeptide
- Biological effects accumulate over the course and persist months after treatment

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7. Side Effects & Safety Profile

- **Excellent safety record across 35+ years** of institutional research
- No significant adverse events reported in published clinical studies
- Mild injection site reactions (standard)
- No hepatotoxicity, nephrotoxicity
- No hormonal disruption (normalizes rather than disrupts hormone levels)
- **Theoretical:** Telomerase activation in cancer cells — theoretical concern for patients with active malignancy; contraindicated
- **Not for active cancer patients**
- **Pregnancy:** Insufficient data; avoid

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8. Stacking Notes

Stack Partner	Rationale
NAD+	DNA repair fuel (NAD+/PARP) + DNA repair activation (Epithalon telomerase) = comprehensive genomic longevity
GHK-Cu	Gene expression normalization (GHK-Cu) + telomere elongation (Epithalon) = most complete anti-aging peptide stack available
Selank	Sleep-focused stack: Selank for anxiety/BDNF + Epithalon for deep sleep/melatonin
MOTS-c	Metabolic longevity + genomic longevity = comprehensive aging intervention
Semax	Neuroprotection + telomere health; cognitive anti-aging stack
VIP	Circadian regulation from two angles: VIP for SCN synchronization + Epithalon for melatonin

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9. Epithalon vs. Other Longevity Peptides

Feature	Epithalon	GHK-Cu	MOTS-c	Thymalin
Telomere elongation	■■ Proven	Indirect	■	Indirect
Sleep/melatonin	■■ Best	■	■	■
Cancer prevention (data)	■■ Human study	■ Gene expression	Animal	■ Human
Gene expression reset	■ Good	■■ Remarkable	Metabolic	Immune
Lifespan extension (animal)	■■ 15–25%	Moderate	■ Good	■ Good
Immune support	Moderate	Moderate	Moderate	■■ Best
Research base	■■ Extensive	■■ Extensive	■ Good	■■ Extensive

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10. FAQ — For Rep Use

Q: Does activating telomerase cause cancer?

A: This is the most common concern. Cancer cells do use telomerase to become "immortal," but Epithalon's telomere activation is in a different context — it normalizes rather than aberrantly activates telomerase. Importantly, 35+ years of research shows REDUCED cancer incidence with Epithalon use, not increased. The tumor suppressor gene upregulation counterbalances any proliferative concern. However, active cancer is still a contraindication as a precaution.

Q: How quickly will sleep improve?

A: Most clients report noticeably improved sleep depth and morning freshness within the first 3–5 days of a 10-day course. Melatonin levels begin normalizing quickly. This is often the first noticeable benefit.

Q: How often should it be used?

A: Standard protocol is 2–3 courses per year, each lasting 10–20 days. Annual use is the minimum for sustained telomere/longevity benefit. Some practitioners favor 3–4 courses annually for clients with significant aging markers.

Q: Is there any human data on longevity — not just animals?

A: Yes — the 12-year prospective study (n=266) showed significant reduction in all-cause mortality, cancer incidence, and cardiovascular events in Epithalon-treated vs. untreated control groups. This is remarkable human longitudinal data for a peptide.

Q: Can it be taken with melatonin supplements?

A: Yes — they're complementary. Epithalon stimulates endogenous melatonin production from the pineal; exogenous melatonin adds to this. Together they reinforce circadian restoration. Many longevity protocols combine both.

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MOTS-c — Mavyn Rep Training Sheet

SKUs: MS10 (10mg) | MS15 (15mg) | Category: Mitochondrial-Derived Peptide / Metabolic / Anti-Aging

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1. What Is MOTS-c?

MOTS-c (Mitochondrial Open Reading Frame of the 12S rRNA-c) is a mitochondria-derived peptide (MDP) — a revolutionary class of signaling molecules encoded not by nuclear DNA but by mitochondrial DNA. It was discovered in 2015 by Dr. Pinchas Cohen and his team at the USC Leonard Davis School of Gerontology.

MOTS-c is a 16-amino acid peptide: MRWQEMGYFY RKMSMGRSGG (single-letter code). It is encoded by the 12S ribosomal RNA gene in the mitochondrial genome — a highly conserved region, suggesting fundamental evolutionary importance.

MOTS-c circulates in the bloodstream and acts as a "mitochondrial hormone" — communicating mitochondrial status to the nucleus and to systemic tissues. It is considered one of the most important anti-aging molecular discoveries of the last decade.

Mechanism of Action (Molecular/Cellular Level)

AMPK Activation (Central Mechanism):

MOTS-c is a potent activator of AMP-activated protein kinase (AMPK) — the master metabolic regulator often called the "exercise enzyme" or "longevity switch." AMPK activation:

- Increases glucose uptake and insulin sensitivity in muscle
- Promotes fatty acid oxidation (fat burning)
- Inhibits anabolic pathways that drive cellular senescence
- Activates autophagy (cellular self-cleaning)
- Inhibits mTORC1 (the aging/growth driver that must be modulated for longevity)

Folate-Methionine Cycle Inhibition → AMPK:

The mechanistic pathway: MOTS-c inhibits AICAR transformylase (an enzyme in purine synthesis that uses methylenetetrahydrofolate). This leads to accumulation of AICAR (5-Aminoimidazole-4-carboxamide ribonucleotide) — a direct AMPK activator. This is the unique molecular pathway linking mitochondrial status to nuclear AMPK signaling.

Nuclear Translocation & Gene Expression:

Under cellular stress (exercise, hypoxia, nutrient restriction), MOTS-c translocates from mitochondria to the nucleus, where it directly modulates gene expression — particularly stress response genes (ATF1, NRF2) and metabolic genes. This "mitochondria-to-nucleus retrograde signaling" is a newly discovered mechanism of cellular communication.

Insulin Sensitization:

MOTS-c dramatically improves insulin sensitivity in skeletal muscle — the primary site of glucose disposal. Mechanism: AMPK → GLUT4 translocation to cell membrane → enhanced glucose uptake without insulin signaling. This means it can improve metabolic function even in insulin-resistant states.

Anti-Inflammatory Action:

MOTS-c suppresses NF-κB-mediated inflammatory gene expression. In aging-related chronic inflammation ("inflammaging"), MOTS-c reduces circulating IL-6, TNF-α, and other SASP (senescence-associated secretory phenotype) factors.

Exercise Mimetic:

MOTS-c produces transcriptional profiles similar to aerobic exercise at the molecular level. It activates many of the same AMPK-dependent pathways as endurance exercise — earning it the description of an "exercise mimetic."

Age-Related Decline:

Critically, MOTS-c levels decline with age — and MOTS-c concentrations correlate with human longevity. In a landmark study, centenarians (100+ year lifespan) had significantly higher circulating MOTS-c levels than younger control groups.

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2. Clinical Research & Trials

Study 1: Discovery & Metabolic Effects — Key Paper

- **Citation:** Lee C, Zeng J, Drew BG et al., *Cell Metabolism*, 2015. Vol 21:443-454.
- **Findings:** MOTS-c injected into obese mice on high-fat diet:
- Prevented obesity development
- Improved insulin sensitivity by >50% vs. controls
- Normalized blood glucose to near-lean-mouse levels
- Reduced fat mass significantly without reducing food intake (metabolic activation, not appetite suppression)
- AMPK and AICAR pathway confirmed as mechanism

Study 2: Exercise Performance — Aging Model

- **Citation:** Reynolds JC et al., *Nature Aging*, 2021.
- **Design:** MOTS-c administration in old mice (equivalent to ~70 year old humans)
- **Results:** Aged mice administered MOTS-c demonstrated exercise capacity equivalent to young controls. Muscle mitochondrial function normalized. Inflammatory markers reduced. Strength and endurance significantly improved.
- **Significance:** This is one of the first demonstrations of actual reversal (not just slowing) of age-related functional decline in exercise capacity.

Study 3: Longevity Correlation — Human Data

- **Citation:** Zempo H et al., *Journal of Clinical Endocrinology & Metabolism*, 2021.
- **Population:** 287 Japanese participants aged 16–98; plus centenarian cohort
- **Findings:** Circulating MOTS-c levels significantly correlated with longevity. Centenarians had MOTS-c levels equivalent to much younger individuals. MOTS-c levels declined with age across the cohort, then reversed or stabilized in super-agers (90+).
- **Implication:** MOTS-c is a potential biomarker of healthy aging AND a therapeutic target for extending healthspan.

Study 4: Osteoporosis / Bone Metabolism

- **Citation:** Ming W et al., *Cell Metabolism*, 2016.
- **Findings:** MOTS-c directly promotes osteoblast differentiation and bone mineral density in aging mice. Ovariectomy-induced osteoporosis (menopause model) partially prevented by MOTS-c supplementation. Mechanism: AMPK-mediated regulation of osteoblast metabolism.

Study 5: Alzheimer's Disease Model

- **Animal model:** 5xFAD transgenic mice
- **Results:** MOTS-c reduced amyloid-β plaque burden, improved spatial memory, and reduced neuroinflammation markers. Mechanism: mitochondrial function restoration in neurons + NF-κB suppression.

Study 6: MOTS-c vs. Exercise in Obese Subjects

- **Design:** Comparison of transcriptional profiles (human muscle biopsy) in MOTS-c-treated vs. exercise-trained obese subjects
- **Findings:** Significant overlap in gene expression profiles — particularly oxidative phosphorylation genes, AMPK targets, and mitochondrial biogenesis genes. MOTS-c recapitulates a significant portion of the molecular exercise signature.

Human Clinical Trials

- Phase 1/2 trials being conducted by researchers at USC and affiliated institutions
- As of 2024–2025: early clinical results indicate safety and dose-dependent AMPK activation in human subjects
- Full trial publications pending

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3. Primary Uses & Benefits

Goal	Mechanism
Metabolic optimization	AMPK activation, insulin sensitization
Anti-aging / longevity	Mitochondrial restoration, inflammaging reduction
Fat loss / body composition	Fatty acid oxidation, AMPK-mediated fat burning
Exercise performance	Mitochondrial biogenesis, energy production
Muscle preservation (aging)	AMPK, mitochondrial function in muscle
Bone health	Osteoblast differentiation, bone density
Cognitive health	Neuronal mitochondrial support, amyloid-β reduction
Immune regulation	NF-κB suppression, SASP reduction

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4. Symptoms & Conditions It Targets

- Insulin resistance / type 2 diabetes / prediabetes
- Obesity / metabolic syndrome
- Mitochondrial dysfunction (fatigue, poor exercise tolerance)
- Age-related metabolic decline
- Sarcopenia (age-related muscle loss)
- Osteoporosis (especially menopause-related)
- Cognitive decline / early neurodegeneration
- Chronic low-grade inflammation
- Poor post-exercise recovery
- Desire to mimic exercise benefits (sedentary clients, mobility-limited)

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5. Key Selling Points (Rep Talking Points)

- ■ **Encoded in your own mitochondrial DNA** — not a foreign compound; MOTS-c is your body's own longevity signal, just replenished as it declines with age
- ■ **Exercise in a peptide** — activates the same molecular pathways as aerobic exercise; ideal for clients who can't exercise adequately (injury, age, illness)
- ■ **Longevity biomarker in humans** — centenarians have high MOTS-c; this is one of the strongest longevity correlates discovered in human data
- ■ **AMPK = the master switch** — the same pathway activated by metformin, rapamycin, and caloric restriction — the three best-validated longevity interventions; MOTS-c activates it physiologically
- ■ **Bone + metabolic + cognitive in one** — rare multi-system anti-aging action from a single peptide; particularly valuable for menopausal women (metabolic + bone + brain triple benefit)

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6. Common Protocols

Anti-Aging / Metabolic Protocol

- **Dose:** 5–10mg SubQ, 3–5x/week
- **Cycle:** 8–12 weeks on, 4–6 weeks off
- **Goal:** Metabolic optimization, longevity, inflammation reduction

Athletic Performance Protocol

- **Dose:** 10–15mg SubQ, 3–5x/week (or pre-training on training days)
- **Cycle:** 8-week cycles aligned with training blocks
- **Goal:** Exercise capacity, recovery, body composition

High-Dose Anti-Aging Protocol

- **MS15:** 15mg SubQ, 3–4x/week
- **Cycle:** 12 weeks, 6 weeks off
- **For:** Significant metabolic dysfunction, post-menopausal women, clients with high inflammatory burden

Reconstitution

- Reconstitute with bacteriostatic water: 1–2 mL per vial
- 10mg + 1mL = 10mg/mL; 15mg + 1.5mL = 10mg/mL
- SubQ injection, abdomen or thigh

Storage

- Lyophilized: refrigerate; stable 12–18 months
- Reconstituted: refrigerate; use within 30 days

Half-Life

- ~2 hours plasma half-life; AMPK activation persists 12–24 hours; transcriptional effects longer

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7. Side Effects & Safety Profile

- **Excellent tolerability in animal studies and early human trials**
- No significant adverse events reported in published research
- Mild injection site reactions (standard)
- No hepatotoxicity, nephrotoxicity documented
- No hormonal disruption at standard doses
- Potential for hypoglycemia if stacked with metformin or insulin (additive insulin sensitization) — monitor blood glucose in diabetic clients
- **Pregnancy:** Insufficient data; avoid

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8. Stacking Notes

Stack Partner	Rationale
NAD+	AMPK (MOTS-c) + sirtuin fuel (NAD+) = complete longevity metabolic stack; among the most powerful anti-aging combinations
Retatrutide	MOTS-c's AMPK activation + Retatrutide's triple incretin = comprehensive metabolic synergy; insulin sensitivity amplified
Epithalon	DNA longevity (Epithalon telomere) + metabolic longevity (MOTS-c AMPK); comprehensive anti-aging program
Ipamorelin / HGH	GH/IGF-1 anabolism + MOTS-c AMPK catabolism balance = optimal body composition
BPC-157	Recovery + metabolic optimization combined
Semax	Brain energy (Semax BDNF) + neuronal energy supply (MOTS-c mitochondrial)

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9. MOTS-c vs. Other Metabolic Peptides

Feature	MOTS-c	Humanin	SS-31
Origin	Mitochondrial DNA	Mitochondrial DNA	Synthetic
Primary mechanism	AMPK activation	IGF-1R, STAT3	Inner mitochondrial membrane
Metabolic effect	■■ Strong	Moderate	Moderate
Neuroprotection	■ Good	■■ Strong	Moderate
Longevity data	■■ Centenarian correlation	■ Good	Animal only
Clinical trials	Early phase	Early phase	Early phase

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10. FAQ — For Rep Use

Q: How is MOTS-c different from other peptides?

A: It's the only peptide in this portfolio encoded by mitochondrial DNA — not nuclear DNA. It represents a fundamentally different type of signaling: mitochondria communicating their status to the rest of the body. It's not treating a symptom; it's restoring a fundamental aging signal.

Q: Is it basically the same as taking metformin?

A: They share AMPK activation, but MOTS-c works through a different upstream pathway (mitochondrial-to-nuclear signaling vs. complex I inhibition). MOTS-c also has effects outside of AMPK (nuclear gene regulation, anti-inflammatory signaling) that metformin lacks. They may actually be synergistic in some longevity protocols.

Q: Who benefits most?

A: Anyone 40+ looking for metabolic anti-aging. Particularly impactful for: insulin-resistant individuals, menopausal women (metabolic + bone + cognition), athletes wanting metabolic edge, and clients with fatigue/poor exercise tolerance.

Q: How long before measurable results?

A: Metabolic markers (fasting glucose, insulin, HbA1c) may improve within 4–8 weeks. Exercise tolerance improvements often noticed within 2–4 weeks. Anti-aging/longevity effects are cumulative over months.

Q: Is there human clinical trial data?

A: Yes — early phase clinical trials are ongoing at USC and affiliated institutions. Published animal data is extensive, and the human longevity correlation study (centenarian data) is compelling. Full Phase 2/3 clinical data is anticipated within 2–3 years.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

MELANOTAN I (MT-I / Afamelanotide) — Mavyn Rep Training Sheet

SKU: MT1 | Strength: 10mg | Category: Melanocortin Agonist / Photoprotective

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1. What Is Melanotan I?

Melanotan I (MT-I), also known as afamelanotide, is a synthetic analogue of α-melanocyte-stimulating hormone (α-MSH). Its chemical structure is: [Nle4, D-Phe7]-α-MSH. The modifications (norleucine at position 4, D-phenylalanine at position 7) make it more potent and significantly longer-acting than endogenous α-MSH, while maintaining high selectivity for MC1R.

Afamelanotide is approved as a pharmaceutical drug (brand name: **Scenesse**) by the European Medicines Agency (EMA) and the U.S. FDA for prevention of phototoxic reactions in patients with **Erythropoietic Protoporphyria (EPP)** — making it one of the few peptides in this guide with full regulatory approval.

Mechanism of Action (Molecular/Cellular Level)

Melanocortin 1 Receptor (MC1R) — Primary Target:

MC1R is expressed predominantly on melanocytes (skin pigment cells). MT-I binds and activates MC1R with ~10x greater potency than native α-MSH due to its structural modifications.

Melanin Production Pathway:

MC1R activation → cAMP via adenylyl cyclase → PKA → MITF (Microphthalmia-Associated Transcription Factor) upregulation → transcription of melanogenesis enzymes:

- **Tyrosinase** (rate-limiting enzyme)
- **TRP-1 and TRP-2** (downstream melanin synthesis)
- → **Eumelanin production** (brown/black pigment — the UV-protective form)

Unlike natural tanning (which requires UV damage to trigger melanogenesis), MT-I activates the pathway directly — pigmentation develops WITHOUT prior UV exposure.

UV Protection Mechanism:

Eumelanin produced by MT-I activation:

- Absorbs UV-A, UV-B, and UV-C radiation
- Acts as a radical scavenger (antioxidant), neutralizing reactive oxygen species generated by UV
- Reduces UV-induced DNA damage (cyclobutane pyrimidine dimers — CPDs)
- Reduces inflammatory cytokine release from UV-exposed keratinocytes (IL-1β, TNF-α)

DNA Repair Enhancement:

Beyond pigmentation, MC1R signaling directly enhances nucleotide excision repair (NER) — the primary mechanism for repairing UV-induced DNA damage. MC1R → p53 pathway modulation → enhanced repair enzyme recruitment at DNA damage sites.

Anti-Inflammatory Effects:

MT-I (via MC1R and other melanocortin receptors) inhibits NF-κB, suppresses inflammatory cytokines in skin and systemically, and modulates mast cell activation in the skin.

Selectivity Profile (key difference from Melanotan II):

- MT-I is highly selective for MC1R
- Melanotan II is non-selective: activates MC1R, MC3R, MC4R, MC5R — hence the sexual arousal and appetite effects of MT-II
- MT-I does NOT significantly activate MC4R → no sexual side effects, no significant appetite suppression

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2. Clinical Research & Trials

Study 1: FDA/EMA Approved Indication — Erythropoietic Protoporphyria (EPP)

- **Drug:** Scenesse (afamelanotide 16mg implant)
- **Design:** Multiple Phase 2 and Phase 3 RCTs across European and US centers
- **Population:** EPP patients (genetic disorder: severe phototoxic pain with any sun exposure)
- **Dosing:** 16mg biodegradable implant subcutaneous every 2 months
- **Results (Phase 3, n=93):** Treated patients spent significantly more time in direct sunlight without pain — median 69 minutes vs. 40 minutes in placebo. Quality of life scores dramatically improved. Reduced phototoxic reactions.
- **Citation:** Langendonk JG et al., *New England Journal of Medicine*, 2015. Vol 373:48-59.

Study 2: Polymorphous Light Eruption (PLE)

- **Design:** Double-blind RCT, n=48 PLE patients
- **Dosing:** MT-I (afamelanotide) 20mg SubQ implant before spring sun exposure
- **Results:** Only 12.5% of treated patients had PLE reaction vs. 58.3% placebo. Significant protective effect.
- **Citation:** Biolcati G et al., *British Journal of Dermatology*, 2015.

Study 3: Skin Cancer Prevention — Solar Keratosis

- **Design:** Phase 2, n=55 patients with multiple actinic keratoses
- **Dosing:** MT-I 20mg SubQ every 2 months
- **Results:** 76% reduction in new actinic keratosis development compared to placebo. No increase in melanoma risk. Quality and texture of skin improved.
- **Citation:** Clinuvel Pharmaceuticals Phase 2 data, 2019.

Study 4: Vitiligo Treatment

- **Design:** Open-label pilot + narrow-band UVB combination, n=55 patients
- **Protocol:** MT-I + NB-UVB phototherapy
- **Results:** 73% of treated patients achieved significant repigmentation (>50% area repigmented) vs. 33% with NB-UVB alone. MT-I accelerates melanocyte migration into depigmented areas.
- **Citation:** Lim HW et al., Clinuvel Pharmaceuticals data.

Study 5: Cosmetic Tanning — Dose-Finding Studies

- **Design:** Phase 1/2, healthy volunteers (n=28)
- **Dosing:** MT-I 0.08–0.16 mg/kg SubQ
- **Results:** Consistent melanin density increase (colorimetric measurement) after 1–2 weeks. Dose-dependent tanning without UV. No nausea, vomiting, or sexual side effects (MC4R-selective advantage).
- **Citation:** Hadley ME et al., *Pigment Cell Research*, 1996; multiple dose-finding publications.

Study 6: Anti-Inflammatory Skin Conditions

- **Conditions studied:** Atopic dermatitis, psoriasis (pilot data)
- **Findings:** MC1R activation reduced skin inflammation markers, mast cell density in lesional skin, and pruritus scores. Mechanism: NF-κB suppression in keratinocytes and mast cells.

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3. Primary Uses & Benefits

Goal	Mechanism
Sunless/accelerated tanning	MC1R → eumelanin production
Sun protection	Eumelanin UV absorption + antioxidant
EPP treatment (FDA approved)	Photoprotection via melanin shield
Skin cancer risk reduction	DNA repair enhancement + UV damage reduction
Vitiligo treatment	Melanocyte activation in depigmented areas
Anti-aging skin	UV damage prevention + anti-inflammatory
Light sensitivity disorders	PLE, solar urticaria protection

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4. Symptoms & Conditions It Targets

- Fair skin / sun-sensitive skin types (Fitzpatrick I–III)
- Erythropoietic protoporphyria
- Polymorphous light eruption
- Solar urticaria
- Vitiligo (adjunct with phototherapy)
- Actinic keratosis (precancerous lesions)
- High UV exposure occupations/lifestyles
- Desire for natural-looking tan without UV damage

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5. Key Selling Points (Rep Talking Points)

- ■ **FDA and EMA approved** — Scenesse is an approved drug for EPP; MT-I is the same active molecule, giving it the strongest regulatory validation of any tanning peptide
- ■ **Safe tanning without sun damage** — produces natural eumelanin without requiring UV radiation; the tan is real and protective
- ■ **No sexual side effects** — MC1R selective vs. Melanotan II's non-selective melanocortin activation; clients who don't want libido/erection effects should use MT-I
- ■ **Anti-aging skin protection** — UV is the #1 driver of skin aging; MT-I's melanin shield and DNA repair enhancement are genuinely preventive
- ■ **Vitiligo and skin condition applications** — growing clinical evidence for repigmentation in vitiligo; expands client population beyond cosmetic tanning

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6. Common Protocols

Beginner Tanning Protocol

- **Dose:** 0.5–1mg SubQ, every other day for 2 weeks (loading)
- **Maintenance:** 0.5–1mg, 2–3x/week
- **Results expected:** Visible pigmentation in 2–4 weeks
- **Enhancement:** Minimal UV exposure (10–15 min sun) during loading accelerates results but is NOT required

Intermediate Protocol

- **Dose:** 1–2mg SubQ every other day
- **Cycle:** 2–3 week loading, then 1–2x/week maintenance to hold tan
- **Goal:** Deeper, faster tan

Therapeutic Protocol (EPP, PLE, Vitiligo)

- **Reference dose:** 16–20mg implant every 2 months (per Clinuvel/Scenesse protocol)
- OR: SubQ injections of 1–2mg 3x/week for 4–6 weeks
- Vitiligo: combine with NB-UVB phototherapy for best results

Reconstitution

- Reconstitute with bacteriostatic water: 1–2 mL per 10mg vial
- 10mg + 1mL = 10mg/mL; 10mg + 2mL = 5mg/mL
- Typical dose volume: 0.1–0.4 mL

Storage

- Lyophilized: refrigerate; stable 12–24 months
- Reconstituted: refrigerate; use within 30 days; protect from light

Half-Life

- MT-I plasma half-life: ~1.5–2 hours
- Melanogenic effect persists 2–4 weeks after dosing ceases (melanin already deposited)

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7. Side Effects & Safety Profile

- **Generally excellent tolerability** (confirmed in FDA approval process)
- Mild injection site reaction (redness, swelling — transient)
- Nausea: occasionally mild, dose-dependent; uncommon at doses <2mg
- Flushing, facial warmth: reported occasionally
- **No libido/sexual effects** (unlike MT-II) — MC1R selective
- No appetite suppression (no MC4R activation)
- **Mole/nevus monitoring:** Melanocortin activation may cause existing moles to darken — pre-existing atypical nevi should be monitored; not a risk for new melanoma formation at standard doses

- **Not recommended:** Personal or family history of melanoma (theoretical caution, though MT-I actually enhances DNA repair which is anti-carcinogenic)
- **Not for use in:** Pregnancy; active skin malignancy

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8. Stacking Notes

Stack Partner	Rationale
Melanotan II	Not typically stacked — they target same receptors; choose one based on whether sexual/appetite effects are desired
GHK-Cu	Skin health synergy: MT-I drives pigmentation and UV protection; GHK-Cu drives collagen synthesis and elasticity — comprehensive skin anti-aging
Epithalon	Anti-aging synergy: Epithalon for DNA/telomere protection; MT-I for UV DNA damage prevention
BPC-157	If treating skin conditions with inflammatory component

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9. Melanotan I vs. Melanotan II — Key Comparison

Feature	Melanotan I (MT-I)	Melanotan II (MT-II)
Receptor selectivity	MC1R (selective)	MC1R, MC3R, MC4R, MC5R
Tanning potency	High (therapeutic)	Very high (cosmetic)
Sexual arousal effect	■ None	■ Yes (MC4R)
Appetite suppression	■ Minimal	■ Yes
Nausea	Rare	More common
FDA approved	■ Yes (Scenesse)	■ No
Best for	Medical/safe tanning	Cosmetic tanning + libido
Half-life	~1.5–2 hrs	~1–2 hrs

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10. FAQ — For Rep Use

Q: Is Melanotan I the same as Melanotan II?

A: Related but different. Both are synthetic α-MSH analogues, but MT-I is MC1R-selective (tanning without sexual effects) while MT-II activates multiple melanocortin receptors, causing both tanning and libido/erection effects. MT-I is also the one with FDA approval.

Q: Will it cause melanoma?

A: There is no evidence of increased melanoma risk, and strong mechanistic evidence that MT-I reduces UV-induced DNA damage — a key cause of melanoma. However, clients with a history of melanoma should consult their oncologist before use.

Q: How long does the tan last?

A: The melanin deposited during treatment persists for several weeks after stopping. With maintenance dosing (1–2x/week), the tan can be maintained indefinitely.

Q: Does it require sun exposure to work?

A: No — MT-I activates melanogenesis without UV stimulus. However, a small amount of UV exposure can accelerate the depth and evenness of the tan. The protective effect of the melanin formed is equivalent whether developed with or without UV.

Q: Who is the ideal client?

A: Fair-skinned clients (Fitzpatrick I–III) seeking safe tanning without UV damage, anyone with light-sensitive skin conditions, vitiligo patients, and clients interested in skin cancer prevention.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

MELANOTAN II (MT-II) — Mavyn Rep Training Sheet

SKU: ML10 | Strength: 10mg | Category: Melanocortin Agonist / Tanning / Sexual Function

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1. What Is Melanotan II?

Melanotan II is a synthetic cyclic heptapeptide analogue of α-melanocyte-stimulating hormone (α-MSH): cyclo-[Nle4, D-Phe7]-α-MSH(4-10). It was developed at the University of Arizona under the direction of Dr. Victor Hruby and Dr. Mac Hadley, originally with the intention of creating a "sunless tanning agent" to reduce melanoma risk in fair-skinned individuals.

Unlike Melanotan I (afamelanotide), MT-II is a **non-selective melanocortin receptor agonist** — it activates MC1R, MC3R, MC4R, and MC5R, producing a broad spectrum of effects beyond skin pigmentation. The MC4R activation is responsible for its significant effects on sexual arousal and erectile function, and the MC3R/MC5R activation contributes to appetite modulation and metabolic effects.

Mechanism of Action (Molecular/Cellular Level)

MC1R Activation — Tanning:

- Same pathway as MT-I: MC1R → cAMP → PKA → MITF → tyrosinase → eumelanin
- Activates melanocytes to produce dark, UV-protective eumelanin without UV exposure
- More potent than MT-I for tanning (lower doses achieve equivalent pigmentation)

MC4R Activation — Sexual Function:

MC4R is expressed in:

- Hypothalamus (paraventricular nucleus, arcuate nucleus)
- Spinal cord (lumbosacral dorsal horn — critical for erection/arousal)
- MC4R agonism triggers nitric oxide (NO) release in the corpus cavernosum via the central nervous system pathway, independent of the traditional PDE5/cGMP mechanism
- This explains why MT-II causes erection/arousal even in men who don't respond to PDE5 inhibitors (Viagra, Cialis)
- In women: increased clitoral engorgement, lubrication, and subjective arousal via analogous spinal/central pathways

MC3R Activation — Appetite & Metabolic:

- MC3R expressed in hypothalamus, peripheral metabolic tissues
- Activation modulates energy balance, reduces appetite (synergistic with GLP-1 class?)
- Contributes to the body fat reduction some users report

MC5R Activation — Exocrine & Metabolic:

- MC5R in exocrine glands, adipose, muscle
- Roles in sebaceous gland secretion regulation and peripheral metabolic functions
- Less clinically prominent than MC1R/MC4R effects

Duration of Effect:

MT-II's cyclic structure makes it highly resistant to enzymatic degradation — significantly longer plasma half-life and receptor residence time than linear α-MSH analogues.

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2. Clinical Research & Trials

Study 1: Erectile Dysfunction — Phase 2 Trial (PT-141/Bremelanotide)

- **Note:** PT-141 (bremelanotide) is a structural derivative of MT-II (7-peptide fragment) that was developed into a pharmaceutical. Clinical data for PT-141 is directly relevant.
- **Design:** Randomized, double-blind, crossover trial, n=20 men with moderate-severe ED
- **Dosing:** PT-141 0.025 mg/kg SubQ
- **Results:** 80% of men achieved erections sufficient for intercourse vs. 0% placebo. Significant improvement in rigidity and duration vs. placebo. Critically: this includes men who failed PDE5 inhibitors.
- **Citation:** Wessells H et al., *Journal of Urology*, 1998; subsequent trials leading to FDA approval of Vyleesi (bremelanotide).

Study 2: Female Sexual Dysfunction (FSAD)

- **Drug:** PT-141 (bremelanotide) — derived from MT-II
- **Design:** Randomized, double-blind, placebo-controlled, n=327 premenopausal women with hypoactive sexual desire disorder (HSDD)
- **Results:** Significantly more satisfying sexual events (SSEs) per month: 2.5 vs. 1.7 placebo. Desire scores (FSFI) improved. Sexual distress scores reduced.
- **Note:** FDA-approved as Vyleesi (bremelanotide) for HSDD in premenopausal women, 2019
- **Citation:** Simon JA et al., *Obstetrics & Gynecology*, 2019.

Study 3: Tanning Dose-Finding (Arizona University)

- **Design:** Phase 1/2 trials, healthy volunteers
- **Dosing:** 0.01–0.025 mg/kg SubQ
- **Results:** Consistent, dose-dependent melanin increase. Tan developing without UV over 1–2 weeks. Lower doses needed than MT-I for equivalent tanning. Nausea and facial flushing noted at higher doses.
- **Citation:** Hadley ME et al., original University of Arizona publications; Dorr RT et al., *Journal of Investigative Dermatology*, 1996.

Study 4: Appetite Suppression / Obesity

- **Design:** Phase 1 dose-escalation in obese individuals (n=10)
- **Results:** Significant reduction in caloric intake at 0.1 mg/kg. Weight loss observed over 4 weeks. Mechanism: MC3R/MC4R hypothalamic appetite suppression.
- **Note:** Not developed as obesity drug due to sexual side effect profile; these effects are still observed at tanning doses

Study 5: Erectile Dysfunction — PDE5 Non-Responders

- **Population:** Men with ED who failed sildenafil (Viagra)
- **Results:** MT-II produced erections in 73% of PDE5 non-responders — demonstrating a completely independent mechanism through the central melanocortin pathway.
- **Clinical implication:** Clients with vascular or neurological ED who don't respond to conventional treatments may respond to melanocortin pathway activation.

Study 6: Neuroprotective Effects (Animal)

- **Findings:** MC4R activation in brain has anti-apoptotic properties; MT-II showed neuroprotective effects in cerebral ischemia models (animal), possibly related to central melanocortin's anti-inflammatory pathway.

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3. Primary Uses & Benefits

Goal	Mechanism
Sunless tanning	MC1R → eumelanin
Erectile dysfunction	MC4R → central NO activation
Female sexual arousal	MC4R → central arousal pathway
Libido enhancement	Hypothalamic MC4R
Appetite suppression	MC3R/MC4R hypothalamic
Body composition	Fat loss via appetite + metabolic MC3R
UV skin protection	Eumelanin formation

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4. Symptoms & Conditions It Targets

- Erectile dysfunction (especially PDE5 non-responders)
- Hypoactive sexual desire disorder (HSDD) in women
- Low libido (men and women)
- Fair skin / desire for safe tanning
- Light sensitivity / sun-reactive skin
- Weight management (secondary effect)
- Sexual dysfunction from medications (SSRIs, antihypertensives)

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5. Key Selling Points (Rep Talking Points)

- ■ **Dual tanning + sexual function** — unique combination; clients seeking tan often discover the libido/erection benefit is an added value
- ■ **Works when Viagra fails** — central melanocortin pathway is completely independent of PDE5; 73–80% response rate in ED including PDE5 non-responders
- ■ **Female sexual dysfunction** — bremelanotide (PT-141, the MT-II derivative) is FDA-approved for HSDD in women; directly applicable clinical validation
- ■ **Appetite reduction as bonus** — clients using for tanning often report spontaneous reduction in appetite/cravings; body composition benefit
- ■ **Very potent at low doses** — lower dose needed than MT-I for equivalent tanning; highly cost-effective per mg

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6. Common Protocols

Tanning Protocol

- **Beginner:** 0.25–0.5mg SubQ every other day; start low to assess nausea tolerance
- **Intermediate:** 0.5–1mg SubQ every other day (loading phase 2–3 weeks)
- **Maintenance:** 0.5–1mg, 2–3x/week to maintain tan
- **Enhance with:** Minimal sun/UV exposure during loading (accelerates pigmentation)

Sexual Function Protocol

- **Dose:** 0.025–0.1 mg/kg body weight (typically 1.5–3mg for 70kg person)
- **Timing:** 1–3 hours before sexual activity (SubQ)
- **Onset:** 30–60 minutes to erection/arousal effect
- **Frequency:** As needed; not recommended daily (receptor desensitization)

Reconstitution

- Reconstitute with bacteriostatic water: 2 mL per 10mg vial = 5mg/mL
- Common dose measurement: 0.1 mL = 0.5mg; 0.2 mL = 1mg
- SubQ injection, abdomen

Storage

- Lyophilized: refrigerate; stable 12–24 months
- Reconstituted: refrigerate; use within 30 days; protect from light

Half-Life

- ~1–2 hours plasma; biological effects last 6–12 hours (sexual effects: several hours; tanning: accumulates over days)

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7. Side Effects & Safety Profile

- **Nausea** (most common, 30–60% of users at standard doses): dose-dependent; typically mild, peaks 30–90 min post-injection, resolves in 1–3 hours. Minimized by: starting very low (0.25mg), evening dosing, eating beforehand
- **Facial flushing**: Vasodilatory effect; transient (30–60 min)
- **Spontaneous erections** in men (expected effect; can occur unpredictably at higher doses — caution re: timing)
- **Yawning** (unusual but reported; central melanocortin receptor effect)
- **Fatigue/drowsiness**: At higher doses; typically mild
- **Appetite suppression**: Expected; not side effect per se
- **Blood pressure**: Mild transient increase or decrease possible
- **Moles/freckles**: May darken during tanning protocols — monitor pre-existing nevi
- **Avoid in:** Cardiovascular disease (BP effects), melanoma history (theoretical caution), pregnancy
- **Drug interactions:** Theoretical caution with PDE5 inhibitors (additive hypotension/erection effect — can be excessive); avoid high doses combined

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8. Stacking Notes

Stack Partner	Rationale
Not with MT-I	Same receptors; no benefit to combining; choose one
GHK-Cu	Collagen + skin aging; MT-II handles melanin; GHK-Cu handles structural skin health
PT-141 / bremelanotide	Essentially MT-II derivative — same therapeutic niche

Ipamorelin	GH pulsing supports body composition goals alongside MT-II's appetite effects
BPC-157	Gut protection for clients experiencing GI side effects

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9. MT-I vs MT-II — Quick Reference

Feature	MT-I	MT-II
Receptor profile	MC1R selective	MC1R + MC3R + MC4R + MC5R
Tanning	■ Strong	■■ Very strong
Sexual/libido	■ None	■■ Very strong
Appetite suppression	■ Minimal	■ Moderate
Nausea risk	Low	Moderate-high
FDA approved	■ Yes (Scenesse)	■ No (derivative PT-141 = Vyleesi, approved)
Dose needed for tan	Higher	Lower
Safety profile	Excellent	Generally good, higher side effect rate

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10. FAQ — For Rep Use

Q: Why do some people get erections from MT-II when they didn't expect it?

A: MC4R activation in the spinal cord and hypothalamus triggers erection via the central nervous system pathway — independent of being sexually stimulated. This is a pharmacological effect, not dependent on psychological arousal. Clients should be warned when starting at higher doses.

Q: Is Vyleesi the same as MT-II?

A: Vyleesi contains bremelanotide (PT-141), which is the cyclic 7-amino acid core fragment of MT-II. They work through the same MC4R mechanism. MT-II has additional receptor activity. Clinically, their sexual function effects are nearly identical.

Q: Can women use it for sexual dysfunction?

A: Yes — bremelanotide (the MT-II derivative) is FDA-approved specifically for premenopausal women with HSDD. MT-II is used off-label for the same purpose with comparable efficacy based on mechanism.

Q: Will it cause melanoma?

A: No evidence of melanoma causation. Existing nevi (moles) may darken during use — baseline dermatological check and monitoring recommended for anyone with multiple or atypical moles. The eumelanin produced actually provides UV protection.

Q: How do I manage nausea?

A: Start at 0.25mg and titrate up over 1–2 weeks. Take in the evening (sleep through the nausea window). Having a small meal beforehand helps. Ginger tea or ondansetron (if prescribed) can assist in sensitive individuals.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

SELANK — Mavyn Rep Training Sheet

SKU: SK10 | Strength: 10mg | Category: Nootropic / Anxiolytic Peptide

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1. What Is Selank?

Selank is a synthetic heptapeptide (7 amino acids: Thr-Lys-Pro-Arg-Pro-Gly-Pro) developed by the Institute of Molecular Genetics at the Russian Academy of Sciences. It is an analogue of the endogenous tetrapeptide tuftsin (Thr-Lys-Pro-Arg), with three additional amino acids (Pro-Gly-Pro) added to stabilize the molecule and extend its bioactivity.

Mechanism of Action (Molecular/Cellular Level)

Enkephalinase Inhibition: Selank inhibits enkephalinase (neprilysin), the enzyme responsible for degrading endogenous enkephalins (opioid peptides). By preventing their breakdown, Selank increases the availability of Met-enkephalin and Leu-enkephalin in the brain, producing anxiolytic and mood-stabilizing effects without classical opioid dependency or respiratory depression.

Serotonergic Modulation: Selank increases expression of serotonin transporter (SERT) mRNA and modulates 5-HT receptor sensitivity. Studies in rodents show a 1.5–2x increase in tryptophan hydroxylase activity following administration, leading to elevated serotonin synthesis.

BDNF Upregulation: Selank significantly increases Brain-Derived Neurotrophic Factor (BDNF) in the hippocampus and cortex — a key driver of neuroplasticity, memory consolidation, and mood regulation. BDNF deficiency is associated with depression, anxiety, and cognitive decline.

GABAergic Potentiation: Selank modulates GABA-A receptor sensitivity, increasing inhibitory tone without causing sedation or tolerance at therapeutic doses. This partially explains its anti-anxiety effects distinct from benzodiazepines.

Immune Regulation via IL-6 / T-Helper Cells: As a tuftsin analogue, Selank modulates the innate immune system, influencing macrophage activity, IL-6 expression, and T-helper cell ratios. This gives it a unique dual action: neuroprotective AND immunomodulatory.

NO Synthase Inhibition: Selank inhibits nitric oxide synthase in stress-exposed animals, reducing oxidative neuronal damage and glutamate excitotoxicity during acute stress.

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2. Clinical Research & Trials

Study 1: Anxiolytic Efficacy in Generalized Anxiety Disorder (GAD)

- **Institution:** Serbsky National Medical Research Center, Moscow
- **Design:** Randomized, double-blind, placebo-controlled trial
- **Population:** 62 patients with GAD (DSM-IV criteria)
- **Dosing:** Intranasal Selank 400 mcg/day (200 mcg/nostril) for 14 days vs. placebo; secondary arm compared to Medazepam 10mg/day
- **Results:** Hamilton Anxiety Scale (HAM-A) scores reduced by 48% in Selank group vs. 22% in placebo. No significant difference in efficacy vs. Medazepam, but Selank group showed **zero sedation, zero withdrawal symptoms, and zero cognitive impairment** at follow-up.
- **Citation:** Semenova TP et al., *Eksperimental'naya i Klinicheskaya Farmakologiya*, 2010.

Study 2: Cognitive Enhancement & Memory

- **Design:** Crossover study in healthy volunteers (n=40)
- **Dosing:** Intranasal Selank 250–500 mcg, single dose
- **Results:** Significant improvements in short-term memory recall, attention switching, and working memory tasks within 20 minutes of administration. Effects persisted 4–6 hours. No impairment of psychomotor coordination (unlike benzodiazepines).
- **Citation:** Zozulya AA et al., *CNS Drug Reviews*, 2001.

Study 3: Immune Modulation in Immunocompromised Patients

- **Design:** Open-label pilot, n=28, patients with recurrent viral infections
- **Dosing:** Selank 900 mcg/day intranasally for 5 days
- **Results:** Significant normalization of IgA and IgM levels. IL-6 expression increased transiently then normalized. T-helper/suppressor ratio improved toward normal range in 82% of patients.
- **Citation:** Uchakina ON et al., *Immunologiya*, 2008.

Study 4: Anti-Stress / Anti-Depressant Properties

- **Animal Model:** Rats subjected to chronic unpredictable mild stress (CUMS model)
- **Dosing:** 300 mcg/kg IP daily for 10 days
- **Results:** Forced swim test immobility reduced by 60% vs. control (comparable to fluoxetine). BDNF mRNA expression in hippocampus increased 2.3x. Cortisol analogue (corticosterone) levels normalized significantly.
- **Citation:** Semenova TP et al., *Bulletin of Experimental Biology and Medicine*, 2009.

Study 5: Selank + Learning Enhancement

- **Population:** Medical students during exam period (n=30, open-label)
- **Dosing:** 300 mcg intranasally TID for 10 days
- **Findings:** Self-reported improvements in focus, recall, and exam performance. No adverse effects. Sleep quality improved (Pittsburgh Sleep Quality Index scores improved by 35%).

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3. Primary Uses & Benefits

Goal	Mechanism
Anxiety reduction	Enkephalinase inhibition, GABAergic potentiation
Cognitive enhancement	BDNF upregulation, serotonin modulation
Stress resilience	HPA axis modulation, cortisol normalization
Immune support	Tuftsin-like immunomodulation, IL-6 regulation
Depression adjunct	BDNF, serotonin, enkephalin system
Focus & productivity	Working memory, attention, alertness enhancement
Neuroplasticity support	BDNF-mediated synaptogenesis

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4. Symptoms & Conditions It Targets

- Generalized Anxiety Disorder (GAD)
- Social anxiety / performance anxiety
- Chronic stress / burnout / HPA dysregulation
- Brain fog, poor focus, memory issues
- Mild-to-moderate depression (adjunct therapy)
- Immune dysfunction / recurrent infections
- Cognitive decline (early/preventive)
- Sleep disturbance secondary to anxiety

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5. Key Selling Points (Rep Talking Points)

- ■ **No sedation, no dependency** — anxiolytic effects without benzodiazepine risks; no withdrawal, no addiction
- ■ **Dual action: brain + immune** — rare peptide that improves mood AND immune function simultaneously
- ■ **Fast onset** — noticeable anxiolytic and cognitive effects within 20–40 minutes of nasal administration
- ■ **BDNF booster** — one of the few peptides proven to upregulate BDNF, the brain's "growth hormone"
- ■ **Clinically validated in Russia** — approved as a pharmaceutical-grade anxiolytic (brand name: Selank) with two decades of institutional research

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6. Common Protocols

Beginner Protocol

- **Dose:** 200–300 mcg intranasally (1–2 sprays per nostril), once daily in the morning
- **Cycle:** 2–3 weeks on, 1 week off
- **Goal:** Baseline anxiety management, focus support

Intermediate Protocol

- **Dose:** 400–600 mcg/day, split AM and midday
- **Cycle:** 4 weeks on, 1–2 weeks off
- **Goal:** Cognitive enhancement + anxiety reduction during high-stress periods

Advanced Protocol

- **Dose:** 600–900 mcg/day (200–300 mcg, 3x/day)
- **Cycle:** 4–6 weeks; some users run continuous low dose (300 mcg/day) with periodic breaks
- **Goal:** Chronic anxiety management, immune support, neuroplasticity enhancement

Reconstitution

- Reconstitute with bacteriostatic water (0.9% benzyl alcohol saline) or sterile water
- Use 1–2 mL bac water per 10mg vial
- Intranasal delivery is preferred; subcutaneous injection also effective
- Stable 30 days refrigerated post-reconstitution

Storage

- Lyophilized: room temperature up to 6 months; refrigerate for 12+ months
- Reconstituted: refrigerate, use within 30 days; do not freeze

Half-Life

- ~8 minutes plasma half-life; functional effects last 4–8 hours (receptor-level persistence)

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7. Side Effects & Safety Profile

- **Very well tolerated** in clinical studies
- Mild transient nasal irritation (intranasal route)
- Rare: mild fatigue or slight drowsiness at higher doses
- No hepatotoxicity, nephrotoxicity, or cardiac effects reported
- No hormonal disruption
- **Not recommended:** Pregnancy, nursing (insufficient data)
- **Drug interactions:** Theoretical caution with MAOIs; no documented interactions with SSRIs but clinical monitoring advised

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8. Stacking Notes

Stack Partner	Rationale
Semax	Complementary nootropic stack — Semax boosts ACTH/BDNF via different pathway; together = cognitive synergy
BPC-157	Gut-brain axis support + neuroplasticity; Selank + BPC is a popular cognitive/recovery duo
Ipamorelin	Adding GH pulse support for deeper recovery and sleep quality
NAD+	Mitochondrial + cognitive support; complementary without interaction
Epithalon	Anti-aging + sleep normalization alongside Selank's anxiolytic effects

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9. Comparison to Similar Peptides

Feature	Selank	Semax	Dihexa
Primary action	Anxiolytic / cognitive	Cognitive / stimulant	Neuroplasticity
BDNF effect	Strong upregulation	Strong upregulation	HGF/c-MET pathway
Anxiety	■ Reduces	Neutral/mild stimulant	Neutral
Energy	Neutral	■ Stimulating	Neutral
Research base	Strong (clinical RCTs)	Strong (clinical)	Animal only
Best for	Anxiety + brain fog	Focus + drive	Cognitive repair

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10. FAQ — For Rep Use

Q: Is Selank addictive?

A: No. It does not act on classical opioid or benzodiazepine receptors in an addictive manner. Long-term users show no withdrawal, no tolerance escalation.

Q: Can it be used alongside an SSRI?

A: Many users do. There are no documented dangerous interactions, but advise clients to consult their prescribing physician. Selank may complement SSRI therapy through different mechanisms.

Q: How quickly does it work?

A: Most users report noticeable calmness and mental clarity within 20–45 minutes of intranasal administration.

Q: Is intranasal or injection better?

A: Intranasal is the standard delivery method — excellent bioavailability via nasal mucosa. SubQ injection also works but intranasal is preferred for ease.

Q: How is it different from CBD or traditional anxiety supplements?

A: Selank works at the enkephalin and BDNF level — it's a precision peptide with clinical trial data, not a herbal/OTC supplement. The mechanism is fundamentally different and more specific.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

SEMAX — Mavyn Rep Training Sheet

SKU: XA5 | Strength: 5mg | Category: Nootropic / Neuroprotective Peptide

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1. What Is Semax?

Semax is a synthetic heptapeptide derived from the N-terminal fragment of ACTH (adrenocorticotropic hormone): Met-Glu-His-Phe-Pro-Gly-Pro. It was developed in the 1980s by the Institute of Molecular Genetics of the Russian Academy of Sciences and is approved as a pharmaceutical drug in Russia and Ukraine for stroke recovery, cognitive decline, and peptic ulcer disease.

Unlike ACTH itself, Semax has no adrenal-stimulating activity — the structural modifications eliminate cortisol/steroid-stimulating effects while preserving and amplifying the neurotropic actions.

Mechanism of Action (Molecular/Cellular Level)

BDNF & NGF Upregulation: Semax dramatically increases expression of BDNF (Brain-Derived Neurotrophic Factor) and NGF (Nerve Growth Factor) in the hippocampus, frontal cortex, and striatum. BDNF mRNA expression has been shown to increase 2–4x within hours of administration. BDNF drives synaptogenesis, neuronal survival, and long-term potentiation (LTP) — the cellular basis of learning and memory.

Dopaminergic Modulation: Semax increases dopamine turnover in the mesolimbic and mesocortical pathways (the reward and executive function circuits). This accounts for its motivating, drive-enhancing, and focus-sharpening properties — users often describe a "clean stimulant" effect without jitteriness.

Serotonin System: Semax modulates serotonergic transmission, particularly in stress-response circuits, contributing to mood stabilization.

Neuroprotection — Glutamate Excitotoxicity: Semax reduces NMDA-receptor overactivation during ischemia, hypoxia, and oxidative stress — protecting neurons from excitotoxic death. This is the basis for its clinical use in stroke recovery.

Melanocortin Receptor Binding: Semax binds MC4R and MC5R (melanocortin receptors) in brain tissue, activating downstream cAMP/PKA pathways that promote neuronal growth, myelination, and anti-inflammatory signaling.

Anti-Inflammatory CNS: Inhibits NF-κB-mediated neuroinflammation; reduces microglial activation markers (TNF-α, IL-1β). This makes it relevant not just for cognition but for conditions involving chronic neuroinflammation (e.g., post-COVID brain fog, TBI recovery).

HIF-1α Activation: Under hypoxic conditions, Semax activates Hypoxia-Inducible Factor 1-alpha, improving oxygen utilization efficiency at the neuronal level — relevant for altitude, cognitive fatigue, and brain injury.

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2. Clinical Research & Trials

Study 1: Ischemic Stroke Recovery — Phase II/III Clinical Trial

- **Institution:** Pirogov Russian National Research Medical University
- **Design:** Randomized controlled trial, n=187 patients with acute ischemic stroke
- **Dosing:** Intranasal Semax 0.1% solution (1mg/day) for 5 days, initiated within 6 hours of stroke

- **Results:** 34% reduction in neurological deficit scores (NIH Stroke Scale) vs. control at 30 days. Significantly improved recovery of motor function and speech. Mortality reduction trend noted (not statistically significant in this cohort).
- **Citation:** Skvortsova VI et al., *Journal of Neurological Sciences*, 2003; registered as pharmaceutical treatment in Russia.

Study 2: Optic Nerve Disease / Glaucoma

- **Design:** Open-label clinical study, n=64 patients with open-angle glaucoma and progressive optic nerve atrophy
- **Dosing:** Intranasal Semax 0.1% for 10-day courses, 3 courses/year
- **Results:** 72% of patients showed stabilization or improvement in visual field parameters. NGF levels in tear fluid increased significantly, indicating local neurotrophic effect on optic nerve.
- **Citation:** Maksimova MV et al., Russian ophthalmology registry data.

Study 3: Cognitive Enhancement in Healthy Volunteers

- **Design:** Double-blind crossover, n=50 healthy adults aged 25–45
- **Dosing:** 600 mcg intranasal single dose
- **Results:** Statistically significant improvements in: Stroop color-word test (attention/inhibition), n-back working memory task (+18% accuracy), digit span test. Effects onset ~30 min, peak 1–2 hours, duration 6–8 hours.

Study 4: ADHD-Like Symptoms

- **Design:** Pilot open-label, n=34 children with attention deficit symptoms (Russia)
- **Dosing:** 200 mcg/day intranasally for 14 days
- **Results:** Teacher-rated attention scores improved in 79% of participants. No adverse events. Dopaminergic modulation proposed as mechanism.

Study 5: Post-COVID Cognitive Dysfunction ("Brain Fog")

- **Design:** Observational, n=22 patients with persistent cognitive symptoms 3+ months post-COVID
- **Dosing:** 600 mcg/day for 14 days
- **Results:** 17/22 patients reported subjective improvement in memory, word retrieval, and focus. Neuroinflammation marker reduction (CRP, IL-6) observed in 8/10 who had labs drawn.

Study 6: Animal Models — Alzheimer's & Neurodegeneration

- **Model:** APP/PS1 transgenic mice (Alzheimer's model)
- **Findings:** Semax reduced amyloid-beta plaque accumulation, improved spatial memory (Morris water maze), and normalized BDNF levels to near-wild-type values. Microglial activation markers reduced significantly.

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3. Primary Uses & Benefits

Goal	Mechanism
Cognitive enhancement	BDNF/NGF, dopamine, NMDA modulation
Stroke recovery / neuroprotection	Excitotoxicity protection, neurogenesis
Focus & executive function	Dopaminergic mesocortical pathway
Memory consolidation	LTP enhancement, hippocampal BDNF
Brain fog (post-illness, chronic stress)	Neuroinflammation reduction, neurotrophins

Mood & drive	Serotonin/dopamine dual modulation
Optic nerve health	NGF delivery to visual pathways
TBI / concussion recovery	Neuroprotection, anti-inflammatory

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4. Symptoms & Conditions It Targets

- Cognitive decline (age-related, disease-related)
- ADHD symptoms / attention deficits
- Chronic fatigue + brain fog
- Post-stroke neurological recovery
- Post-COVID cognitive dysfunction
- Anxiety with cognitive dulling
- Low motivation / anhedonia
- Memory problems (short-term, working memory)
- Optic nerve degeneration / glaucoma

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5. Key Selling Points (Rep Talking Points)

- ■ **Pharmaceutical-grade, clinically approved** — approved drug in Russia/Ukraine for stroke and cognitive disorders with decades of institutional study
- ■ **Strongest BDNF booster in the peptide class** — 2–4x upregulation of BDNF is remarkable and drives measurable neuroplasticity
- ■ **Clean dopamine enhancement** — focus and drive without stimulant side effects (no racing heart, no crash, no addiction)
- ■ **Neuroprotective AND cognitive-enhancing** — dual action makes it ideal for both recovery clients AND high-performers
- ■ **Fast, practical delivery** — intranasal administration, fast onset, no injections required for most users

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6. Common Protocols

Beginner Protocol

- **Dose:** 200–300 mcg intranasally, once daily (morning)
- **Cycle:** 2 weeks on, 1 week off
- **Goal:** Cognitive baseline support, test tolerance

Intermediate Protocol

- **Dose:** 400–600 mcg/day, AM (single dose or split AM/midday)
- **Cycle:** 3–4 weeks on, 1–2 weeks off
- **Goal:** Performance enhancement, memory, focus

Advanced / Therapeutic Protocol

- **Dose:** 600–1200 mcg/day
- **Stroke recovery:** 1mg/day for 5–14 days (acute); maintenance 600 mcg/day
- **Cycle:** 4–6 weeks; some practitioners run 2-month cycles for neurodegeneration
- **Goal:** TBI recovery, significant cognitive restoration

Reconstitution

- Reconstitute with sterile/bacteriostatic water; 1–2 mL per 5mg vial
- Intranasal delivery standard (nasal spray bottle preferred)
- SubQ injection also effective at same doses

Storage

- Lyophilized: up to 12 months refrigerated; 3–6 months room temperature (stable peptide)
- Reconstituted: refrigerate, use within 30 days

Half-Life

- Plasma half-life: ~2–4 minutes; functional neurological effect duration: 6–12 hours (receptor-mediated persistence)

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7. Side Effects & Safety Profile

- Excellent tolerability across decades of clinical use
- Most common: mild headache or irritability at higher doses (typically dose-dependent, resolves)
- Rare: transient fatigue post-dose (paradoxical in some users)
- Nasal irritation with intranasal route (mild, transient)
- No reported hepatotoxicity, nephrotoxicity, or cardiac effects
- No hormonal disruption
- **Note:** Stimulating effects may interfere with sleep if taken late in day — recommend morning/midday only
- **Avoid in:** Active psychosis, uncontrolled bipolar disorder (dopaminergic stimulation may exacerbate mania)
- **Pregnancy/nursing:** Insufficient data; avoid

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8. Stacking Notes

Stack Partner	Rationale
Selank	The classic "cognitive stack" — Semax for drive/focus + Selank for anxiety relief = balanced cognitive enhancement
BPC-157	Gut-brain axis, vagus nerve support, neuroprotection from different angle
Ipamorelin	GH support → IGF-1 → brain tissue repair; synergistic for recovery
Epithalon	Telomere protection + pineal gland support; Semax handles cognition while Epithalon handles longevity
NAD+	Cellular energy + neuroprotection: NAD+ fuels the neurons Semax is repairing

GHK-Cu	Tissue remodeling + nerve regeneration; Semax drives neurotrophins, GHK-Cu supports structural repair

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9. Comparison to Similar Peptides

Feature	Semax	Selank	Dihexa	P21
Focus/drive	■■ Strong	Neutral	Neutral	Neutral
BDNF effect	■■ Very strong	■ Strong	Via HGF	■
Anxiolytic	Mild	■■ Strong	No	No
Neuroprotection	■■ Proven	Moderate	Strong	Strong
Clinical data	■ Approved drug	■ Clinical trials	Animal only	Animal only
Stimulant-like	Mild yes	No	No	No

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10. FAQ — For Rep Use

Q: Is Semax like Adderall?

A: Not really. Semax enhances dopaminergic transmission through receptor modulation rather than flooding the synapse with dopamine like Adderall. Users describe sharper focus without the jitteriness, heart rate elevation, or crash. No addiction potential documented.

Q: How long until I feel it?

A: Most users notice improved clarity and focus within 30–60 minutes of intranasal administration. Full benefit builds over 1–2 weeks of consistent use as BDNF levels accumulate.

Q: Can it be used long-term?

A: Yes — it's prescribed in Russia for months at a time. Most protocols use cycles (3–4 weeks on, 1–2 weeks off) as a best practice, though continuous use at lower doses is common.

Q: Is it safe with nootropic supplements?

A: Generally yes. Stacks well with racetams, lion's mane, and other nootropics. The mechanisms don't typically overlap dangerously.

Q: Why hasn't the FDA approved it?

A: It's a research peptide in the US — not FDA-approved, but used widely in clinical wellness practices. The clinical evidence base is robust; it simply hasn't been submitted through the FDA process by a US pharmaceutical company.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

VIP (Vasoactive Intestinal Peptide) — Mavyn Rep Training Sheet

SKU: VIP10 | Strength: 10mg | Category: Neuropeptide / Anti-Inflammatory / Autonomic

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1. What Is VIP?

Vasoactive Intestinal Peptide (VIP) is a 28-amino acid neuropeptide naturally produced throughout the body — primarily in the hypothalamus, gut enteric nervous system, pancreas, lungs, immune cells, and peripheral neurons. It is one of the most pleiotropic peptides known: a single molecule that regulates inflammation, vasodilation, circadian rhythm, immune tolerance, gut motility, pulmonary function, and hormone secretion.

Despite its name, VIP's clinical relevance extends far beyond vascular effects. It is a master regulatory signal at the intersection of the nervous, immune, and endocrine systems.

Mechanism of Action (Molecular/Cellular Level)

VPAC1 & VPAC2 Receptor Binding: VIP exerts its effects through two G-protein-coupled receptors: VPAC1 (expressed broadly on immune cells, gut, brain) and VPAC2 (more selective: brain, smooth muscle, T-cells). Binding triggers adenylyl cyclase → cAMP → PKA signaling cascade with powerful downstream anti-inflammatory and regulatory effects.

Immune Modulation — Th1/Th2/Treg Balance:

- Suppresses Th1 inflammatory cytokines: IL-2, IFN-γ, TNF-α, IL-6
- Promotes Th2/regulatory responses: IL-4, IL-10, TGF-β
- Directly induces regulatory T-cell (Treg) differentiation — the "immune tolerance" cells that prevent autoimmune overactivation
- Potently suppresses macrophage activation and dendritic cell maturation
- This makes VIP a major player in autoimmune conditions, mast cell disorders, and MCAS

Vasodilation: VIP relaxes vascular smooth muscle through cAMP-mediated pathway, independent of NO. Reduces pulmonary vascular resistance particularly effectively — clinical applications in pulmonary hypertension.

Circadian Rhythm Regulation: VIP is the primary neurotransmitter of the suprachiasmatic nucleus (SCN) — the brain's master clock. VIP neurons synchronize the circadian pacemaker, coordinating sleep-wake cycles, hormone release timing, and metabolic rhythms. VIP deficiency → circadian dysfunction, poor sleep architecture, hormone dysregulation.

Gut Function: Relaxes GI smooth muscle, regulates intestinal secretion, modulates ENS (enteric nervous system) signaling. Anti-inflammatory in the gut — reduces intestinal inflammation via suppression of mucosal immune activation.

Neuroprotection: VIP promotes neuronal survival, inhibits apoptosis in oxidative/excitotoxic conditions, and stimulates BDNF and activity-dependent neuroprotective protein (ADNP) secretion from astrocytes.

Pulmonary: Bronchodilatory, reduces airway inflammation, modulates mast cell degranulation in the lungs. Historically investigated as a treatment for pulmonary arterial hypertension (PAH).

CIPA (Cytokine-Induced Pulmonary Inflammation): VIP is the most potent endogenous suppressor of cytokine storm in lung tissue — directly relevant to COVID-related pathology.

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2. Clinical Research & Trials

Study 1: Pulmonary Arterial Hypertension — Phase 2 RCT

- **Design:** Randomized, double-blind, placebo-controlled, n=96 patients with PAH
- **Dosing:** Inhaled VIP 200 mcg 4x/day via nebulizer for 24 weeks
- **Results:** Significant improvement in 6-minute walk distance (+46m vs. +9m placebo). Pulmonary vascular resistance decreased significantly. No systemic hypotension at inhaled doses. Exercise capacity improved in 68% of treatment group.
- **Citation:** Petkov V et al., *Chest*, 2003; subsequent expanded trial by Universität Wien.

Study 2: Rheumatoid Arthritis — Phase 1/2

- **Design:** Open-label dose-escalating trial, n=18 patients with active RA on DMARDs
- **Dosing:** SubQ VIP 25–100 pmol/kg twice daily for 12 weeks
- **Results:** DAS28 (disease activity score) improved significantly in 78% of patients. TNF-α, IL-1β, IL-6 serum levels decreased 40–60%. Treg count increased significantly. No serious adverse events.
- **Citation:** Gomariz RP et al., *Annals of the Rheumatic Diseases*, 2006; Juarranz Y et al., multiple publications.

Study 3: Sepsis & Systemic Inflammation

- **Design:** Animal model (LPS-induced sepsis in mice)
- **Results:** VIP treatment reduced 24-hr mortality from 100% → 35%. Serum TNF-α reduced 80%. IL-6 reduced 90%. Bacterial clearance unimpaired (selective anti-inflammatory without immunosuppression).
- **Citation:** Ottaway CA et al.; multiple groups including Bhora FH lab.

Study 4: Inflammatory Bowel Disease (IBD)

- **Design:** Animal model (DSS colitis) and in vitro human tissue studies
- **Results:** VIP significantly reduced colon inflammation scores, improved mucosal integrity, reduced mucosal TNF-α and IL-12. Treg induction in lamina propria T-cells.
- **Citation:** Multiple publications, Abad C et al., *Journal of Pharmacology and Experimental Therapeutics*.

Study 5: MCAS / Mast Cell Activation

- **Mechanism:** VIP suppresses mast cell degranulation via VPAC2 receptor — cAMP-mediated inhibition of IgE-triggered histamine release
- **Clinical Observation:** Multiple clinical reports of VIP use in MCAS patients with significant symptom reduction
- **Note:** Strong mechanistic rationale; formal RCT not yet published

Study 6: Autism Spectrum Disorder (ASD) — Nasal VIP

- **Design:** Phase 1/2, n=30 children with ASD
- **Dosing:** Intranasal VIP (Aviptadil) 8 or 16 mcg/day for 16 weeks
- **Results:** Significant improvements in social communication scores (ADOS, Vineland). Proposed mechanism: VIP deficiency in SCN/amygdala circuits → circadian and social processing deficits.
- **Citation:** Clinical trial data, Sherri Tenpenny et al.; ongoing research.

Study 7: COVID-19 ARDS — Aviptadil (IV VIP)

- **Drug:** Aviptadil (synthetic VIP) — emergency FDA IND granted

• **Design:** Randomized controlled trial in critically ill COVID-19 ARDS patients

• **Results:** Aviptadil group showed 47% survival advantage at 60 days vs. placebo. Lung function preservation significantly better. VIP's suppression of cytokine storm and protection of Type 2 pneumocytes proposed as mechanism.

• **Citation:** ClinicalTrials.gov NCT04311697; Perelman D et al., NEJM Evidence, 2022 pre-publication data; NeuroRx/Relief Therapeutics trial data.

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3. Primary Uses & Benefits

Goal	Mechanism
Autoimmune modulation	Treg induction, Th1 suppression, cytokine downregulation
Pulmonary health / PAH	Vasodilation, bronchodilation, mast cell stabilization
MCAS treatment	Mast cell degranulation inhibition via VPAC2
Circadian rhythm / sleep	SCN synchronization, master clock regulation
Gut health / IBD	Mucosal anti-inflammatory, ENS regulation
Neuroprotection	ADNP/BDNF stimulation, anti-apoptotic
Systemic inflammation	Cytokine storm suppression
Rheumatoid arthritis	Joint inflammation reduction, Treg promotion

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4. Symptoms & Conditions It Targets

• Mast cell activation syndrome (MCAS)
• POTS / dysautonomia with mast cell component
• Pulmonary arterial hypertension
• Rheumatoid arthritis and other autoimmune conditions
• Inflammatory bowel disease (Crohn's, UC)
• Chronic systemic inflammation / "cytokine storm" states
• Circadian rhythm disorders / chronic poor sleep
• Long COVID / post-viral inflammatory syndrome
• Asthma with inflammatory component
• Autism spectrum (emerging research)

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5. Key Selling Points (Rep Talking Points)

• ■ **Master anti-inflammatory neuropeptide** — addresses inflammation at the cytokine level across multiple organ systems simultaneously

• ■ **Treg induction = immune re-education** — doesn't just suppress; it promotes immune tolerance — the root fix for autoimmune conditions

• ■ **MCAS game-changer** — growing community of MCAS/dysautonomia patients reporting significant relief; one of the few peptides with a clear mechanistic fit

• ■ **COVID/Long COVID support** — Aviptadil (VIP) has emergency FDA authorization data for ARDS; directly targets the cytokine storm mechanism

- ■ **Circadian + sleep restoration** — SCN regulation is a unique niche; for clients with circadian disruption, VIP addresses a root cause no sleep supplement touches

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6. Common Protocols

Intranasal Protocol (Most Common)

- **Dose:** 50–100 mcg intranasally, 1–2x/day
- **Cycle:** 4–8 weeks; some chronic condition clients use ongoing
- **Route:** Nasal spray (10–20 mcg/spray; titrate up)

Inhaled Protocol (Pulmonary Focus)

- **Dose:** 200 mcg via nebulizer, 3–4x/day (per PAH trial protocols)
- **Use:** PAH, severe asthma, pulmonary inflammation

SubQ / IV Protocol (Clinical Setting)

- **Dose:** 25–100 pmol/kg (clinical RA trial dosing) — generally low absolute doses
- **Note:** IV typically in clinical/hospital setting only

Reconstitution

- Reconstitute with sterile or bacteriostatic water; VIP is sensitive to degradation
- Use within 7–14 days reconstituted (less stable than many peptides)
- Refrigerate; protect from light; avoid repeated freeze-thaw

Storage

- Lyophilized: refrigerate; stable 12+ months
- Reconstituted: refrigerate; use within 14 days; very sensitive to degradation

Half-Life

- Plasma half-life: ~1–2 minutes (rapidly degraded by proteases); tissue effects much longer via receptor-mediated signaling; intranasal delivery bypasses first-pass degradation efficiently

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7. Side Effects & Safety Profile

- **Generally well tolerated at therapeutic doses**
- Most common: mild facial flushing, transient drop in blood pressure (vasodilatory effects — monitor in hypotensive clients)
- Nasal: mild irritation, sneezing (intranasal route)
- Rare: nausea, transient diarrhea (GI smooth muscle relaxation)
- High IV doses: significant hypotension (clinical setting protocol)
- **Avoid in:** Severe hypotension, pregnancy (smooth muscle relaxation, insufficient safety data)
- No hepatotoxicity, no renal toxicity documented

8. Stacking Notes

Stack Partner	Rationale
BPC-157	Gut healing + VIP's ENS modulation = comprehensive IBD/gut repair stack
Selank	Neuroinflammation + systemic inflammation; complementary anti-inflammatory pathways
Epithalon	VIP for circadian regulation + Epithalon for pineal/melatonin restoration = powerful sleep/anti-aging synergy
GHK-Cu	Tissue remodeling complement to VIP's anti-inflammatory signaling
NAD+	Cellular energy support for the immune system VIP is regulating

9. FAQ — For Rep Use

Q: Who is the ideal client for VIP?

A: Clients with MCAS, autoimmune conditions, long COVID symptoms, IBD, PAH, or severe circadian dysfunction. Also high-inflammatory individuals (high hs-CRP, frequent illness, joint pain) who haven't responded to standard anti-inflammatories.

Q: Is VIP related to the "VIP" social status term?

A: No — named "vasoactive intestinal peptide" for its discovery site (intestine) and original identified action (vasodilation). The naming is coincidental.

Q: How does it help MCAS?

A: VIP directly suppresses mast cell degranulation through VPAC2 receptors. It reduces histamine release, leukotrienes, and prostaglandins from mast cells without suppressing healthy immune function.

Q: Is it related to COVID treatments?

A: Yes — Aviptadil (synthetic VIP) received FDA emergency authorization for clinical trials in COVID ARDS. It showed significant survival benefit by suppressing the cytokine storm and protecting lung epithelial cells.

Q: Can it be taken with antihistamines for MCAS?

A: Yes — complementary. Antihistamines block histamine receptors after release; VIP prevents the release upstream. Together, they address MCAS from multiple angles.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

GLOW STACK — Mavyn Rep Training Sheet

SKU: GLOW | Strength: 70mg total | Category: Recovery / Collagen / Repair Stack

Composition: BPC-157 (10mg) + GHK-Cu (50mg) + TB-500 (10mg)

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1. What Is the GLOW Stack?

GLOW is Mavyn's foundational recovery and tissue repair stack — a targeted combination of three of the most extensively researched healing peptides in the portfolio. It is the "core" of the KLOW stack, minus the KPV component, designed for clients whose primary goals are physical recovery, anti-aging skin/tissue quality, and musculoskeletal repair without requiring the additional gut-protection and NF-κB immunomodulation that KPV provides.

The three components are:

- **BPC-157 (10mg):** Tissue repair, tendon/ligament, nerve regeneration, gut healing, angiogenesis
- **GHK-Cu (50mg):** Collagen synthesis and remodeling, anti-fibrotic, anti-aging gene expression, hair/skin
- **TB-500 (10mg):** Systemic angiogenesis, actin mobility, satellite cell activation, muscle repair

Why These Three Specifically?

This combination was not arbitrary — it represents the **three most complementary phases of the healing cascade**:

BPC-157: The master healing signal. Activates FAK-paxillin, NO synthase, VEGFR2, and FGFR2 to initiate and direct healing at injury sites. Drives local angiogenesis, cell migration, and tissue-specific repair (tendon, muscle, nerve, gut). The "catalyst."

TB-500: The systemic mobilizer. Releases G-actin from the sequestered pool → cell motility → migration of endothelial cells, satellite cells, and stem cells to injury sites via the bloodstream. Drives large-scale angiogenesis and satellite cell recruitment across the whole body. The "delivery system."

GHK-Cu: The quality controller. Ensures that the new tissue being built (collagen, ECM) is organized and functional rather than random scar tissue. Simultaneously removes damaged collagen (MMP activation) and builds new, organized collagen (lysyl oxidase activation). The anti-fibrotic, copper-powered "rebuilder."

Together, the three form a complete repair loop:

1. BPC-157 initiates healing and recruits local repair machinery
2. TB-500 brings systemic vascular supply and cell reinforcements
3. GHK-Cu ensures the rebuilt tissue is high quality, not scarred

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2. Mechanism Summary for Each Component

BPC-157 (10mg) in GLOW

Full mechanism in the BPC-157 individual training sheet. GLOW highlights:

- **FAK-paxillin activation:** Cell migration + proliferation at all injury sites
- **VEGF/angiogenesis:** Local blood supply restoration
- **NO modulation:** Inflammation-to-healing transition
- **GI mucosal protection:** Some gut benefit even without KPV

- **Dose (10mg):** This is a high-loading vial — provides extended supply at standard 250–500 mcg/day injection dosing (~20–40 days of daily doses)

TB-500 (10mg) in GLOW

Full mechanism in TB-500 individual training sheet. GLOW highlights:

- **Systemic reach:** Unlike BPC-157, TB-500 distributes throughout the body, reaching injuries deep in muscle, joint, and connective tissue
- **Satellite cell mobilization:** Critical for muscle fiber repair
- **Anti-inflammatory (systemic):** NF-κB suppression across tissues
- **Vascular network building:** Especially important in avascular tissue (tendons, cartilage) that heal slowly due to poor blood supply

GHK-Cu (50mg) in GLOW — The Dominant Component

Full mechanism in GHK-Cu individual training sheet. At 50mg, GHK-Cu is the largest volume component and the defining character of the GLOW stack.

- **Collagen type I & III synthesis:** 300–400% increase in fibroblast cultures at therapeutic concentrations
- **Elastin and glycosaminoglycans:** Full ECM restoration
- **MMP/TIMP modulation:** Organized remodeling (remove old → replace with new)
- **Anti-fibrotic via TGF-β1 suppression:** Prevents scar formation — the key outcome advantage
- **4,000+ gene expression normalization:** The "anti-aging" layer of GLOW
- **50mg dose:** Provides months of anti-aging dosing (at 1–2mg/day = 25–50 days of daily injections)

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3. Research Basis for the Combination

Synergy Evidence

BPC-157 + TB-500 Combination Studies:

Several animal studies have directly examined the combination:

- **Rat Achilles tendon transection model:** BPC-157 + TB-500 combination showed additive improvement in tensile strength vs. either alone. Histology: significantly more organized collagen fibers and less fibrous adhesion tissue in the combined group.
- **Mechanism synergy confirmed:** TB-500's VEGFR2 upregulation + BPC-157's independent angiogenic signaling produce additive vascular bed formation.

GHK-Cu + Healing Peptide Synergy (Theoretical + In Vitro):

- GHK-Cu's TGF-β1 suppression is anti-fibrotic — prevents scar formation
- BPC-157 and TB-500 direct organized cell growth
- Together: growth direction + quality control = superior tissue architecture vs. either alone

Clinical Observation:

The BPC-157 + TB-500 + GHK-Cu combination has been widely used in clinical settings (IV wellness clinics, sports medicine) with consistent reports of:

- Faster return to full training post-injury
- Reduced re-injury rates
- Improved tissue quality on imaging (MRI, ultrasound)

- Skin and joint improvements concurrent with musculoskeletal healing

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4. Primary Uses & Benefits

Application	Mechanism
Athletic injury recovery	BPC-157 + TB-500 tissue/vascular + GHK-Cu collagen
Tendinopathy treatment	TB-500 tendon vascularization + BPC-157 tenocyte signaling
Anti-aging (skin, joints)	GHK-Cu 50mg skin/collagen + BPC-157/TB-500 tissue maintenance
Post-surgical healing	All three accelerate wound closure + tissue quality
Joint degeneration	GHK-Cu cartilage + TB-500 synovial vascular + BPC healing
Muscle strain recovery	TB-500 satellite cells + BPC-157 myogenic signal
Skin appearance	GHK-Cu dominant — collagen, elastin, wrinkle reduction
Hair growth (systemic)	GHK-Cu hair follicle enlargement via SubQ delivery

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5. Key Selling Points (Rep Talking Points)

- ■ **The gold standard injury recovery stack** — BPC-157 + TB-500 is the most widely discussed and used peptide combination in sports medicine and clinical wellness; GLOW formalizes it with the addition of GHK-Cu for quality outcomes
- ■ **GHK-Cu at 50mg is the differentiator** — most individual protocols use 1–2mg/day; GLOW's 50mg GHK-Cu provides a full anti-aging collagen remodeling course in a single vial — the skin/anti-aging benefit is "built in"
- ■ **Reduces re-injury risk** — by improving tissue quality and architecture during healing (not just healing speed), clients come back stronger and less prone to recurrence
- ■ **Accessible entry point to stack protocols** — GLOW is lower cost than KLOW; appropriate for clients who need recovery/anti-aging but don't have the gut/autoimmune profile that justifies KLOW's premium
- ■ **Versatile client pool** — athletes, aging adults, post-surgical patients, and cosmetic/anti-aging clients all benefit; broadest appeal of any single Mavyn product

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6. Common Protocols

Standard GLOW Recovery Protocol

Loading Phase (2 Weeks, Daily):

- BPC-157 component: 250–500 mcg SubQ daily
- TB-500 component: 2.5–5mg SubQ, 2x/week
- GHK-Cu component: 1–2mg SubQ daily

(Or combined SubQ injection if supplied as blended vial)

Maintenance Phase (Weeks 3–8):

- BPC-157: 250 mcg EOD
- TB-500: 2–2.5mg, 1x/week

- GHK-Cu: 1mg daily

Anti-Aging / Cosmetic Protocol

- **Focus on GHK-Cu component:** 1–2mg/day SubQ
- **BPC-157 + TB-500:** 2–3x/week for systemic tissue maintenance
- **Duration:** 8–12 weeks; 2–3 courses/year
- **Goal:** Skin collagen, joint maintenance, anti-aging

Pre-Competition / Preventive Protocol

- **For athletes entering intensive training block:**
- GLOW 3x/week throughout training block
- Maintains tissue resilience, reduces injury risk
- GHK-Cu anti-inflammatory + BPC/TB-500 ongoing repair

Reconstitution

- Reconstitute with bacteriostatic water: 2–3 mL for 70mg vial
- Roll gently; do not shake
- BPC-157 and GHK-Cu are very stable; TB-500 moderately stable; refrigerate immediately
- Use within 30 days

Storage

- Lyophilized: refrigerate; stable 12–18 months
- Reconstituted: refrigerate; use within 30 days

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7. Side Effects & Safety Profile

- **Very well tolerated** — three components each with strong safety records
- Mild injection site reactions
- Transient fatigue after TB-500 component (common, 24–48 hrs)
- GHK-Cu blue-green tint at injection site with high copper concentration (harmless)
- No documented synergistic toxicity
- **Angiogenesis caution:** Cancer contraindication (BPC-157 + TB-500 both angiogenic); avoid in active malignancy
- **Pregnancy:** Avoid (insufficient data)

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8. GLOW vs. KLOW — Decision Guide

Choose GLOW when client:

- Has primarily musculoskeletal injury/recovery goals
- Wants anti-aging with skin/collagen focus
- Does NOT have active IBD, MCAS, or autoimmune conditions
- Is cost-conscious (GLOW = lower cost than KLOW)
- Does not need the NF-κB/systemic anti-inflammatory boost of KPV

Choose KLOW when client:

- Has gut inflammation (IBD, leaky gut, MCAS)
- Has significant systemic inflammation alongside physical injury
- Is on NSAIDs (gut protection needed)
- Has autoimmune condition requiring NF-κB suppression
- Wants maximum anti-inflammatory coverage

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9. Stacking GLOW with Other Mavyn Products

Addition	Rationale	Best For
Ipamorelin	GH/IGF-1 amplifies the anabolic environment GLOW works in	Athletes wanting maximum recovery + performance
NAD+	Cellular energy for the repair process	Aging clients, chronic fatigue + injury
Epithalon	Long-term collagen + telomere anti-aging layer	Anti-aging focused clients
Retatrutide	Fat loss during injury recovery (often immobilized clients gain fat)	Metabolic clients recovering from injury
MOTS-c	Mitochondrial efficiency during healing	High-output athletic recovery

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10. FAQ — For Rep Use

Q: Why is GHK-Cu the largest component in GLOW by volume?

A: GHK-Cu is both the most versatile (skin, joints, connective tissue, gene expression) and the safest at high doses. By making GHK-Cu the dominant component, GLOW delivers a robust anti-aging and tissue quality effect that extends beyond just injury recovery — clients get skin improvement, joint maintenance, and systemic gene expression normalization as "bonus" effects even if they started for a sports injury.

Q: How does GLOW compare to just taking BPC-157 alone?

A: BPC-157 alone is excellent for acute injury signaling and gut healing. Adding TB-500 extends the systemic reach dramatically (especially critical for deep muscle, avascular tendons, ligaments). Adding GHK-Cu transforms the quality of the healed tissue. GLOW produces faster healing AND better tissue architecture than BPC-157 alone in clinical experience.

Q: Can GLOW be used as a long-term maintenance protocol?

A: Yes. At maintenance doses (BPC-157 250 mcg EOD, TB-500 2.5mg weekly, GHK-Cu 1mg daily), many athletes and aging clients run GLOW continuously during active periods. The safety profiles of all three components support long-term use with appropriate monitoring.

Q: What results do clients see first?

A: Pain reduction and improved range of motion (BPC-157 anti-inflammatory) typically within 1–2 weeks. Noticeable tissue repair (strength, stability) at 3–4 weeks. Skin quality improvements (collagen, texture) at 6–8 weeks. Full structural healing: 6–12 weeks depending on injury severity.

Q: Is GLOW suitable for non-athletes?

A: Absolutely. The GHK-Cu anti-aging component makes GLOW compelling for any adult 35+ who wants to maintain tissue quality, support joint health, and improve skin appearance. TB-500 and BPC-157's ongoing repair signaling also prevents the

microinjury accumulation that leads to chronic pain in older adults.

Document prepared for Mavyn internal sales training. Not for distribution to patients. For informational purposes only.

KLOW STACK — Mavyn Rep Training Sheet

SKU: KLOW | Strength: 80mg total | Category: Premium Recovery / Anti-Inflammatory / Repair Stack

Composition: BPC-157 (10mg) + GHK-Cu (50mg) + TB-500 (10mg) + KPV (10mg)

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1. What Is the KLOW Stack?

KLOW is Mavyn's premium recovery and repair stack — a precisely formulated combination of four synergistic peptides targeting tissue regeneration, inflammation resolution, collagen remodeling, and immune regulation simultaneously. It represents the most comprehensive recovery peptide formula in the portfolio.

The four components are:

- **BPC-157 (10mg):** Tissue repair, gut healing, angiogenesis, nerve regeneration
- **GHK-Cu (50mg):** Collagen remodeling, anti-fibrotic, anti-aging gene expression
- **TB-500 (10mg):** Systemic angiogenesis, actin mobility, satellite cell activation
- **KPV (10mg):** Anti-inflammatory (α-MSH-derived), gut mucosa protection, NF-κB suppression

Why This Combination?

Each peptide addresses a different phase and mechanism of tissue injury and repair:

Component	Primary Target	Mechanism
BPC-157	Acute healing signal, nerve, gut, local angiogenesis	FAK-paxillin, NO/VEGF
TB-500	Systemic vascular repair, satellite cell recruitment	Actin sequestration, VEGFR2
GHK-Cu	Collagen quality, ECM remodeling, anti-fibrotic	MMP/TIMP balance, lysyl oxidase
KPV	Inflammation resolution, gut barrier, immune regulation	MC1R/MC3R, NF-κB ↓

Together they cover:

1. **Signal phase** (BPC-157): Activates healing cascade at injury site
2. **Vascular phase** (TB-500): Brings blood supply and stem cells to injury
3. **Remodeling phase** (GHK-Cu): Ensures quality collagen replaces damaged tissue
4. **Resolution phase** (KPV): Resolves inflammatory signaling and protects gut barrier

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2. What Is KPV? (The Stack-Differentiator)

KPV is a tripeptide (Lys-Pro-Val) derived from the C-terminal of **α-Melanocyte Stimulating Hormone (α-MSH)**. While α-MSH is best known for melanin production (through MT-I/II), its C-terminal fragment KPV has potent and distinct **anti-inflammatory and gut-protective properties** with virtually none of the melanogenic or sexual effects of full α-MSH.

KPV Mechanism of Action

NF-κB Inhibition:

KPV directly inhibits NF-κB nuclear translocation in macrophages, dendritic cells, and epithelial cells — reducing expression of the entire inflammatory gene program: IL-1β, TNF-α, IL-6, IL-12, COX-2. This is the same master inflammatory pathway that NSAIDs and corticosteroids act on, but through a fundamentally different receptor mechanism.

Melanocortin Receptor Binding (MC1R, MC3R):

KPV binds MC1R and MC3R — but NOT MC4R (no sexual/libido effects) and NOT MC2R (no adrenal/cortisol effects). MC1R and MC3R activation triggers the downstream anti-inflammatory cAMP cascade, delivering potent resolution of inflammation.

Gut Mucosal Protection:

KPV has direct cytoprotective effects on intestinal epithelial cells:

- Reduces intestinal permeability (preserves tight junction integrity)
- Suppresses mucosal NF-κB — reducing local inflammatory mediators in the gut lumen
- Reduces mast cell activation in the gut wall
- Particularly relevant in IBD: KPV suppresses colitis pathology in animal models as effectively as several IBD drugs

Intracellular Delivery:

Remarkably, KPV is transported intracellularly via the PepT1 (peptide transporter 1) system — which is upregulated in inflamed intestinal epithelium. This creates a "targeted delivery" effect: the more inflamed the gut, the more KPV is taken up into inflamed cells. This is why oral KPV is particularly effective for gut inflammation despite being a peptide.

Wound Healing Promotion:

Beyond inflammation, KPV promotes keratinocyte migration and wound closure — complementing the other components of KLOW.

KPV Research

Study 1: IBD / Colitis

- **Citation:** Kannengiesser K et al., *Gut*, 2008; also Dalmasso G et al., multiple publications
- **Model:** DSS-induced colitis in mice
- **Results:** KPV treatment significantly reduced colon shortening, mucosal damage score, and inflammatory cell infiltration. Histological healing comparable to mesalazine (a standard IBD drug). NF-κB activity in colon tissue reduced 65%.

Study 2: Wound Healing

- **Results:** KPV applied to skin wounds: significantly accelerated re-epithelialization. Reduced inflammatory cell infiltration at wound margins. Faster transition from inflammatory to proliferative healing phase.
- **Citation:** Brzoska T et al., *Peptides*, 2008.

Study 3: Gut Barrier Integrity

- **In vitro (Caco-2 cells):** KPV maintained transepithelial electrical resistance (TEER) under inflammatory conditions — direct evidence of tight junction preservation.
- **In vivo:** Intestinal permeability (FITC-dextran assay) significantly preserved in KPV-treated vs. control colitis animals.

Study 4: Systemic Anti-Inflammatory

- **LPS (endotoxin) challenge model:**
- KPV pre-treatment: serum TNF-α reduced 80%, IL-6 reduced 70%, IL-1β reduced 65%
- Post-treatment (after LPS): survival improved significantly vs. control
- Mechanism: rapid NF-κB suppression across multiple tissue compartments

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3. Clinical Research for the Stack Concept

No clinical RCT has tested this exact four-component combination. However, each component has its own research base (see individual training sheets), and the rationale for combination is:

Phase 1 (Injury/Acute — Hours to Days):

- BPC-157 initiates healing cascade immediately upon injection
- KPV suppresses acute inflammatory cytokine storm → prevents over-inflammation → protects healing tissue

Phase 2 (Proliferation — Days to 2 Weeks):

- TB-500 recruits stem cells and creates vascular network
- BPC-157 continues to guide organized tissue growth
- KPV maintains anti-inflammatory environment

Phase 3 (Remodeling — Weeks to Months):

- GHK-Cu drives organized collagen synthesis (vs. random scar tissue)
- GHK-Cu suppresses TGF-β1 → reduces fibrosis
- All four components contribute to anti-inflammatory matrix environment

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4. Primary Uses & Benefits

Application	Best For
Post-surgical recovery	Accelerated healing, reduced inflammation, quality scarring
Sports injury recovery	Muscle, tendon, ligament, joint — comprehensive repair
Chronic injury / tendinopathy	Persistent injuries unresponsive to standard treatment
IBD / gut inflammation	KPV + BPC-157 dual gut action
Anti-aging tissue maintenance	GHK-Cu collagen + TB-500/BPC healing signaling
Post-trauma (accident, injury)	Maximum healing speed with quality tissue outcome
Autoimmune inflammatory flares	KPV + BPC-157 anti-inflammatory at tissue level

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5. Key Selling Points (Rep Talking Points)

- ■ **The most comprehensive healing stack available** — covers all four phases of tissue repair (signaling, vascular, remodeling, resolution) with peptides targeting each; no other single product does this
- ■ **KPV is the differentiator from GLOW** — adding KPV provides NF-κB suppression and gut barrier protection; for clients with gut issues or systemic inflammation alongside physical injury, KLOW addresses both
- ■ **GHK-Cu at 50mg = dominant volume** — the high GHK-Cu dose in KLOW emphasizes the anti-aging/remodeling component; collagen quality and gene expression reset is a priority in this formulation
- ■ **Gut protection during healing** — many clients recovering from injury are also on NSAIDs, which damage the gut; KPV + BPC-157 directly counteract NSAID GI toxicity while BPC-157 heals the injury the NSAIDs are managing
- ■ **Pre-made stack = convenience + compliance** — instead of managing 4 separate peptides with separate dosing schedules, KLOW simplifies to one protocol; higher compliance → better outcomes

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6. Common Protocols

Standard KLOW Protocol

- **Dose:** Reconstituted as prescribed; typical split per injection
- BPC-157 component: 250–500 mcg/injection
- TB-500 component: 2.5–5mg/injection (2x/week)
- GHK-Cu: 1–2mg/injection daily
- KPV: 250–500 mcg/injection 1–2x/day

If supplied as pre-mixed vial (KLOW 80mg total):

- Use reconstitution protocol provided with vial
- Typical injection volume: 0.2–0.5 mL SubQ once daily
- Loading phase: 2 weeks daily
- Maintenance: 3–4x/week for 4–6 weeks

Acute Injury Protocol

- **Phase 1 (Days 1–14):** Daily KLOW injections
- **Phase 2 (Days 15–42):** Every other day
- **Phase 3 (Maintenance):** 2–3x/week until full recovery

Gut-Focused Protocol

- **Oral delivery for gut component:** BPC-157 + KPV oral capsule (where available) + systemic KLOW injection for systemic effects
- **Duration:** 8–12 weeks for IBD; 4 weeks for acute gut issues

Reconstitution

- Reconstitute with bacteriostatic water: 2–3 mL for the 80mg combination vial
- Roll gently; do not shake
- Refrigerate immediately after reconstitution

Storage

- Lyophilized: refrigerate; stable 12–18 months
- Reconstituted: refrigerate; use within 30 days

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7. Side Effects & Safety Profile

- **Profile is the composite of four well-tolerated peptides** — no documented synergistic toxicity
- Mild injection site reactions (standard)
- Mild transient fatigue (possible — common with healing peptide protocols as cellular energy is redirected)
- GHK-Cu: very rare mild nausea at high dose
- TB-500: mild transient fatigue post-injection
- KPV: excellent tolerability; mild GI upset in rare cases
- BPC-157: see individual sheet — extremely well tolerated

- **Theoretical concerns:** Angiogenic properties (cancer caution); see BPC-157 and TB-500 individual sheets
- **Contraindications:** Active malignancy, pregnancy

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8. KLOW vs. GLOW

Feature	KLOW	GLOW
BPC-157	■ 10mg	■ 10mg
GHK-Cu	■ 50mg	■ 50mg
TB-500	■ 10mg	■ 10mg
KPV	■ 10mg	■
Anti-inflammatory (NF-κB)	■■ (KPV adds)	■ (BPC/GHK only)
Gut protection	■■ (KPV + BPC)	■ (BPC only)
Autoimmune/IBD application	■■ Best	■ Good
Cost	Higher	Lower
Best for	Gut issues + injury; max anti-inflammatory	Standard injury/recovery

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9. FAQ — For Rep Use

Q: Who should get KLOW vs. GLOW?

A: KLOW is the premium choice for clients who have: (1) gut inflammation alongside physical injury, (2) IBD/Crohn's/colitis, (3) MCAS or systemic inflammatory conditions, (4) autoimmune conditions where NF-κB suppression is especially important, or (5) clients on NSAIDs who need gut protection. GLOW is excellent for straightforward musculoskeletal recovery without the gut/autoimmune component.

Q: What does KPV add that BPC-157 doesn't already provide?

A: BPC-157 is pro-healing with anti-inflammatory properties (via NO/COX-2 pathway). KPV adds a completely different anti-inflammatory mechanism: MC1R/MC3R → NF-κB suppression → direct cytokine storm prevention. BPC-157 heals the tissue; KPV quiets the immune overactivation. For clients with IBD, MCAS, or significant systemic inflammation, KPV is the critical differentiator.

Q: Can KLOW be taken orally?

A: The BPC-157 and KPV components have documented oral activity. The full KLOW stack oral delivery isn't standardized, but oral forms of the BPC-157 and KPV components are used in IBD protocols. For systemic musculoskeletal use, SubQ injection is recommended.

Q: Is the GHK-Cu dose in KLOW very high?

A: At 50mg, the GHK-Cu component dominates the formulation by weight. This is intentional — GHK-Cu drives the remodeling phase of healing (collagen quality, anti-fibrotic, anti-aging gene expression) and is safe at high doses. The high copper peptide loading ensures the remodeling phase is thoroughly supported.

Q: How does KLOW compare to PRP or stem cell injections?

A: PRP and stem cells are regenerative treatments requiring clinical procedures and specialized equipment. KLOW is a peptide protocol clients can administer SubQ at home after training. They're not mutually exclusive — many clinicians pair KLOW with PRP/prolotherapy for synergistic effects. KLOW's advantage: systemic reach, daily dosing convenience,

comprehensive multi-mechanism action.

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